

PreMed:

For pre-med, the proper anchor is AAMC's competency model, AMCAS guidance, MSAR school requirements, AACOM/AACOMAS instructions, and each school's own prerequisites, not a generic biology-major checklist. AAMC now frames applicants through 17 competencies, and prerequisites vary materially by school, which is why your app needs a transcript-to-requirement engine instead of pretending "one year of everything" solves it.

For dedicated and shelves, ground the blueprint in USMLE Step 2 CK's official content outline and physician-task framework, then map clerkship modules to NBME's discipline-specific shelf outlines. NBME currently provides outlines across medicine, surgery, pediatrics, OB/GYN, psychiatry, family medicine, neurology, ambulatory care, and more.

Use these as the two "God Blueprint" prompts for Noctyrium or another build agent.

PREMED GOD BLUEPRINT: U.S. + INTERNATIONAL + SGU PREMEDICAL OPERATING SYSTEM

You are building the most comprehensive, adaptive, audit-ready Pre-Med Blueprint possible for use inside an education and application-planning platform.

This is not a generic "how to get into medical school" article. Do not output a lecture, long essay, or static checklist.

Build a navigable, modular Pre-Med Operating System that can be installed into an app as structured categories, tasks, trackers, timelines, requirements, dashboards, alerts, and decision tools.

The primary audience is U.S. pre-medical students applying to MD and DO programs. It must also support:

- * International students pursuing U.S. medical education
- * Caribbean/international medical pathways
- * SGU undergraduate, post-baccalaureate, premedical, BSc/MD, and direct-entry pathway students
- * Nontraditional students
- * Career changers
- * Students from any undergraduate major
- * Students beginning in high school, college, post-bacc, gap year, or after graduation

Do not assume the user is a biology major. The system must support humanities, business, psychology, engineering, public health, social science, nursing, international, and other majors while intelligently identifying missing medical-school prerequisites.

The system must be evidence-backed, source-labeled, versioned, and conservative when information is school-specific or uncertain. Never invent requirements, admissions statistics, deadlines, prerequisite policies, AP policies, online-course policies, community-college policies, international-credit policies, or letters-of-recommendation rules.

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CORE PRODUCT OBJECTIVE

Build a full Pre-Med Command Center that allows a student to:

1. Import or manually enter an academic record.
2. Upload or parse a DARS report, degree audit, transcript, or DARS-equivalent.

3. Select an intended major, minor, university, country, graduation year, target application cycle, and target pathway.
4. Identify graduation requirements separately from medical-school prerequisites.
5. Map completed coursework against:
 - * Degree requirements
 - * General education requirements
 - * Major requirements
 - * Medical-school prerequisites
 - * MCAT-relevant coursework
 - * Recommended upper-level science coursework
 - * Individual target-school requirements
6. Track grades, science GPA, cumulative GPA, trends, repeated courses, withdrawals, pass/fail coursework, transfer credit, AP/IB/A-Level/CAPE credit, post-bacc work, and graduate coursework.
7. Track clinical experience, nonclinical service, research, shadowing, leadership, employment, tutoring, teaching, advocacy, athletics, creative work, military experience, caregiving, and meaningful life experiences.
8. Track letters of recommendation, contacts, deadlines, evaluator relationships, waivers, and submission status.
9. Build a school list using fit logic rather than prestige worship, humanity's oldest group project.
10. Guide the student through AMCAS, AACOMAS, TMDSAS where applicable, SGU and international pathways, secondaries, interviews, financial aid, acceptances, waitlists, and matriculation.
11. Generate a realistic personalized readiness score, gap analysis, and next-best-action list.
12. Preserve a longitudinal evidence vault so every activity, story, reflection, patient interaction, achievement, hardship, leadership moment, and application-worthy insight is captured while it is still fresh.



PART I: PLATFORM ARCHITECTURE

Create these top-level navigation categories:

1. Mission Control
2. Academic Map
3. Degree Audit and Course Planner
4. Medical School Prerequisites
5. MCAT Readiness
6. Experiences and Competencies
7. Clinical Exposure
8. Community Service and Advocacy
9. Research and Scholarly Work
10. Leadership, Teaching, and Employment
11. Letters of Recommendation
12. Personal Narrative Vault
13. School Discovery and Fit Engine
14. Application Cycle Center
15. Secondary Essay Lab
16. Interview Preparation
17. Financial Planning
18. Gap Year and Reapplication Strategy
19. International and SGU Pathways
20. Wellness, Time, and Burnout Safeguards

21. Data Sources, Audit Trail, and Change Log

Each category must contain structured cards, subtasks, evidence fields, tags, deadlines, alerts, linked resources, source records, and performance indicators.

Do not populate the interface with massive paragraphs. Every item must be actionable, collapsible, filterable, searchable, and connected to other categories.

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PART II: USER ONBOARDING AND PROFILE ENGINE

Create an onboarding flow that asks for:

Identity and Academic Context

- * Full legal name
- * Preferred name
- * Citizenship(s)
- * Permanent residence
- * State of legal residence
- * Country of undergraduate institution
- * Undergraduate institution
- * Major(s)
- * Minor(s)
- * Expected graduation date
- * Current academic year

- * Transfer history
- * Community college history
- * Dual enrollment history
- * AP, IB, A-Level, CAPE, CLEP, or equivalent credit
- * Study-abroad coursework
- * International coursework
- * Prior degrees
- * Post-baccalaureate coursework
- * Graduate coursework
- * Intended medical-school entry year
- * Whether the student is traditional, gap-year, career-changer, post-bacc, international, or reapplicant

Medical Pathway Intent

Allow multiple selections:

- * U.S. MD
- * U.S. DO
- * Texas MD/DO pathway
- * Canadian medical school
- * Caribbean MD
- * SGU pathway
- * International medical school
- * MD/PhD
- * MD/MPH

- * MD/MBA
- * Military medicine
- * Primary care focused
- * Academic medicine
- * Undecided

Academic Import Options

- * Upload transcript PDF
- * Upload DARS report
- * Upload degree audit
- * Upload CSV of courses
- * Manual entry
- * Institution-specific planner import
- * International transcript conversion notes
- * Mark uncertain items for advisor review

Never treat OCR or parsed transcript data as final. Require user confirmation before locking entries.



PART III: DEGREE AUDIT AND COURSE-MAPPING ENGINE

Build a structured academic engine that separates four distinct things:

1. Degree completion requirements
2. Major and minor requirements
3. Medical-school prerequisites
4. MCAT preparation requirements

The system must prevent the common failure mode where a student finishes their major but accidentally lacks a school-specific prerequisite. A tremendous achievement, in the same way as building a beautiful bridge that stops three feet before the riverbank.

For every course, track:

- * Institution
- * Course code
- * Course title
- * Department
- * Term
- * Credits
- * Grade
- * Grading basis
- * Lab included
- * Lecture credits
- * Lab credits
- * In-person, hybrid, or online
- * Community college versus four-year institution
- * AP/IB/A-Level/CAPE/transfer designation
- * Repeated course status
- * Withdrawn or incomplete status

- * Medical-school subject equivalency
- * MCAT subject mapping
- * Degree requirement mapping
- * Target-school acceptance confidence
- * Source of policy
- * Date policy was last verified
- * Advisor-review flag

Create prerequisite groups:

- * Introductory biology with lab
- * General chemistry with lab
- * Organic chemistry with lab
- * Physics with lab
- * Biochemistry
- * Mathematics
- * Statistics
- * English/writing-intensive coursework
- * Psychology
- * Sociology
- * Genetics
- * Cell biology
- * Molecular biology
- * Physiology
- * Anatomy
- * Microbiology

- * Immunology
- * Neuroscience
- * Public health
- * Ethics
- * Behavioral science
- * Communication/writing
- * Upper-division biology
- * Other school-specific coursework

For each target school, label a requirement as:

- * Required
- * Recommended
- * Accepted alternative
- * School-specific
- * Unknown or awaiting verification
- * Not required
- * Potential concern due to online format
- * Potential concern due to community-college format
- * Potential concern due to AP/IB substitution
- * Potential concern due to international coursework

Do not create universal hard rules for online labs, AP credit, or community-college credit. These policies differ by school and must be stored per target institution with a source citation, policy date, and confidence score.

PART IV: GPA, ACADEMIC PERFORMANCE, AND RECOVERY SYSTEM

Build calculators for:

- * Cumulative GPA
- * BCPM/science GPA
- * AACOMAS science GPA approximation
- * Undergraduate GPA
- * Post-bacc GPA
- * Graduate GPA
- * Trend GPA by semester
- * Trend GPA by academic year
- * Recent 30-credit GPA
- * Recent 45-credit GPA
- * Repeated-course impact
- * Projected GPA after planned courses
- * Credit-weighted grade scenarios
- * “What grade do I need?” planner
- * Academic-risk alerts
- * Upward-trend analysis

Create visual dashboards showing:

- * Overall GPA trajectory
- * Science GPA trajectory

- * Course difficulty over time
- * Grade distribution
- * Credits completed versus credits remaining
- * Prerequisite completion percentage
- * High-risk prerequisite performance
- * MCAT readiness based on coursework completion
- * Application readiness based on GPA trend plus experiential evidence

Create recovery pathways for:

- * One poor semester
- * Several poor semesters
- * Low science GPA
- * Low cumulative GPA
- * Late discovery of medicine
- * Weak prerequisite performance
- * Academic misconduct history requiring professional guidance
- * Failed or repeated courses
- * Long academic gap
- * International transcript uncertainty
- * Need for post-bacc or special master's program consideration

Do not promise that any GPA or activity profile guarantees admission.

PART V: EXPERIENCE AND COMPETENCY ENGINE

Build a tracker mapped to the AAMC premed competency framework.

For every activity, collect:

- * Activity name
- * Organization
- * Role
- * Location
- * Start date
- * End date
- * Hours per week
- * Weeks per year
- * Total hours
- * Paid versus unpaid
- * Clinical versus nonclinical
- * Direct patient contact level
- * Leadership level
- * Population served
- * Supervisor
- * Verification contact
- * Key responsibilities
- * Impact
- * Challenge faced
- * Reflection

- * Competencies demonstrated
- * Potential application category
- * Most meaningful candidate flag
- * Story-bank links
- * Proof documents
- * Confidentiality warning
- * Last updated date

Create competency mapping across:

Professional Competencies

- * Service orientation
- * Social skills
- * Cultural awareness
- * Teamwork
- * Oral communication
- * Ethical responsibility
- * Reliability and dependability
- * Resilience and adaptability
- * Capacity for improvement

Thinking and Reasoning Competencies

- * Critical thinking
- * Quantitative reasoning

- * Scientific inquiry
- * Written communication

Science Competencies

- * Living systems
- * Human behavior
- * Social systems

For every competency, generate:

- * Evidence strength
- * Number of distinct experiences supporting it
- * Depth score
- * Reflection quality score
- * Missing-evidence warning
- * Suggested future experiences
- * Potential interview stories
- * Potential personal-statement stories
- * Risk of redundancy

Do not reduce a person to hour totals. Hours matter, but reflective depth, longitudinal commitment, impact, responsibility, and coherent motivation matter more.

PART VI: CLINICAL, SERVICE, RESEARCH, AND SHADOWING TRACKERS

Clinical Experience

Track:

- * Paid clinical work
- * Volunteer clinical work
- * Hospital volunteering
- * EMT
- * Scribing
- * CNA/MA/PCT work
- * Hospice
- * Free clinic work
- * Patient navigation
- * Public health outreach
- * Telehealth support
- * International clinical exposure
- * Direct versus indirect patient contact

Flag ethical risks:

- * Performing tasks outside training
- * Misrepresenting role
- * Breaches of confidentiality
- * Medical tourism concerns

- * Unsupervised patient care
- * Inflated clinical claims

Nonclinical Service

Track:

- * Food insecurity
- * Homelessness outreach
- * Refugee/immigrant support
- * Youth mentoring
- * Disability advocacy
- * Prison education
- * Domestic violence support
- * Community organizing
- * Crisis support
- * Faith-based service when appropriate and respectful
- * Longitudinal service to underserved populations

Research

Track:

- * Basic science
- * Clinical research
- * Public health

- * Quality improvement
- * Education research
- * Humanities research
- * Case reports
- * Posters
- * Abstracts
- * Publications
- * Data analysis
- * IRB status
- * PI relationship
- * Authorship status
- * Research hours
- * Deliverables
- * Skills acquired

Shadowing

Track:

- * Specialty
- * Physician
- * Setting
- * Inpatient/outpatient
- * Community/academic
- * Procedure exposure
- * Reflection

- * Lessons about medicine
- * Hours
- * Virtual versus in-person designation

Create warnings against collecting random specialty hours merely to decorate an application. The app should prioritize informed career exploration and meaningful reflection.

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PART VII: LETTER OF RECOMMENDATION SYSTEM

Build a secure Letter Tracker with:

- * Recommender name
- * Credentials
- * Institution
- * Relationship
- * Course or role context
- * Dates known
- * Strength estimate
- * Whether recommender can write specifically
- * Letter type
- * Target application system
- * Deadline
- * Request date
- * Resume provided

- * Personal statement provided
- * Transcript provided
- * Brag sheet provided
- * Follow-up date
- * Submission status
- * Thank-you status
- * Waiver status
- * Confidentiality status

Letter categories:

- * Science faculty
- * Non-science faculty
- * Research mentor
- * Clinical supervisor
- * Physician
- * Employer
- * Committee letter
- * Composite letter
- * Character/community letter where permitted

Build a “Letter Readiness” dashboard that identifies missing letter categories based on each target school’s documented policy.

PART VIII: PERSONAL NARRATIVE VAULT

Create an ongoing narrative capture system.

Every student should be able to quickly log:

- * Patient interaction reflections
- * Academic setbacks
- * Service moments
- * Leadership failures
- * Research discoveries
- * Family experiences
- * Cultural experiences
- * Identity development
- * Moments that clarified motivation for medicine
- * Ethical dilemmas
- * Clinical observations
- * Difficult conversations
- * Growth moments
- * Gratitude reflections
- * Memorable quotes
- * “Why medicine?” fragments

For each entry, tag:

- * Date

- * Setting
- * Experience category
- * Competencies demonstrated
- * Emotion
- * Insight
- * Future use: personal statement, most meaningful activity, secondary, interview, scholarship, reapplication explanation
- * Confidentiality safety flag
- * Patient-identifying-information check

The system must never encourage disclosure of protected health information.

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PART IX: SCHOOL DISCOVERY AND FIT ENGINE

Build a school database layer for:

- * U.S. MD programs
- * U.S. DO programs
- * Canadian programs where applicable
- * Texas pathway institutions where applicable
- * SGU and international pathway options
- * Caribbean schools only when clearly labeled by accreditation, eligibility, financing, clinical-training, and residency-risk context

For each school, track:

- * School name
- * Degree type
- * Accreditation status
- * Campus
- * State/country
- * Public/private status
- * In-state preference
- * International applicant policy
- * Canadian applicant policy
- * Prerequisite coursework
- * Recommended coursework
- * Online course policy
- * Community college policy
- * AP/IB policy
- * MCAT policy
- * GPA and MCAT ranges where officially available
- * Mission
- * Service orientation
- * Research orientation
- * Rural/urban focus
- * Primary care focus
- * Diversity initiatives
- * Special pathways
- * Dual-degree programs
- * Financial aid information

- * Tuition and estimated cost
- * Secondary fee
- * Application deadline
- * Early decision option
- * Letters policy
- * CASPer/Preview/interview requirements where applicable
- * Technical standards
- * Interview format
- * Admissions contact
- * Source URL
- * Source type
- * Last verified date
- * Confidence score

Use school-fit logic based on:

- * Residency
- * Academic metrics
- * Mission alignment
- * Geographic preferences
- * Financial considerations
- * Family/support constraints
- * Research interests
- * Service history
- * Rural/urban fit
- * International applicant eligibility

- * Diversity of school list

- * Application risk

Do not label schools “safety schools.” Medical school admissions are not a grocery-store parking lot. Use:

- * Aspirational

- * Competitive

- * Likely competitive

- * Mission-aligned reach

- * High uncertainty

- * Data insufficient

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PART X: APPLICATION CYCLE ENGINE

Create separate workflow modules for:

AMCAS

Track:

- * Personal information

- * Schools attended

- * Coursework

- * Work and activities
- * Most meaningful experiences
- * Letters
- * Personal statement
- * Medical schools
- * Transcript submission
- * Verification
- * Fee assistance eligibility
- * Application status
- * Secondaries
- * Interview invitations
- * Acceptance protocols
- * Waitlists
- * Commit to Enroll/Plan to Enroll where applicable

AACOMAS

Track:

- * Academic history
- * Coursework
- * Experiences
- * Achievements
- * Supporting information
- * Evaluations
- * Program materials

- * Verification
- * Supplemental requirements
- * School-specific deadlines
- * AACOMAS traffic guidelines

TMDSAS

Create a distinct workflow, not a watered-down AMCAS clone.

SGU and International Pathways

Track:

- * Entry route
- * Academic prerequisites
- * Required documents
- * MCAT requirements where applicable
- * International credential evaluation
- * Visa/immigration considerations
- * Financial planning
- * Clinical training implications
- * Accreditation and eligibility verification
- * Residency and licensure considerations
- * Country-specific documentation

Build a master application timeline:

- * 24 months before applying
- * 18 months before applying
- * 12 months before applying
- * 9 months before applying
- * 6 months before applying
- * Application opening
- * Submission window
- * Verification
- * Secondary phase
- * Interview phase
- * Acceptance/waitlist phase
- * Matriculation phase

All dates must be dynamic and must pull from official current-cycle sources.

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PART XI: SECONDARY ESSAY LAB

Build a school-specific secondary essay engine.

For each prompt:

- * School
- * Cycle year

- * Prompt text
- * Character/word limit
- * Source
- * Last verified date
- * Theme tags
- * Deadline
- * Draft versions
- * Feedback
- * Final version
- * Submission status

Common prompt categories:

- * Why us?
- * Diversity and inclusion
- * Challenge/adversity
- * Service
- * Leadership
- * Gap year
- * Academic discrepancy
- * COVID-era impact
- * Rural medicine
- * Underserved populations
- * Research
- * Cultural humility
- * Ethical challenge

- * Future goals
- * Other impactful experiences

Build a reusable-story matrix so the user can adapt authentic experiences without copy-pasting robotic sludge across applications.

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PART XII: INTERVIEW PREPARATION SYSTEM

Create modules for:

- * Traditional interviews
- * MMI
- * Panel interviews
- * Group interviews
- * Virtual interviews
- * Asynchronous/video interviews
- * Ethics stations
- * Healthcare policy questions
- * Teamwork stations
- * Conflict scenarios
- * Role-play stations
- * “Why medicine?”
- * “Why this school?”
- * Resume walkthrough

- * Gap-year explanation
- * Weaknesses and failures
- * Research explanation
- * Current events in health care
- * Questions for interviewers
- * Post-interview thank-you tracking

Create an interview performance tracker:

- * Answer clarity
- * Structure
- * Authenticity
- * Concision
- * Evidence use
- * Professionalism
- * Ethical reasoning
- * Nonverbal communication
- * Confidence
- * Filler-word count
- * Red-flag language
- * Feedback themes

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PART XIII: SGU AND INTERNATIONAL PATHWAY MODULE

Build a dedicated SGU/international pathway section that supports:

- * SGU undergraduate pathways
- * SGU post-baccalaureate science preparation
- * SGU BSc/MD or direct-entry pathway context where applicable
- * International secondary-school systems
- * CAPE, IB, A-Level, BTEC, and other credential mapping
- * International transcript and grading-scale interpretation
- * Country-specific premedical routes
- * U.S. versus international prerequisites
- * MCAT rules based on citizenship and pathway
- * Visa and financing planning
- * Clinical-placement and residency-planning education
- * Accreditation/source verification
- * Appropriate warnings when pathway outcomes, costs, or licensure implications require individualized due diligence

Do not market any school. Present verified facts, uncertainty, and decision criteria.

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PART XIV: DASHBOARDS AND ALERTS

Create these dashboards:

Mission Control Dashboard

- * Current academic standing
- * Prerequisite completion
- * MCAT readiness
- * Experience balance
- * Letter readiness
- * School-list readiness
- * Application-cycle readiness
- * Upcoming deadlines
- * Highest-impact next actions
- * Risk alerts

Academic Dashboard

- * GPA graphs
- * Course planner
- * Degree completion
- * Prerequisite completion
- * Academic risk flags

Experience Dashboard

- * Hours by category
- * Longitudinal commitment
- * Competency coverage
- * Reflection depth

- * Missing categories
- * Verification completeness

Application Dashboard

- * Primary application completion
- * Transcript status
- * Letter status
- * School list
- * Secondary completion
- * Interview status
- * Financial aid
- * Offer/waitlist status

Readiness Score

Calculate a transparent score using:

- * Academic readiness
- * Prerequisite completion
- * MCAT readiness
- * Clinical exposure
- * Service orientation
- * Research/scholarship
- * Leadership/teaching/employment
- * Competency evidence

- * Letters
- * Narrative maturity
- * School-list realism
- * Application timing

Never make this a black-box admission predictor. Show the weight of every component and label it as a planning tool, not a guarantee.



PART XV: RESOURCE INTEGRATION

Create resource cards with:

- * Resource name
- * Category
- * Official versus third-party status
- * Cost
- * Free/paid designation
- * Best use case
- * Limitations
- * Linked blueprint modules
- * Last reviewed date

Include:

- * AAMC premed competencies resources
- * AMCAS applicant guide
- * AACOMAS application instructions
- * MSAR and official medical-school websites
- * Choose DO Explorer
- * TMDSAS official materials
- * AAMC MCAT resources
- * Khan Academy MCAT material
- * Jack Westin CARS and MCAT resources
- * Anki
- * Premed school advising resources
- * Scholarship and fee-assistance resources
- * Official SGU program and admissions materials

Do not treat third-party resources as official policy sources.

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PART XVI: DATA GOVERNANCE

Every school-specific or policy-specific item must have:

- * Source
- * Source type
- * Publication date if available
- * Date last checked

- * Confidence rating
- * Verified/not verified state
- * Notes field
- * Change log

Use source hierarchy:

1. Official school admissions webpage
2. Official school catalog
3. Official application service guide
4. Official accreditation body
5. Official AAMC/AACOM/TMDSAS source
6. Direct written confirmation from admissions office
7. Reputable database
8. Third-party source, clearly labeled

When sources conflict:

- * Display both
- * Flag conflict
- * Prioritize current official school policy
- * Recommend direct confirmation from admissions when consequential

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FINAL OUTPUT REQUIREMENTS

Generate the blueprint as:

1. A full category hierarchy.
2. Structured database schemas.
3. User onboarding flow.
4. Dashboard specifications.
5. Required fields for every tracker.
6. Automated rule logic.
7. Source and verification framework.
8. Timeline templates.
9. School-fit logic.
10. Application workflow logic.
11. SGU and international adaptations.
12. A “minimum viable student setup” checklist.
13. A “high-performance applicant setup” checklist.
14. A “recovery and reinvention” pathway for students with academic or experiential gaps.

The installed blueprint must feel like a living command center, not a textbook chapter.

CLINICAL DEDICATED YEARS + SHELF EXAM GOD BLUEPRINT

You are building the most comprehensive, adaptive, clinically realistic, exam-aligned study operating system for medical students in clinical rotations, shelf examinations, NBME exams, Step 2 CK preparation, dedicated study periods, and transition-to-residency readiness.

This is not a generic clerkship guide. Do not output a long lecture or a simplistic calendar.

Build a navigable, installable Clinical Performance Operating System composed of structured categories, shelf-specific blueprints, daily workflows, question-bank systems, patient-log integration, knowledge-gap analytics, clinical reasoning drills, error logs, assessment dashboards, and individualized study plans.

The system should support:

- * U.S. MD students
- * U.S. DO students
- * International medical students
- * Caribbean medical students
- * SGU students
- * Students preparing for NBME subject exams
- * Students preparing for Step 2 CK
- * Students preparing for internal school comprehensive exams
- * Students preparing for IFOM Clinical Science where relevant
- * Students in core clerkships, electives, sub-internships, and dedicated periods

Use current official USMLE Step 2 CK content outlines, physician-task specifications, NBME clinical subject exam outlines, NBME sample items, official assessment guidance, and school-specific clerkship requirements where available.

Do not invent shelf weighting, clerkship policies, passing scores, exam dates, grading rules, score conversions, or institutional requirements. These must be editable by school and verified against current official sources.



CORE OBJECTIVE

Build a Clinical Years Command Center that helps a student:

1. Perform well on clerkships.
2. Learn from real patients without losing the educational thread in a sea of notes, scutwork, and inexplicable printer failures.
3. Master shelf-relevant clinical reasoning.
4. Build longitudinal Step 2 CK readiness during rotations.
5. Track required encounters, procedures, presentations, feedback, professionalism, and evaluations.
6. Translate missed questions into durable clinical knowledge.
7. Convert clinical experiences into illness scripts, differential diagnoses, and management algorithms.
8. Identify weak systems, weak tasks, weak disciplines, and weak question styles.
9. Build a dedicated period that is diagnostic, adaptive, and realistic.
10. Protect sleep, recovery, clinical functioning, and mental health during high-load study periods.



PART I: PLATFORM ARCHITECTURE

Create these primary sections:

1. Mission Control
2. Rotation Planner
3. Shelf Exam Blueprints
4. Step 2 CK Master Blueprint
5. Daily Clinical Workflow
6. Patient and Encounter Log
7. Illness Script Library
8. Differential Diagnosis Engine
9. Question Bank Center
10. NBME and Self-Assessment Center
11. Error Log and Weakness Analytics
12. Anki and Spaced-Repetition System
13. Clinical Skills and Presentations
14. Notes, Procedures, and Core Tasks
15. Ethics, Quality, Safety, and Biostatistics
16. Specialty-Specific Playbooks
17. Dedicated Study Planner
18. Residency Readiness and ERAS Evidence Vault
19. Wellness, Sleep, and Recovery
20. Sources, Version Control, and Audit Trail

The blueprint must work in both rotation mode and dedicated mode.

Rotation mode emphasizes clinical performance, daily question blocks, shelf preparation, and patient-derived learning.

Dedicated mode emphasizes diagnostic baseline assessment, prioritized remediation, timed blocks, targeted review, full-length practice, pacing, and exam-day preparation.



PART II: ONBOARDING PROFILE

Collect:

- * Medical school
- * Degree pathway: MD, DO, IMG, SGU, other
- * Clinical curriculum model
- * Rotation order
- * Rotation dates
- * Shelf exam dates
- * School exams
- * Step 2 CK target date
- * Step 2 CK target score
- * Existing Step 1 baseline
- * Prior NBME or shelf scores
- * Question-bank subscriptions
- * Anki deck(s)
- * Current daily study hours
- * Commute time
- * Call schedule
- * Days off

- * Research, family, work, health, or religious obligations
- * Sleep baseline
- * Learning preferences
- * Weakest systems
- * Weakest specialties
- * Most common question errors
- * Target residency interests, if known

Create initial modes:

- * Starting clerkships
- * Mid-clerkship catch-up
- * Final two weeks before shelf
- * Shelf remediation
- * Dedicated Step 2 CK
- * Step 2 CK maintenance during rotations
- * Post-clerkship longitudinal review
- * International exam adaptation

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PART III: OFFICIAL EXAM BLUEPRINT LAYER

Build the system from three linked dimensions:

Dimension A: Organ System

- * Cardiovascular
- * Respiratory
- * Gastrointestinal
- * Renal/urinary
- * Endocrine
- * Reproductive
- * Neurologic
- * Musculoskeletal
- * Dermatologic
- * Hematologic/oncologic
- * Infectious disease
- * Rheumatologic/immunologic
- * Behavioral health
- * Pediatrics/development
- * Geriatrics
- * Multisystem disorders
- * Nutrition
- * Environmental/occupational medicine
- * Toxicology
- * Biostatistics/epidemiology
- * Ethics
- * Quality improvement
- * Patient safety
- * Communication

Dimension B: Clinical Task

- * History taking
- * Physical examination
- * Differential diagnosis
- * Diagnostic testing
- * Interpretation of diagnostic studies
- * Initial management
- * Acute stabilization
- * Longitudinal management
- * Prevention
- * Screening
- * Counseling
- * Risk-factor modification
- * Prognosis
- * Patient safety
- * Ethics/legal issues
- * Communication and shared decision-making
- * Systems-based practice
- * Evidence appraisal

Dimension C: Clinical Discipline

- * Internal Medicine
- * Surgery
- * Pediatrics

- * Obstetrics and Gynecology
- * Psychiatry
- * Family Medicine
- * Emergency Medicine
- * Neurology
- * Ambulatory Care
- * Geriatrics
- * Critical care
- * Preventive medicine
- * Health systems science

Every knowledge object, question, patient encounter, Anki card, note, missed-question concept, and assessment item must be tagged across all three dimensions.

This allows the app to identify whether a student is weak in:

- * Cardiology generally
- * Acute cardiology management specifically
- * Internal-medicine cardiology questions
- * Diagnostic interpretation
- * Preventive cardiology
- * Time management on long vignettes
- * Data interpretation
- * Ethics and patient safety

PART IV: ROTATION-SPECIFIC SHELF BLUEPRINTS

Create dedicated modules for:

1. Internal Medicine
2. Surgery
3. Pediatrics
4. Obstetrics and Gynecology
5. Psychiatry
6. Family Medicine
7. Neurology
8. Emergency Medicine
9. Ambulatory Care
10. Geriatrics
11. Critical Care where applicable
12. Electives and sub-internships
13. Comprehensive Clinical Science Exam
14. Step 2 CK

Each shelf module must include:

- * Official NBME exam alignment
- * Rotation learning objectives
- * School-specific requirements
- * High-yield systems

- * High-yield diagnoses
- * Chief complaints
- * Must-not-miss emergencies
- * Common management algorithms
- * Screening and prevention
- * Ethics and communication issues
- * Procedures/skills if relevant
- * Required patient encounters
- * Question-bank study plan
- * NBME/self-assessment schedule
- * Anki tags
- * Error-log filters
- * Last-two-week review plan
- * Exam-day strategy
- * Shelf-to-Step-2 carry-forward plan

Do not hard-code percentages unless they are taken from the current official exam outline and labeled with date/source.

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PART V: SPECIALTY PLAYBOOK STRUCTURE

For every specialty, build:

A. Core Presentations

Example format:

- * Chest pain
- * Dyspnea
- * Fever
- * Abdominal pain
- * Altered mental status
- * Syncope
- * Headache
- * Vaginal bleeding
- * Pelvic pain
- * Rash
- * Jaundice
- * Weakness
- * Weight loss
- * Joint pain
- * Anxiety
- * Psychosis
- * Suicidality
- * Pediatric fever
- * Failure to thrive
- * Postoperative fever
- * Trauma

B. Illness Scripts

For each condition include:

- * Epidemiology
- * Risk factors
- * Pathophysiology
- * Classic presentation
- * Atypical presentation
- * Key history clues
- * Key exam findings
- * First-line tests
- * Confirmatory tests
- * Differential diagnoses
- * Initial stabilization
- * Definitive management
- * Contraindications
- * Complications
- * Disposition
- * Prevention
- * Follow-up
- * Board-style traps
- * Clinical pearl
- * “Do not miss” status

C. Management Algorithms

Use decision trees for:

- * Emergency stabilization
- * Next best step
- * Best initial test
- * Most accurate test
- * Screening versus diagnostic testing
- * Inpatient versus outpatient disposition
- * Antibiotic selection principles
- * Anticoagulation logic
- * Fluid and electrolyte management
- * Obstetric emergencies
- * Pediatric age-specific management
- * Psychiatric safety assessment
- * Surgical complications
- * Preventive care

D. Chief Complaint Framework

For every major presentation:

1. Is the patient unstable?
2. What immediately lethal diagnosis must be excluded?
3. What is the most likely diagnosis?
4. What data most efficiently separates the top diagnoses?
5. What is the next best step?

6. What intervention cannot wait?

7. Where should the patient go next: home, clinic, floor, ICU, OR, labor and delivery, psychiatric hold, or transfer?

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PART VI: DAILY CLINICAL WORKFLOW

Build a daily workflow that adapts to actual clinical life.

Pre-Round Workflow

- * Identify assigned patients
- * Review overnight events
- * Review vital signs
- * Review intake/output
- * Review labs
- * Review imaging
- * Review medications
- * Review consultants' notes
- * Identify active problems
- * Draft one-liner
- * Build assessment and plan
- * Identify learning question
- * Select one relevant illness script

During-Rounds Workflow

- * Capture feedback
- * Record unfamiliar diagnoses
- * Record medications/procedures to review
- * Record attending teaching points
- * Track presentation quality
- * Track professionalism and team communication observations

Post-Round Workflow

- * Complete patient tasks
- * Convert one clinical learning point into an illness script
- * Complete targeted question bank work
- * Review missed questions
- * Unsuspend or create targeted Anki cards
- * Update error log
- * Log clinical encounter
- * Prepare for tomorrow's patients
- * Protect a defined wind-down and sleep window

Create versions for:

- * Inpatient rotations
- * Outpatient rotations
- * Emergency department

- * Operating room
- * Labor and delivery
- * Psychiatry
- * Night float
- * Call days
- * Post-call days

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PART VII: QUESTION BANK CENTER

Create an integrated question-bank workflow without reproducing copyrighted questions.

Track:

- * Platform
- * Question ID
- * Specialty
- * System
- * Task
- * Difficulty
- * Timed versus untimed
- * Tutor versus test mode
- * Correct/incorrect
- * Confidence level
- * Time spent

- * Reason missed
- * Knowledge gap
- * Misread stem
- * Missed clue
- * Incorrect differential
- * Incorrect test selection
- * Incorrect management step
- * Biostatistics issue
- * Ethics issue
- * Changed answer
- * Fatigue/attention issue
- * Anki created
- * Learning objective completed
- * Re-test date

Create review modes:

- * Daily mixed block
- * Rotation-specific block
- * System-specific remediation
- * Task-specific remediation
- * Wrong-answer redo
- * Marked-question review
- * Incorrect-question only
- * Timed random simulation
- * Shelf final-review blocks

- * Step 2 dedicated blocks

The system must not reward meaningless question volume. It should reward:

- * Careful review
- * Error correction
- * Durable recall
- * Repeat testing
- * Improved reasoning
- * Better pacing
- * Cross-specialty transfer

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PART VIII: ERROR LOG AND ROOT-CAUSE ANALYTICS

For each missed question, force a root-cause classification:

1. Did not know the fact
2. Knew the fact but failed to recognize pattern
3. Missed a discriminating clue
4. Chose an incorrect differential
5. Misinterpreted a test
6. Selected an incorrect next step
7. Misapplied guideline/screening logic
8. Failed to stabilize first

9. Missed contraindication
10. Misread question wording
11. Ran out of time
12. Changed correct answer
13. Fatigue/distraction
14. Overthought
15. Anchoring bias
16. Premature closure
17. Availability bias
18. Ethical/professionalism reasoning error
19. Biostatistics/data interpretation error

Generate:

- * Top five recurring error types
- * Top weak systems
- * Top weak clinical tasks
- * Top weak disciplines
- * Knowledge gaps due for spaced review
- * Concepts missed more than once
- * Concepts never re-tested
- * Improvement by week
- * High-risk “false confidence” areas

PART IX: ANKI AND SPACED-REPETITION SYSTEM

Build a disciplined Anki workflow.

Allow integration with:

- * AnKing Step 2 tags where available
- * Shelf-specific decks
- * Personal cards
- * Missed-question cards
- * Patient-derived cards
- * Clinical-image cards
- * Medication cards
- * Algorithm cards

Create rules:

- * Do not create cards from every fact.
- * Create cards for recurring misses, unstable knowledge, algorithms, dangerous diagnoses, and concepts likely to recur.
- * Use image occlusion for algorithms and anatomy when appropriate.
- * Use cloze cards for succinct management rules.
- * Use bidirectional cards cautiously.
- * Avoid cards that contain vague, contextless trivia.
- * Link cards to source, rotation, system, task, question-bank error, and review date.

Track:

- * New cards/day
- * Reviews/day
- * Mature retention
- * Lapses
- * Backlog
- * Unsuspended cards
- * Cards linked to missed questions
- * Cards linked to patient encounters
- * Overload risk

Build an “Anki triage” tool:

- * Keep
- * Edit
- * Suspend
- * Merge
- * Replace with better card
- * Convert into algorithm/image occlusion
- * Delete because it is clutter masquerading as discipline

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PART X: ASSESSMENT CENTER

Track:

- * Shelf scores
- * NBME Clinical Science Mastery Series
- * School practice exams
- * CCSE
- * UWorld self-assessments
- * Amboss self-assessments
- * Step 2 CK practice scores
- * Free practice materials
- * Predicted score ranges
- * Score trends
- * Weakness domains
- * Error types
- * Assessment date
- * Conditions: rested, sleep deprived, interrupted, timed, untimed

Create a test-readiness framework:

- * Baseline
- * Stabilizing
- * Competitive
- * Target-range
- * Ready with caveats
- * Not ready
- * Data insufficient

Do not produce false precision. Explain score trends, uncertainty, and content gaps.

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PART XI: DEDICATED STEP 2 CK ENGINE

Create a dedicated planner that begins with diagnostic assessment.

Phase 0: Diagnostic Setup

- * Confirm exam date
- * Confirm registration status
- * Confirm test center logistics
- * Gather baseline scores
- * Identify weakest systems/tasks/disciplines
- * Set realistic study hours
- * Establish sleep and exercise floor
- * Select question-bank strategy
- * Set assessment schedule

Phase 1: Repair

Focus on:

- * Major content holes

- * Repeated missed diagnoses
- * Management algorithms
- * Biostatistics
- * Ethics
- * Patient safety
- * Screening
- * Preventive medicine
- * Weak specialties
- * Weak organ systems

Phase 2: Integration

Focus on:

- * Timed mixed blocks
- * Randomized systems
- * Cross-specialty patterns
- * Long-vignette pacing
- * Ambiguous clinical decisions
- * Best-next-step reasoning
- * Acute versus chronic management
- * Risk-benefit reasoning

Phase 3: Simulation

Focus on:

- * Full-length or near-full-length exam conditions
- * Break strategy
- * Nutrition/hydration
- * Fatigue management
- * Pacing
- * Question triage
- * Review strategy
- * Anxiety management

Phase 4: Taper

Focus on:

- * Consolidation
- * High-yield weak areas
- * Logistics
- * Sleep stabilization
- * Avoiding frantic resource switching
- * Avoiding massive new-card loads
- * Maintaining confidence through evidence, not vibes

The planner must adapt for:

- * 2-week dedicated
- * 4-week dedicated

- * 6-week dedicated
- * 8-week dedicated
- * 12-week dedicated
- * No dedicated period
- * Rotation-plus-Study strategy

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PART XII: STEP 2 CK HIGH-YIELD MASTER DOMAINS

Build integrated modules for:

Acute Care

- * Shock
- * Sepsis
- * Trauma
- * Airway emergencies
- * ACS
- * Stroke
- * GI bleed
- * Pulmonary embolism
- * DKA/HHS
- * Hypertensive emergency
- * Obstetric emergencies
- * Pediatric emergencies

- * Suicidality
- * Overdose/toxicology

Longitudinal Care

- * Diabetes
- * Hypertension
- * Hyperlipidemia
- * CKD
- * Heart failure
- * COPD/asthma
- * Cirrhosis
- * HIV
- * Cancer screening
- * Geriatric syndromes
- * Pregnancy care
- * Pediatrics/development
- * Depression/anxiety
- * Substance use disorders

Prevention and Screening

- * Vaccines
- * Cancer screening
- * Prenatal screening
- * STI screening

- * Cardiovascular risk reduction
- * Osteoporosis
- * Depression/suicide screening
- * Domestic violence
- * Substance use
- * Fall prevention
- * Tobacco cessation
- * Obesity and nutrition counseling

Ethics, Safety, and Systems

- * Capacity
- * Consent
- * Confidentiality
- * Mandatory reporting
- * Error disclosure
- * End-of-life care
- * Surrogate decision-making
- * Organ donation
- * Professional boundaries
- * Quality improvement
- * Root-cause analysis
- * Handoffs
- * Infection prevention
- * Medical errors
- * Cost-conscious care

- * Social determinants of health

Biostatistics and Evidence

- * Study designs
- * Bias
- * Confounding
- * Sensitivity/specificity
- * Likelihood ratios
- * PPV/NPV
- * Relative risk
- * Odds ratio
- * NNT/NNH
- * Confidence intervals
- * P values
- * Type I/II errors
- * Power
- * Intention-to-treat
- * Kaplan-Meier
- * Forest plots
- * Screening test interpretation
- * Diagnostic threshold logic

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PART XIII: CLINICAL SKILLS AND PROFESSIONAL PERFORMANCE

Create a non-exam tracker for:

- * Oral presentations
- * Written notes
- * H&P quality
- * SOAP notes
- * Problem representation
- * Assessment/plan quality
- * Procedure exposure
- * Clinical reasoning
- * Differential diagnosis
- * Interprofessional collaboration
- * Communication with patients
- * Communication with families
- * Cultural humility
- * Feedback received
- * Feedback implemented
- * Attendance
- * Timeliness
- * Professionalism concerns
- * Evaluator relationships
- * Honors-level behaviors
- * Clerkship evaluations
- * Letters and residency evidence

Create a “Feedback Conversion” system:

1. Capture feedback exactly.
2. Categorize it.
3. Translate it into a behavioral experiment.
4. Track implementation.
5. Request follow-up feedback.
6. Preserve validated growth evidence for clerkship evaluation and residency applications.

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PART XIV: RESIDENCY-EVIDENCE VAULT

Build an optional longitudinal record for:

- * Meaningful patient interactions
- * Teaching moments
- * Leadership
- * Quality improvement
- * Research
- * Presentations
- * Awards
- * Procedures
- * Honors
- * Evaluator comments
- * Professional growth

- * Specialty exploration
- * Potential ERAS experiences

Use strong privacy protections:

- * No patient identifiers
- * No copied clinical notes
- * No protected health information
- * User confirmation before exporting any reflection

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PART XV: WELLNESS AND PERFORMANCE SAFETY

Build a required performance-sustainability layer.

Track:

- * Sleep duration
- * Sleep regularity
- * Exercise
- * Meals
- * Hydration
- * Caffeine
- * Mood
- * Burnout score

- * Anxiety
- * Study hours
- * Clinical hours
- * Anki backlog
- * Question burden
- * Days without recovery

Flag:

- * Repeated sleep deprivation
- * Escalating stimulant use
- * Study volume without retention
- * Persistent burnout
- * Exam avoidance
- * Severe anxiety
- * Need for academic support, counseling, physician care, or emergency resources

The app must prioritize patient safety and student wellbeing over “grind harder” nonsense.

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PART XVI: RESOURCE LIBRARY

Build resource cards, categorized by purpose:

Official Sources

- * USMLE Step 2 CK content outline
- * USMLE physician tasks/competencies
- * NBME clinical subject exam outlines
- * NBME sample items
- * NBME Clinical Science Mastery Series
- * School clerkship guides
- * Hospital protocols where permitted

Core Learning Resources

- * UWorld
- * AMBOSS
- * NBME self-assessments
- * CMS forms
- * AnKing Step 2
- * OnlineMedEd
- * Divine Intervention
- * Emma Holliday reviews
- * Case Files
- * Step-Up to Medicine
- * Pestana for surgery
- * Blueprints series where useful
- * ACOG/USPSTF/CDC/ACC/AHA or equivalent current guideline references when needed

For each resource include:

- * Best use
- * Weak use
- * Appropriate phase
- * Cost
- * Time burden
- * Shelf relevance
- * Step 2 relevance
- * Risk of redundancy
- * Whether it is official, primary, or third-party

Do not encourage using every resource. Build a minimal effective stack, then add only when a defined gap exists.

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PART XVII: AUTOMATION LOGIC

Build automated prompts such as:

- * “You have missed pulmonary embolism management three times. Review the algorithm, complete five targeted questions, and re-test in seven days.”
- * “Your pediatrics performance is improving, but developmental milestones remain weak.”
- * “You have completed questions but have not reviewed 14 incorrects.”
- * “Your surgery shelf is in 12 days. Shift from broad content acquisition to timed mixed blocks and high-frequency error review.”
- * “Your Anki review burden exceeds your available time. Triage cards before adding more.”

* “Your sleep average has dropped below a safe study baseline for five days. Reduce new-card burden and protect recovery.”

* “You have not documented feedback from your attending this week. Request one actionable improvement point.”

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PART XVIII: FINAL OUTPUT FORMAT

Generate:

1. A complete category hierarchy.
2. A database schema for all trackers.
3. Rotation-specific shelf blueprints.
4. Step 2 CK master blueprint.
5. Daily workflows by clinical setting.
6. Assessment and error-log system.
7. Anki integration rules.
8. Dedicated study templates by duration.
9. Resource-selection logic.
10. Clinical-performance tracker.
11. Wellness safeguards.
12. Residency-evidence vault.
13. Data source and version-control framework.
14. A minimal viable study setup.
15. A high-performance honors-and-Step-2 setup.

16. A remediation plan for a student who is behind, burned out, or repeatedly underperforming.

The result must feel like an adaptive clinical cockpit: it should tell the student what to learn, why it matters, how to practice it, how to measure it, and what to do next.

For implementation, use the official source as the anchor, then layer third-party resources underneath it. Step 2 CK should be mapped to the USMLE outline and clinical tasks, while shelf modules should be keyed to the relevant NBME discipline, not to whatever resource happens to have the loudest mascot. NBME's Clinical Science Mastery Series also covers multiple clerkship disciplines in 50-question forms, making it suitable as a tracked assessment layer rather than mere decorative "practice."

For SGU-specific premed support, include both the post-baccalaureate pathway and direct-entry/international credential logic, but keep school facts editable and source-verified because program requirements evolve. SGU's post-bacc is currently described as a 43-credit, one-year prerequisite-completion program, while its MD pathways use different academic-entry criteria depending on the route.

Step 3:

Use this as the Step 3 God Blueprint master prompt. I updated it for the current Step 3 structure: FIP and ACM, the March 2026 30-minute block format, 13–14 CCS cases, ambulatory/generalist framing, and the newer nutrition emphasis.

USMLE STEP 3 GOD BLUEPRINT: FULL-SPECTRUM SUCCESS SYSTEM

Research Mandate, Curriculum Architecture, Question Bank Engine, CCS Training System, Assessment Logic, and App-Install Blueprint

You are building the most comprehensive, evidence-based, clinically realistic, highly structured USMLE Step 3 preparation blueprint possible.

This is not a generic study schedule. Do not create motivational filler, broad advice, vague “review Internal Medicine” directives, or an oversized lecture disguised as a plan. Medicine has inflicted enough of those on people already.

The final output must function as both:

1. A complete Step 3 preparation guide for an individual learner.
2. A modular “Install Blueprint” architecture for an educational app such as Noctyrium, where content can be installed as structured categories, systems, objectives, tasks, checklists, question-bank queues, flashcard queues, CCS cases, error logs, and readiness dashboards.

The blueprint must be comprehensive enough to guide a learner from post-Step 2 knowledge through independent-practice-level clinical reasoning, patient management, CCS fluency, test-day execution, and Step 3 readiness.

I. CORE OBJECTIVE

Build a complete, highly detailed Step 3 success system that prepares the learner to:

- * Pass Step 3 confidently.
- * Build strong independent-practice clinical reasoning.
- * Recognize unstable patients, triage urgency, and initiate appropriate stabilization.
- * Diagnose efficiently without overtesting.
- * Choose the next best step in management.
- * Manage outpatient, inpatient, emergency, ICU, obstetric, pediatric, geriatric, preventive, psychiatric, and chronic disease scenarios.
- * Apply evidence-based screening, prevention, vaccination, nutrition, counseling, and risk-factor modification.
- * Handle ethics, patient safety, communication, quality improvement, systems-based practice, and population health questions.
- * Interpret biostatistics, abstracts, drug advertisements, epidemiology, diagnostic studies, and clinical literature.
- * Execute Computer-Based Case Simulations confidently and mechanically.
- * Develop a durable clinical framework useful for residency, not merely a temporary test-taking trick.

The learner should finish with a clinical operating system, not a pile of disconnected facts.

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II. SOURCE HIERARCHY AND RESEARCH RULES

Before generating the blueprint, research current sources online.

Use the following hierarchy:

Tier 1: Official Primary Sources

Prioritize:

- * USMLE Step 3 Content Outline and Specifications.
- * USMLE Step 3 Physician Tasks and Competencies.
- * USMLE Step 3 exam format and question-format pages.
- * Official USMLE Step 3 interactive testing experience and CCS tutorial.
- * Official USMLE Step 3 sample questions.
- * USMLE Bulletin of Information.
- * NBME Comprehensive Clinical Medicine Self-Assessment information.
- * Current NBME and USMLE announcements affecting software, format, content, or testing procedure.

Verify current details before presenting them. Do not rely on outdated Step 3 timelines, block lengths, or CCS assumptions.

Tier 2: High-Yield Educational Resources

Evaluate and integrate, where appropriate:

- * UWorld Step 3 QBank.

- * UWorld CCS Cases.
- * CCSCases.com or equivalent CCS practice tools.
- * AMBOSS Step 3 QBank and library.
- * NBME Comprehensive Clinical Medicine Self-Assessments.
- * OnlineMedEd.
- * Boards and Beyond Step 3 / Clinical Medicine material where useful.
- * Divine Intervention podcasts, selectively and only when tied to concrete objectives.
- * Step 3-focused Anki decks.
- * UWorld Biostatistics Review.
- * UWorld self-assessments if currently available.
- * First Aid for Step 3, if current and relevant.
- * Step-Up to Medicine, selectively as a reference source rather than a primary memorization project.
- * MKSAP-style clinical explanations only when they clarify diagnosis and management.

Tier 3: Supplemental Sources

Use cautiously:

- * Residency teaching resources.
- * Specialty society guidelines.
- * USPSTF screening recommendations.
- * CDC immunization schedules.
- * ACOG recommendations.
- * ACC/AHA, ADA, IDSA, GOLD, GINA, KDIGO, ATS, AASLD, ACP, AAFP, AAP, and other specialty guidelines when clinically relevant.

Do not overwhelm the learner with guideline trivia. Translate each guideline into high-yield decision rules.

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III. EXAM MODEL

Construct the blueprint around the actual Step 3 role:

The examinee is functioning as a general, undifferentiated physician assuming independent responsibility for patient care.

The blueprint must repeatedly reinforce:

- * What is emergent?
- * What must be stabilized first?
- * What can be managed outpatient versus inpatient?
- * What requires consultation?
- * What requires monitoring?
- * What is the safest next action?
- * What is the most evidence-based action?
- * What is unnecessary, excessive, harmful, or premature?
- * What follow-up is required after acute management?
- * What counseling, prevention, screening, or medication reconciliation is needed?

Do not teach Step 3 as “Step 2 again.” Step 3 is management, prioritization, longitudinal care, systems thinking, and practical medicine.



IV. REQUIRED BLUEPRINT OUTPUT STRUCTURE

Create the final blueprint in the following sections.

Section A: Step 3 Exam Anatomy

Include:

- * Current two-day structure.
- * Foundations of Independent Practice.
- * Advanced Clinical Medicine.
- * Multiple-choice question format.
- * Drug advertisement questions.
- * Abstract and literature interpretation questions.
- * Sequential item sets.
- * Computer-Based Case Simulations.
- * Timing structure.
- * Block timing.
- * Break strategy.
- * Tutorial strategy.
- * CCS case timing.
- * Test-day meal, caffeine, hydration, sleep, and logistics planning.
- * Exam-day error-prevention checklist.

For every format, include:

- * What the format tests.
- * The common trap.
- * The correct reasoning framework.
- * A repeatable method for answering.
- * A checklist for reviewing mistakes.

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Section B: Master Clinical Reasoning Framework

Build a universal framework called:

The Step 3 Independent Physician Algorithm

For every vignette, train the learner to identify:

1. Clinical setting.

- * Outpatient.
- * Emergency department.
- * Inpatient floor.
- * ICU.
- * Labor and delivery.
- * Telephone encounter.

- * Home care.
- * Follow-up visit.

2. Patient stability.

- * Airway.
- * Breathing.
- * Circulation.
- * Mental status.
- * Sepsis.
- * Hemorrhage.
- * Acute coronary syndrome.
- * Stroke.
- * Obstetric emergency.
- * Toxicologic emergency.
- * Psychiatric emergency.

3. Time horizon.

- * Immediate intervention.
- * Minutes to hours.
- * Same day.
- * Within days.
- * Routine outpatient follow-up.
- * Longitudinal preventive management.

4. Decision type.

- * Diagnose.
- * Stabilize.
- * Test.
- * Treat.

- * Consult.
- * Admit.
- * Discharge.
- * Monitor.
- * Counsel.
- * Prevent.
- * Reassess.

5. The “least wrong next step.”

- * The answer should be appropriate to the patient’s acuity and setting.
- * Avoid jumping to definitive treatment before stabilizing.
- * Avoid ordering low-yield tests before addressing lethal diagnoses.
- * Avoid overtesting stable patients with clear diagnoses.
- * Avoid underreacting to subtle instability.

6. Follow-through.

- * Monitoring.
- * Follow-up interval.
- * Disposition.
- * Prevention.
- * Patient education.
- * Medication reconciliation.
- * Referral.

Include a one-page quick-reference version and a detailed version.

Section C: Full Topic Blueprint by System

Build separate installable categories for:

1. Cardiovascular Medicine
2. Pulmonary and Critical Care
3. Gastroenterology and Hepatology
4. Renal, Electrolytes, and Acid-Base
5. Endocrinology and Metabolism
6. Hematology and Oncology
7. Infectious Disease
8. Rheumatology and Musculoskeletal Medicine
9. Neurology
10. Psychiatry and Addiction Medicine
11. Dermatology
12. Ophthalmology
13. Otolaryngology
14. Obstetrics
15. Gynecology
16. Pediatrics
17. Emergency Medicine
18. Surgery and Perioperative Care
19. Geriatrics and Palliative Care
20. Preventive Medicine and Population Health
21. Ethics, Communication, Patient Safety, and Systems-Based Practice
22. Biostatistics, Epidemiology, Evidence-Based Medicine, Drug Ads, and Abstracts

23. Nutrition and Lifestyle Medicine

24. CCS Mastery

For each category, create:

A. Core Clinical Presentations

Examples:

- * Chest pain.
- * Dyspnea.
- * Syncope.
- * Fever.
- * Abdominal pain.
- * Altered mental status.
- * Headache.
- * Weakness.
- * Vaginal bleeding.
- * Pregnancy complications.
- * Pediatric fever.
- * Weight loss.
- * Edema.
- * Rash.
- * Back pain.
- * Depression.
- * Suicidality.

- * Delirium.
- * Abnormal laboratory result.
- * Medication adverse effect.
- * Preoperative evaluation.

B. Must-Not-Miss Diagnoses

For each presentation, list:

- * Life-threatening diagnoses.
- * Time-sensitive diagnoses.
- * Common diagnoses.
- * Dangerous mimics.
- * Board-favorite red flags.
- * Clinical clues that separate close differentials.

C. Diagnostic Strategy

For each topic, specify:

- * First test.
- * Best initial test.
- * Most accurate test.
- * Confirmatory test.
- * Test to avoid.
- * Test only after stabilization.

- * Test only in specific clinical contexts.
- * When imaging is indicated.
- * When laboratory workup is sufficient.
- * When empiric treatment precedes testing.

D. Management Strategy

For every diagnosis:

- * Immediate stabilization.
- * Initial treatment.
- * Definitive treatment.
- * Hospitalization criteria.
- * ICU criteria.
- * Consultation criteria.
- * Discharge criteria.
- * Monitoring parameters.
- * Follow-up schedule.
- * Counseling requirements.
- * Prevention and recurrence reduction.

E. Medication Knowledge

Include:

- * First-line drug.

- * Alternative drug.
- * Contraindications.
- * Common adverse effects.
- * Dangerous adverse effects.
- * Drug-drug interactions.
- * Pregnancy considerations.
- * Renal dosing considerations.
- * Hepatic dosing considerations.
- * Monitoring requirements.
- * When to stop, switch, escalate, or de-escalate.

F. High-Yield Clinical Algorithms

Create concise algorithms for:

- * ACS.
- * Stroke.
- * Sepsis.
- * DKA/HHS.
- * Hypertensive emergency.
- * Pulmonary embolism.
- * Heart failure exacerbation.
- * COPD exacerbation.
- * Asthma exacerbation.
- * GI bleed.
- * Acute pancreatitis.

- * Cholangitis.
- * Cirrhosis complications.
- * AKI.
- * Hyperkalemia.
- * Hyponatremia.
- * Calcium disorders.
- * Thyroid storm.
- * Myxedema coma.
- * Adrenal crisis.
- * Neutropenic fever.
- * Transfusion reactions.
- * Meningitis.
- * HIV care.
- * Tuberculosis.
- * Cellulitis versus necrotizing infection.
- * Septic arthritis.
- * Acute psychosis.
- * Mania.
- * Suicidality.
- * Alcohol withdrawal.
- * Opioid withdrawal.
- * Obstetric hemorrhage.
- * Preeclampsia/eclampsia.
- * Ectopic pregnancy.
- * Preterm labor.
- * Neonatal fever.

- * Pediatric dehydration.
- * Pediatric respiratory distress.
- * Delirium.
- * Falls.
- * Capacity assessment.
- * End-of-life symptom management.
- * Perioperative risk assessment.

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V. CCS MASTERY SYSTEM

Create an extensive CCS curriculum.

Do not treat CCS as a minor side topic. It is a separate procedural skill.

A. CCS Mental Model

Teach that each case requires:

1. Rapid initial assessment.
2. Correct location of care.
3. Stabilization when necessary.
4. Focused diagnostic workup.
5. Appropriate treatment.
6. Monitoring.

7. Reassessment after time advances.
8. Consultation when appropriate.
9. Preventive counseling and disposition.
10. Correct follow-up.

B. Universal CCS Opening Orders

Create opening-order templates for:

- * Stable outpatient.
- * Unstable emergency department patient.
- * Chest pain.
- * Dyspnea.
- * Sepsis.
- * Altered mental status.
- * Abdominal pain.
- * Trauma.
- * Pregnancy-related emergency.
- * Psychiatric emergency.
- * Pediatric fever.
- * Inpatient deterioration.
- * ICU patient.
- * Chronic disease follow-up.

Include appropriate use of:

- * Vital signs.
- * Pulse oximetry.
- * Cardiac monitoring.
- * IV access.
- * Oxygen.
- * Fluids.
- * NPO status.
- * Laboratory panels.
- * Imaging.
- * ECG.
- * Urinalysis.
- * Pregnancy testing.
- * Blood cultures.
- * Lactate.
- * Toxicology testing.
- * Consultations.
- * Admission orders.
- * Follow-up appointments.
- * Counseling.

C. CCS Case Categories

Build a categorized case bank:

- * Emergency stabilization cases.
- * Common outpatient management cases.

- * Inpatient deterioration cases.
- * Obstetric and gynecologic cases.
- * Pediatric cases.
- * Psychiatric cases.
- * Geriatric cases.
- * Chronic disease and preventive-care cases.
- * Infectious disease cases.
- * Ethical and capacity cases.
- * Surgical and postoperative cases.

For each category, provide:

- * Typical presentation.
- * Opening orders.
- * Critical diagnostic actions.
- * Critical treatment actions.
- * Common harmful errors.
- * Required follow-up.
- * Commonly missed counseling.
- * When to advance the clock.
- * When to stop the case.

D. CCS Error Taxonomy

Create an error tracker for:

- * Failure to stabilize.
- * Wrong site of care.
- * Delayed treatment.
- * Inappropriate treatment.
- * Missing pregnancy test.
- * Missing medication reconciliation.
- * Missing preventive counseling.
- * Missing follow-up.
- * Missing monitoring.
- * Over-ordering.
- * Under-ordering.
- * Failure to consult.
- * Premature discharge.
- * Failure to advance time.
- * Advancing time before treatment.
- * Missing complications.
- * Failure to reassess after intervention.

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VI. QUESTION BANK SYSTEM

Build an adaptive question-bank framework.

A. QBank Prioritization

Rank resources by purpose:

- * Primary learning QBank.
- * Secondary exposure QBank.
- * Reference library.
- * CCS simulator.
- * Biostatistics supplement.
- * Assessment tools.
- * Anki integration tool.

Do not recommend completing every resource blindly. Create efficient pathways:

Essential Pass Path

For time-limited learners.

Strong Pass Path

For learners with 6–10 weeks.

High-Score / Deep Mastery Path

For learners with 10–16+ weeks.

Residency-Compatible Path

For residents studying around clinical duties.

B. Question Review Framework

For every missed or guessed question, force classification into:

1. Knowledge gap.
2. Recognition gap.
3. Differential diagnosis gap.
4. Diagnostic sequencing error.
5. Management sequencing error.
6. Guideline gap.
7. Preventive-care gap.
8. Biostatistics gap.
9. Ethics/communication gap.
10. Reading-comprehension error.
11. Time-management error.
12. Changed answer error.
13. Fatigue error.
14. CCS mechanics error.

For each error, create:

- * Root cause.
- * Correct rule.
- * Trigger phrase.

- * One-sentence takeaway.
- * Anki card candidate.
- * Re-test date.
- * Related topics.
- * Related question IDs.
- * Related CCS cases.

C. Question Review Output Template

For each missed question, generate:

- * Clinical stem summary.
- * Key clue.
- * Correct diagnosis.
- * Why the right answer is right.
- * Why each distractor is wrong.
- * Rule that would have solved it.
- * Closest competing diagnosis.
- * Future trigger phrase.
- * Flashcard.
- * Related system.
- * Priority level.

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VII. ANKI SYSTEM

Build a Step 3 Anki structure focused on clinical decisions rather than encyclopedic recall.

Deck Architecture

Create decks for:

- * Emergency algorithms.
- * Internal medicine management.
- * OB/GYN management.
- * Pediatrics management.
- * Psychiatry and addiction.
- * Surgery and perioperative care.
- * Preventive medicine.
- * Screening and vaccination.
- * Biostatistics.
- * Ethics and communication.
- * Drug adverse effects and contraindications.
- * Antibiotic selection.
- * CCS management sequences.
- * Error-log cards.

Card Types

Create templates for:

1. Diagnosis recognition cards.
2. Next-best-step cards.
3. “First test versus best test” cards.
4. “Stable versus unstable” cards.
5. Medication selection cards.
6. Contraindication cards.
7. Screening guideline cards.
8. Vaccination cards.
9. Biostatistics interpretation cards.
10. CCS opening-order cards.
11. CCS follow-up cards.
12. Mistake-derived cards.

Anki Rules

- * Prefer cards generated from misses and uncertainty.
- * Avoid massive unsustainable decks.
- * Use cloze cards for algorithms.
- * Use image occlusion for tables and flowcharts.
- * Tag cards by system, acuity, task, source, and weakness category.
- * Use filtered decks for urgent weaknesses.
- * Build a “Repeat Offenders” deck for concepts missed more than once.
- * Build a “Last 14 Days” deck for final review.

VIII. BIOSTATISTICS, ABSTRACTS, AND DRUG ADS

Create a separate high-detail curriculum for:

- * Sensitivity.
- * Specificity.
- * Positive predictive value.
- * Negative predictive value.
- * Likelihood ratios.
- * Pretest probability.
- * Posttest probability.
- * Type I error.
- * Type II error.
- * Power.
- * Confidence intervals.
- * P values.
- * Relative risk.
- * Odds ratio.
- * Absolute risk reduction.
- * Relative risk reduction.
- * Number needed to treat.
- * Number needed to harm.
- * Hazard ratio.
- * Incidence.
- * Prevalence.
- * Confounding.

- * Bias.
- * Selection bias.
- * Recall bias.
- * Lead-time bias.
- * Length-time bias.
- * Intention-to-treat analysis.
- * Crossover studies.
- * Case-control studies.
- * Cohort studies.
- * Randomized controlled trials.
- * Meta-analysis.
- * Forest plots.
- * Kaplan-Meier curves.
- * Noninferiority trials.
- * Study validity.
- * Statistical versus clinical significance.

Include:

- * Drug advertisement attack strategy.
- * Abstract attack strategy.
- * Formula sheet.
- * Interpretation table.
- * Timed-practice workflow.
- * Common traps.
- * One-page rapid-review sheet.



IX. PREVENTIVE MEDICINE, SCREENING, AND NUTRITION

Build a dedicated preventative-care operating system.

Include:

- * USPSTF-style screening logic.
- * Age-based screening.
- * Sex-specific screening.
- * Pregnancy-related screening.
- * Cancer screening.
- * Cardiovascular risk reduction.
- * Diabetes prevention.
- * Obesity counseling.
- * Smoking cessation.
- * Alcohol misuse screening.
- * Substance use screening.
- * Depression screening.
- * Suicide risk.
- * IPV screening where appropriate.
- * Vaccination.
- * Travel and occupational prevention when relevant.
- * Nutrition counseling.

- * Exercise prescriptions.
- * Sleep counseling.
- * Medication adherence.
- * Fall prevention.
- * Advance care planning.

For every preventive item, include:

- * Who qualifies.
- * When to start.
- * Screening interval.
- * When to stop.
- * Test used.
- * Follow-up after abnormal result.
- * Counseling language.
- * High-risk populations.
- * Common board traps.

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X. ASSESSMENT AND READINESS SYSTEM

Build a complete readiness framework.

A. Baseline Assessment

At the start, assess:

- * Question-bank accuracy by system.
- * Question-bank accuracy by task.
- * Timing.
- * CCS familiarity.
- * Biostatistics accuracy.
- * Preventive-care accuracy.
- * Major clinical weakness categories.
- * Endurance.
- * Error patterns.

B. Assessment Timeline

Create plans for:

- * 4-week preparation.
- * 6-week preparation.
- * 8-week preparation.
- * 12-week preparation.
- * 16-week preparation.
- * Resident schedule with 45–90 minutes weekdays and longer weekend blocks.

Use NBME Comprehensive Clinical Medicine Self-Assessments as major checkpoints where available.

For every assessment, include:

- * Timing.
- * Conditions.
- * Review process.
- * Score interpretation.
- * Weakness extraction.
- * Study-plan adjustment.
- * Go/no-go reasoning.

C. Readiness Dashboard

Create a dashboard with:

- * Cumulative QBank percentage.
- * Recent QBank percentage.
- * Accuracy by system.
- * Accuracy by task.
- * Accuracy by clinical setting.
- * Accuracy by acuity.
- * Biostatistics accuracy.
- * Preventive medicine accuracy.
- * Ethics/patient safety accuracy.
- * CCS completion rate.
- * CCS critical-error rate.
- * Timed-block completion rate.

- * Assessment trend.
- * Repeat-offender concepts.
- * Days until test.
- * Sleep consistency.
- * Burnout risk.

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XI. DAILY AND WEEKLY STUDY ENGINE

Create an adaptive schedule that adjusts based on available study time.

Daily Structure

Include:

- * Timed questions.
- * Untimed targeted questions when needed.
- * Deep review.
- * Anki.
- * CCS practice.
- * Biostatistics drills.
- * Guideline/prevention review.
- * Error-log review.
- * Weakness remediation.
- * End-of-day planning.

Weekly Structure

Include:

- * Mixed timed blocks.
- * System-specific repair sessions.
- * One CCS cluster.
- * One biostatistics/abstract cluster.
- * One preventive medicine cluster.
- * One assessment or mini-assessment.
- * One cumulative review session.
- * One recovery block.

Study-Time Versions

Create versions for:

- * 1 hour/day.
- * 2 hours/day.
- * 3 hours/day.
- * 4 hours/day.
- * Dedicated study.
- * Resident call schedule.
- * Post-call schedule.
- * Weekend-intensive schedule.

Do not prescribe impossible daily workloads. Include a minimum viable study day for exhausted residents.

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XII. FINAL 14-DAY AND FINAL 72-HOUR PLAN

Create:

Final 14 Days

- * Mixed blocks.
- * CCS repetition.
- * High-yield weakness repair.
- * Preventive medicine.
- * Biostatistics.
- * Ethics and safety.
- * Sleep stabilization.
- * Nutrition and exercise.
- * Caffeine strategy.
- * Travel and logistics.

Final 72 Hours

Include:

- * No new resource acquisition.
- * No panic-driven QBank expansion.
- * Light targeted review.
- * CCS mechanics review.
- * Formula review.
- * Screening/vaccine rapid review.
- * Medication and emergency algorithm review.
- * Sleep protection.
- * Exam-day supplies.
- * Break plan.
- * Meal plan.
- * Backup transportation plan.

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XIII. NOCTYRIUM / APP INSTALL BLUEPRINT FORMAT

The final product must be designed for direct installation into an app.

Do not output the entire blueprint as a single long article.

Instead, output structured data in this hierarchy:

Step 3

|

- └─ Dashboard
- └─ Exam Anatomy
- └─ Clinical Reasoning Core
- └─ Systems
 - | └─ Cardiology
 - | └─ Pulmonology
 - | └─ Gastroenterology
 - | └─ Renal
 - | └─ Endocrinology
 - | └─ Infectious Disease
 - | └─ Neurology
 - | └─ Psychiatry
 - | └─ OB/GYN
 - | └─ Pediatrics
 - | └─ Surgery
 - | └─ Emergency Medicine
 - | └─ Geriatrics
 - | └─ Preventive Medicine
- └─ Biostatistics and Evidence-Based Medicine
- └─ Ethics and Patient Safety
- └─ CCS Command Center
- └─ QBank Queue
- └─ Anki Queue
- └─ Error Log

- |— Assessments
- |— Study Calendar
- |— Rapid Review Sheets
- └— Test-Day Operations

For every module, create:

- * Module title.
- * Short mission statement.
- * Required knowledge objectives.
- * Clinical presentation categories.
- * Diagnostic algorithms.
- * Management algorithms.
- * Question-bank tasks.
- * Anki tasks.
- * CCS tasks where relevant.
- * Assessment checkpoints.
- * Completion criteria.
- * Weakness tags.
- * Linked resources.
- * Estimated time.
- * Priority level.

Use tags such as:

#emergency

#outpatient

#inpatient

#ICU

#pediatrics

#obstetrics

#geriatrics

#prevention

#biostatistics

#ethics

#CCS

#drug-ad

#abstract

#high-yield

#repeat-offender

#must-not-miss

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XIV. OUTPUT REQUIREMENTS

The final response must include:

1. Executive summary.
2. Current exam format summary.
3. Resource stack ranked by purpose.
4. Full system-by-system study blueprint.

5. CCS training curriculum.
6. Question-bank strategy.
7. Anki system.
8. Biostatistics and drug-ad curriculum.
9. Preventive medicine and nutrition curriculum.
10. Assessment and readiness system.
11. Adaptive schedules.
12. Final 14-day plan.
13. App-install data structure.
14. Error-log templates.
15. Rapid-review tables.
16. Clear warnings against outdated test-format assumptions.
17. Source list with direct official citations.
18. Explicit notation of any information that may change and needs re-verification.

The tone should be precise, disciplined, technically detailed, clinically practical, and operational.

Avoid generic encouragement. Avoid resource hoarding. Avoid redundant lists. Avoid pretending that every detail has equal importance.

Prioritize actions that produce score gains, safer clinical reasoning, faster recognition of emergencies, and reliable CCS execution.

One correction worth preserving in whatever system you build: Step 3 now uses 30-minute blocks for examinees testing on the new software, and CCS mechanics deserve dedicated deliberate practice, not an anxious weekend ritual before the exam. Official USMLE

guidance explicitly warns that examinees who do not practice CCS mechanics are disadvantaged.

Step 2:

Step 2 CK Blueprint: The Clinical Decision System

This is the actual Step 2 version of your Step 1 blueprint.

Step 2 is not “Step 1 but with more medicine.” It tests whether you can recognize illness, prioritize danger, choose the correct test, act in the right sequence, manage the patient, prevent future harm, and avoid becoming the physician who orders a CT for every mildly anxious human being.

Officially, Step 2 CK is dominated by Medicine, then Pediatrics, OB/GYN, Psychiatry, and Surgery. It tests diagnosis, diagnostic studies, mixed management, pharmacotherapy, interventions, prevention, patient safety, professionalism, and practice-based improvement. Medicine comprises roughly 55–65% of exam content; Pediatrics 17–27%; OB/GYN 10–20%; Psychiatry 10–15%; and Surgery 5–15%.

As of May 7, 2026, the exam uses 16 blocks of 30 minutes, no more than 20 questions per block, a 9-hour testing day, at least 55 minutes of break time, and a 5-minute tutorial.

I. The Step 2 Mental Model

For every clinical vignette, your brain should run this sequence:

1. Stable or unstable?
2. What kills the patient first?
3. What syndrome is present?
4. What is the most likely diagnosis?
5. What must be ruled out immediately?
6. What is the best next step?
7. What test changes management?
8. What treatment comes first?
9. Where does the patient go next?
 - Home
 - Clinic follow-up
 - Hospital admission
 - ICU
 - Operating room
 - Labor and delivery
 - Psychiatric admission
10. What prevention, counseling, or follow-up is required?

Your default hierarchy:

Stabilize before diagnose.

Treat before confirm when delay is dangerous.

Do not test when the diagnosis is clinical and treatment is obvious.

Do not order the gold-standard test when it is not the best initial test.

Do not confuse outpatient workup with emergency care.

Do not confuse “most likely diagnosis” with “next best step.”

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II. Exam Weighting: What Actually Deserves Your Time

The official Step 2 content blueprint gives unusual importance to areas students lazily relegate to “later.”

Domain Approximate Official Range

Nutrition 15–20%

Ethics, professionalism, patient safety, systems 10–15%

Renal/urinary/reproductive 7–13%

Cardiovascular 6–12%

MSK/skin 6–12%

Behavioral health 5–10%

Blood/immune 5–10%

GI 5–10%

Neuro/senses 5–10%

Respiratory 5–10%

Multisystem 4–8%

Endocrine 3–7%

Pregnancy/childbirth/puerperium 3–7%

Biostatistics/Epi/medical literature 3–5%

The test also explicitly weights patient-care tasks:

Task Approximate Official Range

Diagnosis 16–20%

Laboratory/diagnostic studies 13–17%

Mixed management 12–16%

Pharmacotherapy 8–12%

Clinical interventions 6–10%

Health maintenance/prevention 5–10%

Prognosis/outcome 5–9%

Systems/patient safety 5–7%

Professionalism 5–7%

Practice-based learning 3–5%

That means your study system must train more than disease recognition. A student who knows every vasculitis but repeatedly misses “what should happen next?” is simply collecting rare Pokémon.

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III. Your Resource Stack

Tier 1: Mandatory Core

1. UWorld Step 2 CK

Use as your primary learning instrument.

Purpose:

- * Clinical pattern recognition
- * Management sequencing
- * Learning why distractors are tempting
- * Building “NBME logic”
- * Identifying your actual weaknesses instead of imaginary ones

Rule:

UWorld is not an assessment.

UWorld is your clinical textbook disguised as a question bank.

2. AnKing Step 2 / Clinical Deck

Use for retention, algorithms, antibiotics, contraindications, screening, management thresholds, and recurring mistakes.

Do not make 1,000 custom cards because a single question hurt your feelings.

3. NBME Clinical Mastery Series

Use during each clerkship/shelf.

NBME offers Clinical Science Mastery Series forms for Neurology, Emergency Medicine, Family Medicine, Medicine, OB/GYN, Pediatrics, Psychiatry, and Surgery. They contain 50 questions per form and provide content-area feedback and explanations.

4. NBME Comprehensive Clinical Science Self-Assessments

Use during dedicated Step 2 preparation.

NBME's CCSSA is specifically designed to estimate Step 2 CK readiness, provide content-area feedback, question-level timing data, explanations, and an estimated probability of passing.

5. Official USMLE Sample Questions + Interactive Test

Use during the final month.

The official interactive experience includes single items, sequential sets, scientific abstracts, and audio/video questions. More than 100 sample Step 2 questions are available.



Tier 2: Targeted Reinforcement

Resource Use It For Do Not Use It For

AMBOSS Weak-system repair, tables, difficult differentials, targeted questions Replacing UWorld entirely

First Aid Step 2 CK Rapid review, annotations, final compression Main learning source
Step Up to Medicine Internal medicine framework Reading cover to cover during dedicated
OnlineMedEd First-pass mental frameworks Passive binge-watching
Divine Intervention Targeted weak areas, rapid review Listening without retrieval practice
Emma Holliday Shelf-oriented overview Your only source
Pestana Surgery rapid review Full surgical education
DeVirgilio Deep surgery reasoning Last 3 weeks before the exam
Case Files Clerkship orientation Replacing questions
Sketchy Micro/Pharm Repairing microbiology/pharmacology gaps Rewatching all of Step
1 for comfort

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IV. The Daily Learning Loop

This is the basic operating system during clerkships.

1. Anki reviews
2. Rotation-specific UWorld block
3. Review every incorrect question
4. Review every uncertain question
5. Identify exact failure type
6. Unsuspend relevant AnKing cards
7. Create custom card only if repeated or algorithmic
8. Add mistake to error log
9. Re-test weak concept within 7 days

Daily Volume Targets

During a normal clerkship

20–40 UWorld questions/day

100–250 Anki reviews/day

1 targeted content block/day

1 rapid algorithm review/day

During a heavy clerkship or shelf week

40–60 UWorld questions/day

Anki reviews only, minimal new cards

1 NBME/CMS form every 1–2 weeks

Focused review of weak systems

During dedicated

60–120 questions/day

Mixed timed blocks

Daily Anki reviews

Focused repair of repeated misses

1 assessment every 7–10 days

V. The Error Log System

Every miss goes into one category.

- A. Knowledge gap
- B. Recognition failure
- C. Differential diagnosis failure
- D. Test-selection failure
- E. Management failure
- F. Sequence/timing failure
- G. Screening/prevention failure
- H. Pharmacology failure
- I. Ethics/patient-safety failure
- J. Biostatistics failure
- K. Misread stem
- L. Changed correct answer
- M. Time-pressure failure
- N. Overthinking

For every wrong question, write:

Diagnosis:

Why I missed it:

Correct clinical rule:

Why the tempting distractor was wrong:

What clue I missed:

What I will do next time:

Anki tag:

Re-test date:

Example:

Diagnosis: Acute ascending cholangitis

Why I missed it:

I chose MRCP before antibiotics and biliary decompression.

Correct clinical rule:

Unstable cholangitis requires IV fluids, broad-spectrum antibiotics, and urgent biliary drainage.

Trap:

MRCP helps define anatomy but does not come before stabilization in septic obstruction.

Trigger:

Fever + jaundice + RUQ pain + hypotension/confusion = source control problem.

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VI. The Master Clinical Decision Tree

A. “Best Next Step” Hierarchy

1. ABCs

2. Hemodynamic stabilization

3. Immediate life-saving treatment
4. Bedside confirmation if it changes immediate action
5. Initial diagnostic test
6. Definitive test
7. Definitive treatment
8. Disposition
9. Prevention/follow-up

B. Best Initial Test vs Gold Standard

Situation What Step 2 Usually Wants

Unstable patient Stabilize first

Clinical diagnosis with obvious treatment Treat; avoid useless testing

Need rapid bedside answer ECG, bedside glucose, pregnancy test, pulse ox, ultrasound, etc.

Need confirmation after screening Confirmatory test

Need anatomy before surgery Imaging that guides intervention

Need tissue diagnosis Biopsy, unless contraindicated

Need emergent source control Intervention before complete workup

C. “Do Not Miss” Diagnoses

Build instant reflexes for:

Aortic dissection

MI / STEMI / NSTEMI

Pulmonary embolism

Tension pneumothorax

Massive hemothorax

Cardiac tamponade

Septic shock

Meningitis

Subarachnoid hemorrhage

Cauda equina syndrome

Spinal epidural abscess

Ectopic pregnancy

Ovarian torsion

Testicular torsion

Necrotizing fasciitis

Mesenteric ischemia

GI perforation

Bowel obstruction with ischemia

Acute limb ischemia

Compartment syndrome

Stroke

Status epilepticus

Severe preeclampsia/eclampsia

Postpartum hemorrhage

Postpartum endometritis

Suicidal intent with plan

Delirium

Child abuse

Elder abuse

Intimate partner violence

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VII. Specialty Blueprints

1. Internal Medicine

Medicine is the foundation. The NBME Medicine shelf itself heavily emphasizes diagnosis and management, with cardiovascular, respiratory, GI/nutrition, endocrine, renal, hematology, neurology, and geriatric care all represented.

Cardiology

Must Master

- * ACS: STEMI, NSTEMI, unstable angina

- * Stable angina

- * Acute decompensated heart failure

- * HFrEF vs HFpEF

- * Atrial fibrillation/flutter

- * SVT

- * Ventricular tachycardia

- * Bradyarrhythmias

- * Aortic stenosis

- * Aortic regurgitation

- * Mitral stenosis/regurgitation
- * Endocarditis
- * Pericarditis
- * Cardiac tamponade
- * Aortic dissection
- * Peripheral arterial disease
- * Acute limb ischemia
- * DVT/PE
- * Hypertensive emergency
- * Cardiogenic shock

Cardiology Algorithm

Chest pain

- ECG + troponins
- unstable? treat immediately
- STEMI? cath lab
- NSTEMI? antiplatelet/anticoagulation/risk stratification
- low-risk chest pain? outpatient evaluation

Dyspnea + edema

- consider HF, PE, COPD, pneumonia, renal failure
- BNP, CXR, ECG, echo when appropriate
- hypoxemia/hypotension = stabilize first

Irregularly irregular rhythm

- atrial fibrillation
- unstable = synchronized cardioversion

→ stable = rate control + anticoagulation assessment

Highest-Yield Traps

- * New murmur + fever + IV drug use = endocarditis.
- * Hypotension + JVD + muffled heart sounds = tamponade.
- * Tearing chest pain radiating to back = dissection, not “maybe anxiety.”
- * Acute limb pain + pallor + pulselessness = vascular emergency.
- * New AF with hypotension, ischemic chest pain, altered mental status, pulmonary edema = synchronized cardioversion.

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Pulmonology

Must Master

- * Asthma
- * COPD
- * Pneumonia
- * Hospital-acquired pneumonia
- * Pulmonary embolism
- * Pleural effusion
- * Pneumothorax
- * Tension pneumothorax
- * ARDS

- * Pulmonary hypertension
- * Lung cancer
- * ILD
- * Sarcoidosis
- * OSA
- * Tuberculosis
- * Hemoptysis

Core Algorithm

Dyspnea

- oxygen saturation
- unstable? oxygen + stabilization
- wheeze? asthma/COPD
- fever/infiltrate? pneumonia
- pleuritic pain/tachycardia/hypoxemia? PE
- sudden unilateral absent breath sounds? pneumothorax
- bilateral infiltrates + severe hypoxemia? ARDS

Never Miss

- * Tension pneumothorax: immediate needle/finger decompression, then chest tube.
- * Massive PE with instability: reperfusion therapy consideration.
- * COPD exacerbation: bronchodilators, steroids, oxygen appropriately, antibiotics when indicated.
- * Asthma with silent chest, exhaustion, rising CO₂: impending respiratory failure.



Nephrology

Must Master

- * Prerenal AKI
- * ATN
- * AIN
- * Postrenal obstruction
- * Nephrotic syndrome
- * Nephritic syndrome
- * RPGN
- * CKD
- * Dialysis indications
- * Sodium disorders
- * Potassium disorders
- * Metabolic acidosis/alkalosis
- * Renal tubular acidosis
- * Hypertension
- * Glomerulonephritis
- * Kidney stones
- * Pyelonephritis

Dialysis Rule

AEIOU:

Acidosis refractory

Electrolyte derangement, especially refractory hyperkalemia

Intoxications

Overload refractory to diuresis

Uremic complications

Hyperkalemia Algorithm

ECG changes?

→ IV calcium first

→ shift potassium intracellularly:

insulin + glucose

beta agonist

bicarbonate if acidotic

→ remove potassium:

diuretics

potassium binders

dialysis if needed

?

GI/Hepatology

Must Master

- * Upper GI bleed
- * Lower GI bleed
- * Peptic ulcer disease
- * Variceal bleeding
- * Pancreatitis
- * Cholecystitis
- * Cholangitis
- * Choledocholithiasis
- * Appendicitis
- * Diverticulitis
- * Mesenteric ischemia
- * IBD
- * IBS
- * Celiac disease
- * Cirrhosis
- * Ascites
- * SBP
- * Hepatic encephalopathy
- * Hepatorenal syndrome
- * Viral hepatitis
- * Colon cancer
- * Esophageal cancer

GI Bleed Logic

Unstable bleed

- two large-bore IVs
- fluids/blood products
- type and cross
- correct coagulopathy when appropriate
- IV PPI for suspected upper nonvariceal bleed
- octreotide + ceftriaxone if variceal suspected
- urgent endoscopy after stabilization

Abdominal Pain Danger Signals

Pain out of proportion = mesenteric ischemia

RUQ pain + fever + jaundice = cholangitis

Epigastric pain to back + elevated lipase = pancreatitis

Rigid abdomen/free air = perforation

Migratory periumbilical to RLQ = appendicitis

LLQ pain + fever = diverticulitis

?

Endocrine

Must Master

* DKA

* HHS

- * Hypoglycemia
- * Type 1/2 diabetes management
- * Thyroid storm
- * Myxedema coma
- * Graves
- * Hashimoto
- * Adrenal insufficiency
- * Cushing syndrome
- * Hyperaldosteronism
- * Pheochromocytoma
- * Hypercalcemia
- * Hypocalcemia
- * Osteoporosis
- * Pituitary lesions
- * SIADH
- * Diabetes insipidus

DKA Algorithm

1. IV fluids
2. Check potassium
3. If K low, replace before insulin
4. Start insulin when potassium safe
5. Add dextrose when glucose falls
6. Continue until anion gap closes



Heme/Onc

Must Master

- * Iron deficiency anemia
- * Anemia of chronic disease
- * Hemolytic anemia
- * Sickle cell disease
- * TTP/HUS
- * DIC
- * HIT
- * ITP
- * Leukemia/lymphoma
- * Multiple myeloma
- * Neutropenic fever
- * Tumor lysis syndrome
- * Hypercalcemia of malignancy
- * Spinal cord compression
- * SVC syndrome

Oncology Emergency Reflexes

Neutropenic fever → immediate broad-spectrum antibiotics

Cord compression → steroids + urgent imaging/intervention

Tumor lysis → fluids + uric acid management

Hypercalcemia → IV fluids, then targeted therapy

DIC → treat underlying cause + supportive blood products

?

2. Pediatrics

Pediatrics is not “adult medicine in a smaller sweater.” Age changes the differential, normal values, treatment, and thresholds.

Must Master

Newborn

- * Neonatal jaundice
- * Sepsis
- * Respiratory distress syndrome
- * Transient tachypnea
- * Meconium aspiration
- * Congenital heart disease
- * Hypoglycemia
- * Inborn errors
- * NEC
- * Pyloric stenosis
- * Hirschsprung disease

Infectious

- * Fever in infant
- * Meningitis
- * Bronchiolitis
- * Croup
- * Epiglottitis
- * Bacterial tracheitis
- * Pneumonia
- * Otitis media
- * Strep pharyngitis
- * Kawasaki disease
- * MIS-C concepts as applicable
- * Osteomyelitis
- * Septic arthritis

Development

- * Milestones
- * Autism
- * ADHD
- * Intellectual disability
- * Fragile X
- * Down syndrome
- * Child abuse

* Failure to thrive

High-Yield Algorithms

Infant fever

- age determines danger
- very young infant = serious bacterial infection workup
- unstable/toxic = cultures + empiric antibiotics

Stridor

- croup = barking cough, supportive/steroids
- epiglottitis = toxic, drooling, tripod, secure airway
- bacterial tracheitis = toxic + purulent secretions

Cyanotic newborn

- think congenital heart disease
- prostaglandin E1 if ductal-dependent lesion suspected

Pediatric Red Flags

Bilious vomiting

Projectile nonbilious vomiting

Failure to pass meconium

Weight loss/failure to thrive

Bulging fontanelle

Petechiae with fever

Nonblanching rash

Inconsistent injury history

Delay in seeking care

Regression of developmental milestones

?

3. OB/GYN

NBME places roughly 40–45% of its OB/GYN shelf content in pregnancy, childbirth, and puerperium.

Obstetrics

Must Master

- * Prenatal care
- * Pregnancy dating
- * Prenatal screening
- * Rh incompatibility
- * Gestational diabetes
- * Chronic hypertension in pregnancy
- * Preeclampsia/eclampsia
- * HELLP
- * Placenta previa
- * Placental abruption
- * Vasa previa
- * PPROM

- * Preterm labor
- * Chorioamnionitis
- * Labor dystocia
- * Shoulder dystocia
- * Fetal heart tracing
- * Postpartum hemorrhage
- * Postpartum endometritis
- * Postpartum depression/psychosis

Preeclampsia/Eclampsia Logic

Severe features?

- magnesium sulfate for seizure prophylaxis
- control severe hypertension
- delivery when indicated

Postpartum Hemorrhage: The 4 Ts

Tone: uterine atony

Trauma: laceration/rupture

Tissue: retained placenta

Thrombin: coagulopathy

Fetal Heart Rate Pattern

Early decelerations = head compression

Variable decelerations = cord compression

Late decelerations = uteroplacental insufficiency

Prolonged bradycardia = urgent evaluation/intervention

?

Gynecology

Must Master

- * Ectopic pregnancy
- * Ovarian torsion
- * PID
- * Tubo-ovarian abscess
- * Endometriosis
- * Fibroids
- * Abnormal uterine bleeding
- * PCOS
- * Infertility
- * Contraception
- * Cervical cancer screening
- * Ovarian cancer
- * Endometrial cancer
- * Breast mass workup
- * Menopause

Pelvic Pain Algorithm

Pregnancy test first

→ positive = ectopic until proven otherwise

→ sudden severe unilateral pain = torsion

→ fever/cervical motion tenderness = PID

→ chronic cyclic pain = endometriosis

?

4. Psychiatry

Must Master

* Major depressive disorder

* Persistent depressive disorder

* Bipolar disorder

* Mania/hypomania

* Schizophrenia

* Brief psychotic disorder

* Delirium

* Dementia

* Anxiety disorders

* Panic disorder

* OCD

* PTSD

- * Substance intoxication/withdrawal
- * Alcohol withdrawal
- * Opioid use disorder
- * Eating disorders
- * Personality disorders
- * ADHD
- * Autism
- * Suicide risk
- * Capacity
- * Involuntary admission

Emergency Psychiatry

Suicidal ideation

- assess intent, plan, means, past attempts, supports
- imminent risk = do not leave alone, psychiatric emergency evaluation

Delirium

- acute fluctuating attention disturbance
- find underlying medical cause

Mania

- decreased sleep, grandiosity, pressured speech, risky behavior
- mood stabilizer/antipsychotic, assess safety

Medication Toxicity

Syndrome Hallmark

Serotonin syndrome Hyperreflexia, clonus, autonomic instability

NMS Rigidity, fever, autonomic instability, elevated CK

Anticholinergic toxicity Hot, dry, blind, flushed, urinary retention

Sympathomimetic toxicity Agitation, sweating, hypertension, mydriasis

Opioid overdose CNS depression, respiratory depression, miosis

Alcohol withdrawal Tremor, autonomic hyperactivity, hallucinations, seizures

?

5. Surgery

The Surgery shelf is heavily GI, cardiovascular, respiratory, trauma, and acute management. It assesses diagnosis more than management, but management still matters significantly.

Must Master

Trauma

- * Primary survey

- * Secondary survey

- * Hemorrhagic shock

- * Tension pneumothorax

- * Hemothorax

- * Cardiac tamponade

- * Flail chest

- * Splenic injury

- * Liver injury
- * Pelvic fracture
- * Head trauma
- * C-spine injury

Acute Abdomen

- * Appendicitis
- * Cholecystitis
- * Cholangitis
- * Pancreatitis
- * Bowel obstruction
- * Volvulus
- * Mesenteric ischemia
- * Perforation
- * Diverticulitis
- * Ischemic colitis
- * Hernias

Postoperative

- * Fever by postoperative day
- * Atelectasis
- * Pneumonia
- * DVT/PE
- * Wound infection

- * Anastomotic leak
- * Ileus
- * Urinary retention
- * Delirium

Surgical Rule Set

Peritonitis = surgical abdomen until proven otherwise.

Free air = perforated viscus until proven otherwise.

Unstable trauma = bedside intervention before detailed imaging.

Bowel obstruction + peritoneal signs/lactate/ischemia = surgery.

Acute limb ischemia = urgent revascularization.

Compartment syndrome = fasciotomy.

?

6. Family Medicine and Preventive Care

This is quietly free points if you stop treating it like boring paperwork.

Must Master

- * Cancer screening
- * Vaccination
- * Hypertension management
- * Diabetes screening/prevention

- * Hyperlipidemia
- * Obesity
- * Smoking cessation
- * Alcohol counseling
- * Osteoporosis
- * Fall prevention
- * Depression screening
- * Intimate partner violence
- * STI screening
- * HIV prevention
- * Contraception
- * Geriatric preventive care
- * Adolescent confidentiality

Screening Rule Architecture

For every screening topic, learn:

Population

Age range

Risk factors

Test

Interval

When to stop

When not to screen

Follow-up after abnormality

Do not memorize isolated intervals like a medieval monk copying scripture. Tie screening to population and risk.

?

7. Neurology

Must Master

- * Ischemic stroke
- * Hemorrhagic stroke
- * TIA
- * Seizure/status epilepticus
- * Meningitis
- * Encephalitis
- * Migraine
- * Temporal arteritis
- * Parkinson disease
- * Multiple sclerosis
- * Myasthenia gravis
- * Guillain-Barré
- * Spinal cord compression
- * Cauda equina
- * Peripheral neuropathy
- * Bell palsy

- * Trigeminal neuralgia
- * Brain tumors
- * Subarachnoid hemorrhage

Stroke Algorithm

Acute focal neurologic deficit

- noncontrast head CT
- exclude hemorrhage
- assess thrombolysis/thrombectomy eligibility
- do not delay treatment for excessive intellectual ornamentation

?

8. Dermatology, Rheumatology, Orthopedics

Dermatology

- * Cellulitis vs necrotizing fasciitis
- * SJS/TEN
- * Drug eruptions
- * Psoriasis
- * Atopic dermatitis
- * Contact dermatitis
- * Melanoma
- * Basal/squamous cell carcinoma

- * Herpes zoster
- * Scabies
- * Tinea
- * Erythema multiforme

Rheumatology

- * RA
- * OA
- * SLE
- * Sjögren
- * Scleroderma
- * Polymyositis/dermatomyositis
- * Vasculitides
- * Gout/pseudogout
- * Septic arthritis
- * Reactive arthritis
- * Ankylosing spondylitis

Orthopedics

- * Hip fracture
- * Compartment syndrome
- * Septic joint
- * Osteomyelitis
- * ACL/PCL/meniscus

- * Rotator cuff
- * Carpal tunnel
- * Back pain red flags
- * Cauda equina
- * Spondylolisthesis
- * Slipped capital femoral epiphysis
- * Legg-Calvé-Perthes

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VIII. Ethics, Communication, Patient Safety

This is not a side category. It is 10–15% of official Step 2 content and 10–15% of NBME Comprehensive Clinical Science content.

Ethics Decision Framework

1. Does the patient have decision-making capacity?
2. Is the patient informed?
3. Is the decision voluntary?
4. Is there an immediate threat to the patient or others?
5. Is confidentiality protected?
6. Is there a mandatory reporting obligation?
7. Is a surrogate needed?
8. Is this a communication problem before it is an ethics problem?

Capacity

Capacity requires:

Understanding

Appreciation

Reasoning

Ability to communicate a choice

Common Rules

Scenario Best First Move

Patient refuses treatment with capacity Respect refusal after informed discussion

Patient lacks capacity Identify surrogate / advance directive

Teen requests confidential sexual care Respect confidentiality within legal/safety limits

Suspected child abuse Ensure safety and report

Impaired physician Protect patients and report through proper channels

Medical error Disclose honestly, explain corrective plan

Language barrier Use trained interpreter

Family asks you to hide diagnosis Ask patient preferences first

Patient with suicidal plan Ensure immediate safety, psychiatric evaluation

Patient Safety Topics

* Medication reconciliation

* Wrong-site surgery prevention

- * Falls prevention
- * Handoffs
- * Hospital-acquired infections
- * Root cause analysis
- * Sentinel events
- * Checklists
- * Systems error vs individual blame
- * Near misses
- * Quality improvement cycles

?

IX. Biostatistics and Evidence-Based Medicine

Must Know Cold

Sensitivity

Specificity

Positive predictive value

Negative predictive value

Likelihood ratios

Relative risk

Odds ratio

Absolute risk reduction

Relative risk reduction

Number needed to treat

Number needed to harm

Confidence intervals

P values

Type I and Type II error

Power

Bias

Confounding

Effect modification

Study design

Meta-analysis

Forest plot

Kaplan-Meier curves

Intention-to-treat

Core Formulas

Sensitivity = $TP / (TP + FN)$

Specificity = $TN / (TN + FP)$

PPV = $TP / (TP + FP)$

NPV = $TN / (TN + FN)$

Relative Risk = risk exposed / risk unexposed

Absolute Risk Reduction = control event rate - treatment event rate

NNT = $1 / ARR$

High-Yield Interpretation Rules

High sensitivity → negative test rules out disease.

High specificity → positive test rules in disease.

Wide confidence interval → imprecise estimate.

CI crossing 1 for RR/OR → not statistically significant.

CI crossing 0 for mean difference → not statistically significant.

$P < 0.05$ does not prove clinical importance.

Association does not prove causation.

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X. Nutrition Blueprint

Nutrition is now explicitly coded as 15–20% of Step 2 content, and USMLE enhanced nutrition content beginning in June 2026.

Must Master

- * Obesity counseling
- * Diabetes prevention
- * Hyperlipidemia counseling
- * Hypertension/DASH principles
- * Enteral vs parenteral nutrition
- * Refeeding syndrome
- * Malnutrition
- * Alcohol-related deficiencies
- * Vitamin deficiencies

- * Iron, B12, folate deficiencies
- * Pregnancy nutrition
- * Pediatric nutrition
- * Geriatric nutrition
- * Nutrition in CKD
- * Nutrition in cirrhosis
- * Nutrition in heart failure
- * Eating disorders
- * Bariatric surgery complications

Nutrition Clinical Rules

Use enteral feeding if the gut works.

Consider parenteral nutrition when the gut cannot be used.

Refeeding syndrome risk = malnourished patient + carbohydrate load.

Check and replace phosphate, potassium, magnesium.

Lifestyle counseling should be specific, practical, and patient-centered.

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XI. Anki Blueprint

Tag Architecture

Step2::Medicine::Cardiology

Step2::Medicine::Pulmonary

Step2::Medicine::Renal

Step2::Medicine::GI

Step2::Medicine::Endocrine

Step2::Medicine::HemeOnc

Step2::Pediatrics

Step2::OBGYN

Step2::Psychiatry

Step2::Surgery

Step2::FamilyMedicine

Step2::Neurology

Step2::Dermatology

Step2::Ethics

Step2::PatientSafety

Step2::Biostatistics

Step2::Nutrition

Step2::Screening

Step2::Antibiotics

Step2::Algorithms

Step2::Incorrects

Step2::RepeatMisses

Create a Custom Card Only When

You missed the concept twice.

It is an algorithm.

It is a screening interval.

It is a dangerous red flag.

It is a medication contraindication.

It is an ethics rule.

It is a repeat UWorld/NBME trap.

It is a personal blind spot.

Best Custom Card Formats

Algorithm card

Front:

Unstable upper GI bleed: first five actions?

Back:

2 large-bore IVs, fluids/blood products, type/cross, IV PPI or octreotide if variceal suspected, urgent endoscopy after stabilization.

Contrast card

Front:

How do you distinguish TTP from DIC?

Back:

TTP: thrombocytopenia + MAHA, normal coagulation studies.

DIC: thrombocytopenia + MAHA + prolonged PT/PTT + low fibrinogen.

Trigger card

Front:

Sudden unilateral pelvic pain + nausea in reproductive-age patient?

Back:

Ovarian torsion until proven otherwise. Urgent pelvic ultrasound/Doppler and gynecologic evaluation.

?

XII. Shelf Integration Plan

Use each shelf as a Step 2 brick, not a temporary exam that gets mentally deleted the day after.

Medicine Shelf

Focus:

* Cardiology

* Pulmonary

* GI

* Renal

* Endocrine

* Heme/onc

* ID

* Rheumatology

* Neurology

* Geriatrics

Medicine shelf heavily emphasizes diagnosis and management, with 70–80% inpatient care and 20–30% emergency care.

Surgery Shelf

Focus:

- * Trauma
- * Acute abdomen
- * GI
- * Vascular
- * Perioperative management
- * Fluids/electrolytes
- * Post-op complications
- * Breast
- * Urology
- * Critical care

Pediatrics Shelf

Focus:

- * Newborn
- * Development
- * Vaccines

- * Respiratory
- * Fever
- * Congenital disease
- * GI
- * Renal
- * Child abuse
- * Adolescent medicine

OB/GYN Shelf

Focus:

- * Pregnancy
- * Labor
- * Fetal heart tracings
- * Hypertensive disorders
- * Bleeding
- * Postpartum
- * Gynecologic malignancy
- * Contraception
- * Infertility
- * Pelvic pain

Psychiatry Shelf

Focus:

- * Mood
- * Psychosis
- * Substance use
- * Anxiety
- * Delirium/dementia
- * Suicide
- * Capacity
- * Medication adverse effects

Family Medicine Shelf

Focus:

- * Screening
- * Prevention
- * Chronic disease
- * Vaccines
- * Outpatient treatment
- * Geriatrics
- * Pediatrics
- * Obstetrics
- * Lifestyle counseling

XIII. Assessment Plan

During Clerkships

Week 1:

Baseline specialty UWorld performance review

Mid-rotation:

CMS form or targeted mixed block

10–14 days before shelf:

NBME CMS form

5–7 days before shelf:

Second CMS or school practice exam

Last 3 days:

Targeted error-log repair

Rapid algorithms

Weak-system questions

No massive new resource

During Dedicated

Day 1:

CCSSA baseline

Every 7–10 days:

New CCSSA or equivalent school-approved assessment

After every assessment:

1. Identify weakest 3 content areas

2. Identify weakest 3 task types
3. Review every wrong question
4. Build 5–20 targeted cards
5. Complete focused question blocks
6. Reassess the weakness within 7 days

NBME's current CCSSA provides content-area performance, question-level timing, explanations, longitudinal tracking, and an estimated probability of passing, but it should be treated as a guide rather than prophecy carved into stone tablets.

?

XIV. Eight-Week Dedicated Plan

Weeks 1–2: Diagnose the Problem

Daily:

40–80 mixed timed questions

Anki reviews

1–2 weak-system modules

Error-log cleanup

Assessment:

Baseline NBME/CCSSA

One focused CMS form for weakest major shelf

Priority:

Medicine

OB/GYN

Pediatrics

Ethics/patient safety

Biostatistics

Weeks 3–4: Build Mixed-Block Endurance

Daily:

2 timed mixed blocks

Deep review

Targeted AMBOSS/UWorld block for weakness

Anki

Priority:

Mixed management

Diagnostic studies

Antibiotics

Emergency medicine

Surgery

Psychiatry

Screening

Weeks 5–6: Repair and Compress

Daily:

2–3 mixed timed blocks

Rapid review notes

NBME incorrects

Repeat-miss Anki

Focused weak-area reinforcement

Priority:

Your personal bottom five systems

Ethics

Biostats

OB emergency

Peds emergencies

Cardio/pulm emergencies

Week 7: Simulation

Two long-form simulation days

Timed blocks at real Step 2 pace

Break practice

Caffeine/food testing

No new giant resource

Week 8: Calibration

Official USMLE interactive test

Official sample questions

NBME error-log review

Ethics/patient safety review

Screening review

Nutrition review

Biostats review

Sleep stabilization

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XV. Exam-Day Timing System

You have 30 minutes for up to 20 questions.

Pacing Rule

Question 1–10:

Fast first pass.

Answer obvious questions promptly.

Question 11–20:

Maintain pace.

Mark only genuine uncertainty.

Final minutes:

Return to marked questions.

Do not reopen questions you answered confidently.

Marking Rule

Mark a question only if:

You are between two plausible choices.

You need to reread a critical detail.

You need to calculate something.

You suspect a timing trap.

Do not mark because you feel emotionally attached to every patient vignette. They are fictional and will survive your decisiveness.

Reset After a Bad Block

1. Leave the block mentally.
2. Hydrate.
3. Eat something simple.
4. Walk briefly.
5. Do not perform an autopsy on the prior block.
6. Restart with the next patient.

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XVI. Your High-Yield “Always Review” Lists

Every week, revisit:

ACS

Heart failure

Arrhythmias

PE/DVT

Pneumonia

Asthma/COPD

AKI

Electrolytes

Acid-base

GI bleed

Cirrhosis

Pancreatitis

Cholangitis

DKA/HHS

Thyroid emergencies

Stroke

Seizure

Meningitis

Sepsis

Antibiotics

Cancer emergencies

Preeclampsia

Postpartum hemorrhage

Ectopic pregnancy

Fetal tracings

Newborn jaundice

Pediatric fever

Croup/epiglottitis/bronchiolitis

Suicide

Delirium

Substance withdrawal

Trauma

Acute abdomen

Post-op complications

Screening

Vaccines

Ethics

Capacity

Patient safety

Biostatistics

Nutrition

?

XVII. The One-Sentence Step 2 Rulebook

Stabilize unstable patients before chasing diagnostic elegance.

Treat immediately when waiting for confirmation risks death or permanent injury.

Choose the least invasive test that answers the immediate clinical question.

Use the patient's age, timeline, setting, and stability to narrow the differential.

When a patient is septic, bleeding, ischemic, toxic, suicidal, pregnant with an emergency, or neurologically deteriorating, act first.

When the question asks "next best step," identify where you are in the sequence.

When ethics appears, protect autonomy, safety, confidentiality, and informed decision-making.

When prevention appears, know the population, risk factor, test, interval, and follow-up.

When biostatistics appears, translate the equation into clinical meaning.

When you miss a question, repair the reasoning error, not merely the fact.

This is your Step 2 foundation: UWorld as clinical apprenticeship, Anki as retention machinery, NBME as calibration, shelves as cumulative preparation, and an error log that converts every miss into a future point.

Premed?:

Premed does not have a single canonical content outline like the MCAT. It is a moving, school-specific system of coursework, competencies, experiences, testing, and application logistics, because apparently becoming a physician required building a small admissions operating system first.

This prompt is built around the AAMC's 17 premed competencies, school-specific prerequisite variability, AMCAS activity constraints, AACOMAS requirements, and separate pathways such as TMDSAS, OMSAS, and international/Caribbean routes. AAMC explicitly notes that prerequisite policies, including AP, online, and community-college acceptance, vary by school; the app must therefore treat each school's policy as source-backed data rather than universal truth. AMCAS currently permits up to 15 experience entries, with up to three marked "most meaningful," while AACOMAS centralizes applications for U.S. osteopathic schools.

PREMED GOD BLUEPRINT

Full-Spectrum Undergraduate-to-Medical-School Readiness System

U.S.-Primary, International-Compatible, SGU Undergraduate-Compatible

You are building the most comprehensive, intelligent, evidence-backed PreMed Blueprint System possible.

This is not a generic “take biology, volunteer, then apply” checklist. Build a deeply structured, adaptive operating system that helps undergraduate students, post-baccalaureate students, career changers, international students, Caribbean students, SGU undergraduate students, and U.S. premedical students navigate the entire path from first semester through medical-school matriculation.

The system must be practical, highly detailed, longitudinal, data-driven, personalized, and built for later integration into an application called Noctyrium.

The final product should function as:

1. A premedical curriculum planner.
2. A DARS/Degree-Audit interpreter.
3. A prerequisite tracking system.
4. A GPA and science-GPA forecasting engine.
5. An MCAT readiness and preparation planner.
6. A clinical, volunteer, research, leadership, employment, and service tracker.
7. A competency-development portfolio.
8. A letters-of-recommendation management system.
9. An application-cycle command center.
10. A medical-school list-building and fit-analysis engine.
11. A U.S., Canadian, international, Caribbean, and SGU-compatible pathway planner.
12. A gap-year, post-baccalaureate, SMP, and reinvention strategy tool.
13. A documentation vault for everything an applicant will later need.

Do not make this a lecture or static handbook.

Build it as a modular blueprint installation system where each major category becomes an expandable dashboard, tracker, planner, evidence repository, and decision-support engine.

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CORE DESIGN PRINCIPLE

Premedical success is not simply:

- * GPA
- * MCAT
- * Clinical hours
- * Volunteer hours
- * Research

It is the construction of a credible, coherent, ethical, academically capable, service-oriented physician identity over time.

The system must help students answer:

- * Am I academically prepared?
- * Have I completed the correct prerequisites for my intended schools?
- * What gaps exist in my transcript?
- * Do I have enough meaningful clinical exposure?

- * Have I developed the AAMC premed competencies?
- * Are my experiences substantive or merely decorative?
- * Is my MCAT timeline realistic?
- * Do I need a gap year?
- * What schools are realistic, aspirational, or poor fits?
- * What application systems apply to me?
- * What should I do next week, next semester, next year, and before submission?

The system must avoid simplistic “minimum hours” logic.

Instead, it should evaluate:

- * Depth
- * Duration
- * Longitudinality
- * Responsibility
- * Patient proximity
- * Reflection quality
- * Evidence of service orientation
- * Academic trajectory
- * Fit with school missions
- * Competency coverage
- * Personal narrative coherence
- * Institutional and geographic eligibility

PRIMARY USERS

Build pathways for:

1. Traditional U.S. undergraduate premed students.
2. Non-science majors completing medical prerequisites.
3. Science majors.
4. Students with academic difficulty or low early GPAs.
5. Career changers.
6. Post-baccalaureate students.
7. Students planning one or more gap years.
8. International students applying to U.S. MD or DO programs.
9. Canadian students applying to U.S. schools.
10. Students applying through AMCAS.
11. Students applying through AACOMAS.
12. Students applying through TMDSAS.
13. Students applying through OMSAS or Canadian provincial systems.
14. Students exploring Caribbean medical schools.
15. Students in SGU undergraduate or pathway programs.
16. Students aiming for MD, DO, MD/PhD, dual-degree, public-health, research-heavy, rural-health, primary-care, or international medical pathways.

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BLUEPRINT INSTALLATION CATEGORIES

When the user clicks “Install PreMed Blueprint,” generate the following independent modules.

Do not output one giant lecture.

Each module should be separately navigable.

1. Premed Command Center
2. Degree Audit and Academic Map
3. Prerequisite Intelligence Engine
4. GPA, BCPM, and Academic Trajectory Tracker
5. Major and Career Planning
6. MCAT Readiness System
7. Clinical Exposure Portfolio
8. Nonclinical Service Portfolio
9. Research and Scholarly Work Portfolio
10. Leadership, Teaching, Employment, and Extracurricular Portfolio
11. AAMC Competency Map
12. Reflection and Narrative Vault
13. Letters of Recommendation Manager
14. Medical School Intelligence Database
15. School List Builder and Fit Engine
16. Application Systems Command Center
17. Personal Statement and Writing Studio
18. Secondary Application System

- 19. Interview Preparation System
- 20. Situational Judgment Test System
- 21. Financial Planning and Fee Assistance Module
- 22. Gap Year / Post-Bacc / SMP Strategy Engine
- 23. International and Caribbean Pathways
- 24. SGU Undergraduate and Pathway Track
- 25. Compliance, Documentation, and Verification Vault
- 26. Risk Alerts and Recovery Plans
- 27. Semester-by-Semester Roadmap
- 28. Application-Year War Room

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MODULE 1: PREMED COMMAND CENTER

Create a visual dashboard showing:

- * Current year in school
- * Major
- * Minor
- * Institution
- * Expected graduation date
- * Intended application year
- * Intended matriculation year
- * Citizenship or residency category
- * Target pathways:

- * U.S. MD
- * U.S. DO
- * Texas MD/DO
- * Canadian MD
- * Caribbean MD
- * International medical school
- * MD/PhD
- * Dual degree
- * Cumulative GPA
- * Science GPA
- * BCPM GPA
- * Trend GPA:
 - * freshman year
 - * sophomore year
 - * junior year
 - * senior year
 - * post-baccalaureate
- * MCAT status
- * Clinical hours
- * Nonclinical service hours
- * Research hours
- * Shadowing hours
- * Leadership hours
- * Teaching hours
- * Paid employment hours
- * Total longitudinal experiences

- * Number of AAMC competencies supported by evidence
- * Letters secured
- * Prerequisite completion percentage
- * Application readiness score
- * School-list readiness score
- * Timeline risk score

Include a “Next Best Action” engine that prioritizes the top 3 actions based on urgency and downstream impact.

Examples:

- * Register for chemistry before the section fills.
- * Add projected volunteer hours.
- * Request a meeting with a research mentor.
- * Repair a prerequisite deficit.
- * Begin MCAT diagnostic preparation.
- * Build a school-specific prerequisite list.
- * Secure a clinical role with direct patient exposure.
- * Schedule a committee-letter meeting.
- * Draft activity reflections before memories evaporate into the void.

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MODULE 2: DEGREE AUDIT AND ACADEMIC MAP

Build a DARS / degree-audit import feature.

Accepted inputs may include:

- * DARS report
- * DegreeWorks report
- * PDF degree audit
- * Transcript
- * CSV course export
- * Manual course entry
- * International transcript course list
- * SGU undergraduate course plan
- * Semester schedule

The system must extract or ask the user to confirm:

- * Institution
- * Degree type
- * Major
- * Minor
- * General education requirements
- * Major requirements
- * Electives
- * Completed courses
- * In-progress courses
- * Planned courses

- * Grades
- * Credit hours
- * Lab status
- * Online status
- * Community college status
- * AP/IB/A-level/dual-enrollment status
- * Transfer institution
- * Repeated coursework
- * Withdrawals
- * Pass/fail coursework
- * Foreign coursework

For every course, generate:

- * Course title
- * Department
- * Course number
- * Credits
- * Grade
- * Term
- * Institution
- * Lab indicator
- * Delivery mode
- * Transfer status
- * AMCAS subject classification candidate
- * AACOMAS classification candidate

- * BCPM inclusion candidate
- * Prerequisite mapping candidate
- * MCAT relevance
- * Recommended upper-level science relevance

The system must never assume that a course satisfies a medical-school requirement merely because the course title sounds correct.

Instead classify it as:

- * Confirmed match
- * Likely match
- * Needs syllabus review
- * School-specific uncertainty
- * Does not satisfy requirement
- * Duplicate / excess coursework

Include a manual override system.

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MODULE 3: PREREQUISITE INTELLIGENCE ENGINE

Build a school-specific prerequisite engine.

Do not use one universal “premed checklist” as final truth.

The engine should create three layers:

Layer A: Common Core Coursework

Track common requirements and recommendations:

- * General Biology I and II with laboratory
- * General Chemistry I and II with laboratory
- * Organic Chemistry I and II with laboratory
- * Physics I and II with laboratory
- * Biochemistry
- * English composition / writing-intensive coursework
- * Mathematics
- * Statistics
- * Psychology
- * Sociology
- * Genetics
- * Cell Biology
- * Molecular Biology
- * Microbiology
- * Physiology
- * Anatomy
- * Immunology
- * Neuroscience
- * Public Health

- * Ethics

- * Spanish or another patient-care language where relevant

Layer B: School-Specific Requirements

For every target school, capture:

- * Required courses

- * Recommended courses

- * Required labs

- * Minimum credits

- * Time limits on prerequisite coursework

- * Online-course policy

- * Online-lab policy

- * Community-college policy

- * AP-credit policy

- * IB-credit policy

- * Dual-enrollment policy

- * International-coursework policy

- * Canadian-coursework policy

- * Pass/fail policy

- * Prerequisite completion deadline

- * Required degree type

- * International applicant eligibility

- * DACA eligibility

- * Canadian applicant eligibility

- * Residency preference
- * State residency preference
- * Mission-specific coursework recommendations
- * Required behavioral science coursework
- * Required statistics coursework
- * Recommended upper-division science coursework

Layer C: Personalized Course Map

Generate:

- * Missing prerequisites
- * At-risk prerequisites
- * Courses needing stronger grades
- * Courses that should not be taken simultaneously
- * Lab-heavy semester warnings
- * MCAT sequencing recommendations
- * Graduation timeline implications
- * Gap-year implications
- * Financial-aid implications
- * Institution-specific course equivalencies

Create a “Can I Apply?” readiness checker for every medical school.

Output:

- * Yes, academically eligible
- * Eligible with conditions
- * Not yet eligible
- * Eligibility uncertain, verify policy
- * Not eligible because of citizenship/residency policy
- * Not eligible because of course-policy conflict

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MODULE 4: GPA, BCPM, AND ACADEMIC TRAJECTORY TRACKER

Build exact GPA tracking.

Calculate:

- * Cumulative GPA
- * Science GPA
- * BCPM GPA
- * Non-science GPA
- * Post-bacc GPA
- * Undergraduate GPA
- * Graduate GPA
- * Trend GPA
- * Last 30-credit GPA
- * Last 45-credit GPA
- * Last 60-credit GPA

- * Prerequisite GPA
- * Chemistry GPA
- * Biology GPA
- * Physics GPA
- * Math GPA
- * Behavioral-science GPA

Include:

- * Grade-replacement simulation
- * Retake simulation
- * Future-semester GPA projection
- * Credit-weighted GPA calculator
- * “What grades do I need?” calculator
- * Academic recovery scenarios
- * Red-flag course detection
- * Withdrawal and repeated-course tracking
- * Academic trend visualization
- * Explanation generator for unusual academic history

Risk flags:

- * Multiple low science grades
- * Repeated prerequisites
- * Downward trend
- * Excessive withdrawals

- * Incomplete courses
- * Low upper-division science performance
- * Unrealistic course loads
- * Delayed prerequisite completion
- * MCAT planned before sufficient content preparation

Do not reduce academic recovery to “take more classes.”

Recommend pathways based on actual profile:

- * Continue undergraduate repair
- * Targeted post-baccalaureate
- * Formal post-baccalaureate
- * DIY post-baccalaureate
- * Special Master’s Program
- * Graduate science program
- * Gap year with academic strengthening
- * Apply broadly with explanation strategy
- * Delay application

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MODULE 5: MAJOR AND CAREER PLANNING

The system must explicitly state:

A student does not need to major in biology, chemistry, or neuroscience to be premed.

Build major planning around:

- * Intellectual fit
- * GPA sustainability
- * Career fallback value
- * Course overlap
- * Time-to-degree
- * MCAT overlap
- * Research access
- * Writing development
- * Service opportunities
- * Medical-school narrative coherence

Allow majors including:

- * Biology
- * Chemistry
- * Biochemistry
- * Psychology
- * Public Health
- * Neuroscience
- * Sociology
- * Philosophy
- * History

- * English
- * Economics
- * Engineering
- * Computer Science
- * Business
- * Anthropology
- * Political Science
- * Education
- * Fine Arts
- * International Relations
- * Nursing
- * Nutrition
- * Kinesiology
- * SGU undergraduate majors
- * Custom major

For every major, generate:

- * Required courses
- * Compatible prerequisite map
- * MCAT relevance
- * Recommended electives
- * Likely schedule bottlenecks
- * Research opportunities
- * Strengths for narrative development
- * Risk areas

- * Suggested minors or certificates
- * Backup career options



MODULE 6: MCAT READINESS SYSTEM

This is not the MCAT blueprint itself.

This module determines whether the student is ready to begin MCAT preparation.

Track:

- * Completed MCAT prerequisite content
- * Biology readiness
- * General chemistry readiness
- * Organic chemistry readiness
- * Biochemistry readiness
- * Physics readiness
- * Psychology readiness
- * Sociology readiness
- * Statistics readiness
- * Research-methods readiness
- * Reading-comprehension baseline
- * CARS baseline
- * Diagnostic score

- * Target score
- * Target-school median score range
- * Exam date
- * Application timeline
- * Available weekly study hours
- * Test-attempt history
- * Accommodation status
- * Test-location planning
- * Budget

Include logic for:

- * Too early to test
- * Academically ready but not strategically ready
- * Ready for diagnostic
- * Ready for content review
- * Ready for full-length phase
- * Ready to schedule exam
- * Exam date conflicts with application cycle
- * Retake risk
- * Gap-year recommendation

Integrate future resource links and tracking fields for:

- * AAMC official materials
- * Jack Westin CARS

- * Khan Academy
- * UWorld
- * Blueprint
- * Kaplan
- * Princeton Review
- * Anki
- * MileDown-style decks
- * JackSparrow-style decks
- * Daily CARS practice
- * Full-length exam history
- * Error logs
- * Question-bank analytics

The system should emphasize that MCAT preparation cannot compensate for missing academic foundation, weak reading habits, or rushed scheduling.

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MODULE 7: CLINICAL EXPOSURE PORTFOLIO

Create a tracker for meaningful exposure to patient care and clinical environments.

Categories:

- * Medical assistant
- * EMT

- * Paramedic
- * CNA
- * Patient care technician
- * Scribe
- * Hospice volunteer
- * Hospital volunteer
- * Clinic volunteer
- * Clinical research coordinator
- * Phlebotomist
- * Nursing assistant
- * Medical interpreter
- * Physical therapy aide
- * Home health aide
- * Free clinic volunteer
- * Public-health clinical outreach
- * International clinical exposure
- * Shadowing
- * Telehealth experience
- * Health-system administrative work
- * Other clinical role

For each experience, track:

- * Organization
- * Supervisor
- * Role

- * Start date
- * End date
- * Weekly hours
- * Total hours
- * Paid or unpaid
- * Direct patient contact
- * Patient population
- * Underserved population exposure
- * Clinical specialty
- * Urban/rural setting
- * Responsibilities
- * Ethical boundaries
- * Memorable patient encounters
- * Lessons learned
- * AAMC competencies demonstrated
- * Verification contact
- * Documentation uploaded
- * Potential activity category
- * Potential most-meaningful designation

Differentiate:

- * Passive observation
- * Clinical exposure
- * Direct patient care
- * Administrative healthcare work

- * Shadowing
- * Research involving patients

Do not falsely classify every hospital-related role as clinical.

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MODULE 8: NONCLINICAL SERVICE PORTFOLIO

Build a serious service tracker.

Track service involving:

- * Food insecurity
- * Homelessness
- * Refugee support
- * Disability advocacy
- * Tutoring underserved students
- * Prison reentry support
- * Domestic violence resources
- * Crisis response
- * Community health education
- * Faith-based service
- * Elder support
- * Youth mentorship
- * Immigration support

- * Rural outreach
- * Public-health outreach
- * Disaster relief
- * Community organizing
- * International service
- * Mutual aid
- * Free legal or social support services

For each experience, capture:

- * Community served
- * Longitudinality
- * Frequency
- * Role
- * Responsibility
- * Direct service versus fundraising
- * Community need addressed
- * Reflection
- * Impact
- * Humility and reciprocity considerations
- * Whether service is sustainable or performative

Flag one-time mission trips, isolated service events, and resume-stacking patterns as weak unless supported by meaningful longitudinal commitment.

MODULE 9: RESEARCH AND SCHOLARLY WORK PORTFOLIO

Track:

- * Basic science research
- * Clinical research
- * Translational research
- * Public health research
- * Health disparities research
- * Behavioral research
- * Computational research
- * Humanities research
- * Quality improvement
- * Case reports
- * Literature reviews
- * Posters
- * Abstracts
- * Oral presentations
- * Publications
- * Preprints
- * Grants
- * Independent projects
- * Thesis work
- * Laboratory technical skills
- * Statistical skills

- * Programming skills
- * IRB exposure
- * Research ethics training

For each project, capture:

- * Principal investigator
- * Institution
- * Project title
- * Research question
- * Methods
- * Study design
- * Data type
- * Contribution level
- * Hours
- * Duration
- * Publication status
- * Authorship position
- * Poster/presentation status
- * Skills gained
- * Challenges overcome
- * Future recommendation-letter potential
- * Competencies demonstrated

Avoid treating research as mandatory for every school.

Instead, use the school-fit database to assess whether research is:

- * Essential
- * Strongly recommended
- * Helpful
- * Not central
- * Mission-specific

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MODULE 10: LEADERSHIP, TEACHING, EMPLOYMENT, AND EXTRACURRICULAR PORTFOLIO

Track:

- * Leadership
- * Teaching assistant roles
- * Tutoring
- * Peer mentorship
- * Resident assistant work
- * Student government
- * Club leadership
- * Athletics
- * Arts
- * Music
- * Writing

- * Entrepreneurship
- * Military service
- * Family caregiving
- * Employment
- * Religious service
- * Community leadership
- * Language skills
- * Technical projects
- * Advocacy
- * Personal achievements
- * Hobbies pursued seriously over time

The system must recognize that paid work, caregiving, family obligations, and nontraditional life experience can be highly meaningful.

Build an “adversity and responsibility” tracker that allows applicants to document:

- * Work needed to support family
- * Caregiving
- * Immigration transitions
- * Financial hardship
- * Disability
- * Academic disruption
- * Military obligations
- * Family illness
- * Housing instability

- * Major life transitions

Do not pressure students to disclose sensitive information publicly.

Provide privacy controls and a “use only if strategically appropriate” toggle.



MODULE 11: AAMC COMPETENCY MAP

Map every experience to the premedical competency framework.

Create categories:

Professional Competencies

- * Service orientation
- * Social skills
- * Cultural competence
- * Teamwork
- * Oral communication
- * Ethical responsibility to self and others
- * Reliability and dependability
- * Resilience and adaptability
- * Capacity for improvement

Thinking and Reasoning Competencies

- * Critical thinking
- * Quantitative reasoning
- * Scientific inquiry
- * Written communication

Science Competencies

- * Living systems
- * Human behavior

For every competency, require:

- * Definition
- * Why it matters
- * Evidence from experiences
- * Evidence from coursework
- * Evidence from employment
- * Reflection example
- * Strength rating
- * Evidence quality rating
- * Missing evidence
- * Recommended developmental activity
- * Potential application story
- * Potential interview story

The dashboard should show:

- * Competencies with strong evidence
- * Competencies with weak evidence
- * Competencies with no evidence
- * Overrepresented competencies
- * Experiences doing too much “checkbox work” but showing little genuine development

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MODULE 12: REFLECTION AND NARRATIVE VAULT

Build a private experience journal.

For each experience, prompt the student to record:

- * What happened?
- * Who was involved?
- * What did I do?
- * What did I notice?
- * What changed in my thinking?
- * What did I misunderstand before?
- * What did I learn about patients, systems, inequity, teamwork, ethics, or medicine?
- * What did I contribute?
- * What would I do differently?

- * What competency does this support?
- * Is this a possible interview story?
- * Is this a possible personal-statement story?
- * Is this a possible most-meaningful activity?

Include date-stamped entries.

Create an “Application Story Bank” with tags:

- * Service orientation

- * Failure

- * Leadership

- * Conflict

- * Ethical dilemma

- * Cultural humility

- * Teamwork

- * Research discovery

- * Clinical insight

- * Adversity

- * Resilience

- * Advocacy

- * Growth

- * Why medicine

- * Why DO

- * Why this school

- * Gap year

- * Academic difficulty
- * Personal identity
- * Patient encounter

The system must preserve authenticity.

Do not generate fake stories, exaggerate responsibilities, or write sentimental nonsense that could belong to any applicant with a stethoscope emoji.

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MODULE 13: LETTERS OF RECOMMENDATION MANAGER

Track potential recommenders:

- * Science professor
- * Non-science professor
- * Research mentor
- * Clinical supervisor
- * Physician supervisor
- * Employer
- * Volunteer coordinator
- * Committee letter advisor
- * Pre-health advisor
- * DO physician
- * MD physician

* Graduate faculty member

For each recommender, track:

* Name

* Institution

* Role

* Contact information

* Relationship length

* Courses taken

* Grades earned

* Research involvement

* Clinical supervision

* Specific examples they can describe

* Strength of relationship

* Likelihood of strong letter

* School-specific eligibility

* Letter deadline

* Requested date

* Follow-up date

* Received date

* Thank-you note status

* Interfolio / AMCAS / AACOMAS / TMDSAS / school submission pathway

Build reminders beginning 6–12 months before application submission.

Add a letter-quality indicator:

* Weak acquaintance

* Adequate

* Strong

* Outstanding

* Unknown

Never recommend asking someone who barely knows the student merely because they have a prestigious title.

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MODULE 14: MEDICAL SCHOOL INTELLIGENCE DATABASE

Build a source-backed school database.

Include:

* School name

* Degree type

* MD / DO / Caribbean MD / international MD

* Location

* State

* Region

* Public/private

- * Mission statement
- * Population served
- * Rural focus
- * Urban focus
- * Primary-care focus
- * Research intensity
- * Community-health emphasis
- * Health-disparities emphasis
- * Faith-based affiliation
- * Military affiliation
- * Historically Black medical school status
- * Hispanic-serving emphasis
- * Tribal/Indigenous emphasis
- * International applicant acceptance
- * Canadian applicant acceptance
- * DACA acceptance
- * State-residency preference
- * Regional preference
- * Early decision availability
- * Application service accepted
- * Required exams
- * PREview requirement
- * Casper requirement
- * Other assessment requirement
- * MCAT policy
- * MCAT validity window

- * GPA screening policy if published
- * Median MCAT if published
- * Median GPA if published
- * Acceptance rate if published
- * Tuition
- * Fees
- * Estimated cost of attendance
- * Scholarships
- * Fee waiver availability
- * Required prerequisite courses
- * Recommended courses
- * Online-course policy
- * Community-college policy
- * AP-credit policy
- * International coursework policy
- * Letters required
- * Secondary essay themes
- * Interview format
- * Curriculum style
- * Grading system
- * Clinical rotation structure
- * Research opportunities
- * Dual-degree options
- * Match outcomes
- * Accreditation status
- * Source URL

- * Source date
- * Confidence rating
- * Last verified date

Every factual field must include:

- * Source
- * Date accessed
- * Date last verified
- * Confidence level
- * Whether the data is school-reported, third-party, estimated, or inferred

Never present stale or unsourced admissions data as current.

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MODULE 15: SCHOOL LIST BUILDER AND FIT ENGINE

Build a school-list tool that evaluates:

Academic Fit

- * GPA
- * Science GPA
- * MCAT
- * Academic trend

- * Course rigor
- * Prerequisite completion
- * Post-baccalaureate performance

Mission Fit

- * Service orientation
- * Rural background
- * Urban health
- * Health disparities
- * Primary care
- * Research
- * Community engagement
- * Military service
- * Faith-based mission
- * Global health
- * Underserved populations
- * Home-state preference
- * Regional connection

Eligibility Fit

- * Residency
- * Citizenship
- * Visa status
- * International applicant policy

- * Canadian policy
- * DACA policy
- * Prerequisite policy
- * MCAT age policy
- * Degree-completion policy

Strategic Fit

- * Application timing
- * Financial feasibility
- * Geographic preferences
- * Family constraints
- * Clinical/research fit
- * Curriculum preferences
- * Grading preference
- * Match aspirations
- * Specialty interests

Output tiers:

- * Strong mission and academic fit
- * Reasonable fit
- * Reach but defensible
- * High-risk reach
- * Low-yield because of residency or mission mismatch
- * Ineligible

- * Needs policy verification

Do not create school lists solely from GPA and MCAT.

Build an “Explain Why This School Is On Your List” field for every school.

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MODULE 16: APPLICATION SYSTEMS COMMAND CENTER

Build separate workflows for:

AMCAS

Track:

- * Biographical information
- * Coursework entry
- * Transcript requests
- * GPA verification
- * Work and activities
- * Most meaningful experiences
- * Personal statement
- * Letters
- * School list
- * Fee Assistance Program

- * PREview
- * Casper
- * Secondary applications
- * Interview invitations
- * Acceptance / waitlist / rejection tracking
- * Choose Your Medical School milestones

AACOMAS

Track:

- * Coursework entry
- * Transcript requests
- * Experiences
- * Achievements
- * Personal statement
- * Osteopathic exposure
- * DO physician letter needs
- * School list
- * Verification
- * Secondary applications
- * Interview status
- * Deposits
- * Traffic-guideline deadlines

TMDSAS

Track:

- * Texas residency
- * Personal characteristics essay
- * optional essay
- * activity entries
- * planned activities
- * school ranking
- * Texas-specific deadlines
- * Texas medical and dental school policies
- * JAMP eligibility when applicable

OMSAS / Canadian Systems

Track:

- * Provincial eligibility
- * ABS-style activity tracking
- * verifiers
- * regional residency
- * school-specific essays
- * MCAT requirements
- * CASPer requirements
- * references
- * GPA formulas

- * bilingual requirements where applicable

International / Caribbean Systems

Track:

- * Degree requirements
- * prerequisite coursework
- * international transcript evaluation
- * English-language testing where needed
- * MCAT policy
- * visa requirements
- * accreditation status
- * residency eligibility implications
- * clinical rotation locations
- * financial aid eligibility
- * U.S. licensure eligibility considerations
- * match outcomes
- * school-specific admissions processes



MODULE 17: PERSONAL STATEMENT AND WRITING STUDIO

Build structured writing systems for:

- * AMCAS personal statement
- * AACOMAS personal statement
- * TMDSAS personal characteristics essay
- * TMDSAS optional essay
- * Canadian autobiographical sketch narratives
- * Secondary essays
- * Diversity essays
- * Challenge essays
- * Adversity essays
- * Gap-year essays
- * “Why us?” essays
- * “Why DO?” essays
- * Research essays
- * Rural-health essays
- * Service essays
- * Reapplicant essays

Build an outline generator based on the student’s actual experience vault.

Require:

- * Central thesis
- * Why medicine
- * Turning points
- * Patient or community insight
- * Evidence of maturity

- * Evidence of service
- * Evidence of reflection
- * Evidence of competence
- * Future direction
- * Avoided clichés
- * School-specific mission alignment

Flag:

- * Resume recitation
- * Hero narratives
- * Saviorism
- * Overdramatization
- * Unsupported claims
- * Excessive trauma disclosure
- * Artificial sentimentality
- * Medical TV influence as primary motivation
- * “I love science and helping people” with no evidence

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MODULE 18: SECONDARY APPLICATION SYSTEM

Build a school-specific secondary tracker.

For each school:

- * Invite date
- * Deadline
- * Fee
- * Essay prompts
- * Character limits
- * Reusable essay themes
- * School mission
- * Required competencies
- * Geographic context
- * Faculty/research interests
- * Clinical affiliations
- * Community programs
- * Submission date
- * Quality-control checklist
- * Proofreading status
- * Final PDF/archive

Create a secondary-essay matrix showing overlap among prompts.

Categories:

- * Diversity
- * Adversity
- * Service
- * Leadership

- * Ethical dilemma
- * Gap year
- * Community
- * Rural medicine
- * Health equity
- * Research
- * Family ties
- * Geographic ties
- * Why this school
- * Why this city
- * Why DO
- * Innovation
- * Challenge
- * Identity
- * Resilience

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MODULE 19: INTERVIEW PREPARATION SYSTEM

Build preparation tracks for:

- * Traditional interview
- * MMI
- * Panel interview
- * Group interview

- * Virtual interview
- * Student interview
- * Faculty interview
- * Blind interview
- * Open-file interview
- * Ethical scenario
- * Role-play
- * Healthcare policy discussion
- * Research discussion
- * Behavioral interview

Include:

- * Personal story bank
- * School-specific mission briefing
- * Mock interview log
- * Feedback tracker
- * Weak-answer detection
- * Filler-word tracking
- * Timing tracker
- * Ethical reasoning framework
- * MMI scenario bank
- * Difficult-question bank
- * Questions to ask programs
- * Post-interview thank-you tracker
- * Waitlist letter-of-intent manager

Key interview prompts:

- * Why medicine?
- * Why this school?
- * Why DO?
- * Tell me about yourself.
- * Tell me about a failure.
- * Tell me about conflict.
- * Tell me about a difficult patient interaction.
- * Tell me about an ethical dilemma.
- * Tell me about an experience with diversity.
- * What have you learned from service?
- * What would you do if you were not accepted?
- * What is a healthcare problem you care about?
- * What does professionalism mean to you?
- * What will you contribute to our class?

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MODULE 20: SITUATIONAL JUDGMENT TEST SYSTEM

Build a school-specific assessment tracker for:

- * AAMC PREview
- * Casper

- * Duet or associated school-fit tools where applicable
- * Other school-specific assessments

Track:

- * Schools requiring the exam
- * Registration deadline
- * Test date
- * Accommodation status
- * Score release timing
- * Preparation resources
- * Practice scenarios
- * Ethical reasoning framework
- * Professionalism framework
- * Communication framework
- * Conflict-resolution framework
- * Equity and cultural humility framework

Do not assume one test applies to every school.

The system must pull current school-specific requirements from verified admissions pages.



MODULE 21: FINANCIAL PLANNING AND FEE ASSISTANCE

Track:

- * Undergraduate tuition
- * MCAT registration
- * MCAT preparation materials
- * PREview fees
- * Casper fees
- * Application fees
- * Secondary fees
- * Transcript fees
- * Interfolio fees
- * Travel
- * Interview attire
- * Deposits
- * Relocation
- * Post-baccalaureate tuition
- * SMP tuition
- * Gap-year income
- * Scholarships
- * Fee assistance
- * Family support
- * Loans
- * Savings target

Build a cycle budget.

Include:

- * Conservative estimate
- * Expected estimate
- * High-cost estimate
- * International applicant estimate
- * Caribbean pathway estimate
- * Number-of-schools sensitivity analysis

Track possible financial-assistance programs where eligible, but never assume eligibility.

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MODULE 22: GAP YEAR, POST-BACC, SMP, AND REINVENTION ENGINE

Build decision pathways for:

- * Apply directly
- * Take one gap year
- * Take two gap years
- * Work clinically before applying
- * Research gap year
- * Service gap year
- * Post-baccalaureate
- * DIY academic enhancer

- * Formal academic enhancer
- * Career changer post-baccalaureate
- * Special Master's Program
- * Master's in public health
- * Master's in biomedical sciences
- * Reapplication plan

Evaluate:

- * Academic readiness
- * MCAT readiness
- * Clinical maturity
- * Service depth
- * Research needs
- * Financial readiness
- * Personal readiness
- * Mental-health and burnout risk
- * Timeline pressure
- * Application competitiveness
- * Longitudinal narrative

The system must normalize delaying application when delay materially improves readiness.

Do not treat immediate application as inherently superior.



MODULE 23: INTERNATIONAL AND CARIBBEAN PATHWAYS

Build specialized support for international applicants.

Track:

- * Country of citizenship
- * Country of residence
- * Undergraduate institution country
- * Degree equivalency
- * WES or credential-evaluation needs
- * English-language testing
- * Visa requirements
- * U.S. MD international acceptance
- * U.S. DO international acceptance
- * Canadian medical-school eligibility
- * Caribbean medical-school pathway
- * UK and Australian pathway considerations
- * Financing restrictions
- * Clinical-rotation eligibility
- * Residency implications
- * Licensing eligibility considerations

For Caribbean pathways, track:

- * Accreditation status
- * Accreditor
- * Recognition pathways
- * U.S. federal-loan eligibility where applicable
- * Clinical rotation network
- * Residency match outcomes
- * Attrition data if published
- * USMLE performance data if published
- * Residency placement details
- * State licensure compatibility
- * Cost
- * Prerequisite requirements
- * MCAT policy
- * Academic-support systems
- * Transfer policies

Do not market Caribbean medicine as interchangeable with U.S. MD or DO pathways.

Present risks, benefits, eligibility restrictions, costs, and residency implications clearly and without sales language.

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MODULE 24: SGU UNDERGRADUATE AND PATHWAY TRACK

Build an SGU-specific premedical planning lane for students in SGU undergraduate programs or associated pathway programs.

Include:

- * Program type
- * Entry pathway
- * Course plan
- * Grenada campus timeline
- * Degree requirements
- * Medical-school prerequisite overlap
- * SGU School of Medicine pathway eligibility
- * External U.S. MD and DO readiness
- * International applicant considerations
- * MCAT preparation timing
- * U.S. clinical-experience planning
- * Research opportunities
- * Letters-of-recommendation strategy
- * International transcript/document handling
- * Summer planning
- * Gap-year planning
- * U.S. application compatibility

The SGU module must be configurable rather than based on assumptions.

It should require:

- * Current catalog year
- * Exact undergraduate program
- * Current course sequence
- * Intended medical-school pathway
- * Citizenship/residency category
- * Target countries
- * Intended application service

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MODULE 25: COMPLIANCE, DOCUMENTATION, AND VERIFICATION VAULT

Build secure storage categories for:

- * Official transcripts
- * Unofficial transcripts
- * Degree audits
- * Course syllabi
- * AP/IB/A-level results
- * Course descriptions
- * Resume/CV versions
- * Certifications
- * BLS/CPR certifications
- * EMT/CNA certifications
- * Research documentation

- * Poster PDFs
- * Publication citations
- * Volunteer verification
- * Employment verification
- * Shadowing verification
- * Contact information
- * Recommendation-letter tracker
- * Immunization records
- * Background-check records
- * Passport and visa documents
- * Financial-aid documents
- * Application PDFs
- * Secondary essay archive
- * Interview notes
- * Acceptance letters
- * Deposit receipts

Each record should have:

- * Upload date
- * Expiration date
- * Verification status
- * Owner
- * Privacy classification
- * Relevance to application
- * Reminder date



MODULE 26: RISK ALERTS AND RECOVERY PLANS

Generate proactive alerts for:

- * Missing prerequisites
- * Low prerequisite GPA
- * Downward GPA trend
- * No clinical exposure
- * No service orientation evidence
- * Shallow research involvement
- * Lack of longitudinal commitment
- * Weak writing evidence
- * No recommenders who know the student well
- * Late MCAT timing
- * Unrealistic application year
- * Ineligible schools on list
- * International eligibility issue
- * Missing transcript
- * Expiring certification
- * Overloaded semester
- * Burnout risk
- * Excessive activity stacking
- * Poor school-list balance

- * Lack of financial planning
- * No backup plan

For every alert, provide:

- * Why this matters
- * Severity
- * Evidence
- * Recommended intervention
- * Timeline
- * Alternative options
- * Whether the issue can be repaired before application
- * Whether application delay is recommended

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MODULE 27: SEMESTER-BY-SEMESTER ROADMAP

Build customizable roadmaps.

Include versions for:

- * First-year undergraduate
- * Sophomore
- * Junior
- * Senior

- * Nontraditional student
- * Career changer
- * Gap-year applicant
- * Reapplicant
- * SGU undergraduate
- * International student
- * Canadian applicant
- * Student with academic repair needs

Every semester should include:

- * Coursework
- * GPA goals
- * Clinical activities
- * Service activities
- * Research
- * Leadership
- * Shadowing
- * Mentorship
- * MCAT milestones
- * Application milestones
- * Letter-building
- * Reflection logging
- * Financial planning
- * Wellness and burnout prevention

Create a workload-balancing model.

Do not recommend simultaneously taking:

- * Multiple high-intensity sciences with labs
- * Heavy leadership commitments
- * High clinical work hours
- * Intensive MCAT preparation
- * Full application writing

unless the student's demonstrated history supports it.

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MODULE 28: APPLICATION-YEAR WAR ROOM

Create an application-year dashboard.

Track:

- * Application opening date
- * Submission target date
- * Transcript request status
- * MCAT score release
- * PREview date
- * Casper date

- * Letter status
- * Personal statement readiness
- * Activity section readiness
- * School-list finalization
- * Secondary turnaround time
- * Interview preparation
- * Financial tracking
- * Acceptance timeline
- * Waitlist strategy
- * Deposit deadlines
- * Traffic-guideline deadlines
- * Reapplication contingency plan

Create weekly priorities:

- * Week 1: finalize transcript data.
- * Week 2: verify activity hours.
- * Week 3: polish personal statement.
- * Week 4: submit primary application.
- * Ongoing: return secondaries rapidly without sacrificing quality.
- * Ongoing: maintain clinical/service commitments.
- * Ongoing: prepare for interviews before invitations arrive.
- * Ongoing: log all communications.

Build a “Do Not Do This” panel:

- * Do not submit inaccurate coursework.
- * Do not exaggerate hours.
- * Do not list unverifiable activities.
- * Do not mass-apply without school-fit logic.
- * Do not use generic secondary essays.
- * Do not wait until interview invitations to begin preparation.
- * Do not abandon meaningful commitments after submitting.
- * Do not apply simply because the calendar says you are supposed to.

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DATA MODEL REQUIREMENTS

Use structured entities.

Student Profile

- * Personal information
- * Residency
- * Citizenship
- * Academic history
- * Career interests
- * Medical-school pathway
- * Application year
- * Major
- * Institution

- * Financial profile
- * Personal constraints
- * Privacy settings

Course Entity

- * Course ID
- * Institution
- * Department
- * Number
- * Title
- * Credits
- * Grade
- * Term
- * Lab
- * Delivery mode
- * Transfer source
- * Prerequisite mappings
- * AMCAS classification
- * AACOMAS classification
- * BCPM designation
- * Verification status

Experience Entity

- * Type

- * Organization
- * Role
- * Dates
- * Hours
- * Supervisor
- * Population served
- * Responsibilities
- * Reflection entries
- * Competencies
- * Verification
- * Application category
- * Most meaningful potential

School Entity

- * Academic requirements
- * Eligibility requirements
- * Mission
- * statistics
- * cost
- * testing requirements
- * application details
- * source documentation
- * update history

Application Entity

* Application service

* School

* essays

* letters

* tests

* deadlines

* payment

* status

* interview notes

* decision

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SCORING SYSTEMS

Build several separate scores.

Do not create one fake “chance of admission” score.

Instead create:

Academic Readiness Score

Based on:

- * GPA

- * BCPM GPA

- * academic trend

- * prerequisite completion

- * course rigor

- * MCAT readiness

Experience Depth Score

Based on:

- * longitudinal clinical exposure

- * nonclinical service

- * leadership

- * research

- * responsibility

- * reflection quality

Competency Coverage Score

Based on AAMC competency evidence.

Application Readiness Score

Based on:

- * transcript readiness

- * letters

- * activities

- * writing

- * tests

- * school list

- * finances

- * deadlines

School Fit Score

Based on:

- * mission alignment

- * eligibility

- * academics

- * geography

- * residency

- * service profile

- * research profile

- * career goals

Risk Score

Based on:

- * missing requirements
- * timeline issues
- * burnout
- * financial risk
- * weak letters
- * insufficient clinical exposure
- * poor school-list strategy

Every score must display its inputs.

Never hide logic behind opaque scoring.

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AI BEHAVIOR REQUIREMENTS

The AI must:

- * Ask for evidence before making conclusions.
- * Distinguish fact from inference.
- * Cite official sources when discussing school policies.
- * Flag uncertain information.
- * Avoid guaranteeing admissions.
- * Avoid telling students they “need” a certain number of hours as a universal rule.
- * Avoid encouraging résumé padding.
- * Avoid generating fabricated patient stories.

- * Avoid telling students to choose a major solely because it “looks good.”
- * Avoid treating service as transactional.
- * Avoid medical advice beyond general wellness and academic-planning guidance.
- * Encourage sustainable, ethical development.

The AI should be direct.

Examples:

Bad:

“You need 500 clinical hours to get into medical school.”

Better:

“Your 75 hours of hospital volunteering show early exposure, but they do not yet demonstrate sustained patient-facing engagement. A longer-term clinical role may strengthen your evidence of commitment.”

Bad:

“Your GPA is too low. Give up.”

Better:

“Your current GPA creates risk for many programs, but the next decision depends on trend, prerequisite grades, MCAT readiness, and whether a targeted academic-enhancement plan can demonstrate durable improvement.”

SOURCE HIERARCHY

Use this source hierarchy:

Tier 1: Official Sources

- * AAMC
- * AMCAS
- * AACOM
- * AACOMAS
- * TMDSAS
- * OMSAS
- * Individual medical-school admissions pages
- * LCME
- * COCA
- * ACGME where relevant
- * SGU official catalog and admissions pages
- * Official university catalogs
- * Official registrar and degree-audit documents

Tier 2: Primary Institutional Data

- * Published admissions statistics
- * School fact sheets
- * Curriculum pages
- * Official outcome reports

- * Published financial-aid documents
- * Accreditation reports

Tier 3: Trusted Educational Sources

- * University pre-health advising offices
- * Official MCAT preparation sources
- * Recognized testing organizations

Do not use Reddit, random forums, or commercial admissions blogs as authoritative sources for requirements.

They may be used only as anecdotal context, clearly labeled as anecdotal.

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OUTPUT REQUIREMENTS

When generating a student's PreMed Blueprint, output:

1. Student snapshot.
2. Academic audit.
3. Missing prerequisites.
4. GPA trajectory analysis.
5. MCAT readiness assessment.
6. Experience portfolio audit.

7. AAMC competency map.
8. Letters-of-recommendation plan.
9. School-list eligibility report.
10. Application-system roadmap.
11. Semester-by-semester action plan.
12. Risk alerts.
13. Suggested gap-year/post-bacc strategy if needed.
14. Immediate next actions.
15. Evidence and sources.

The output must be structured, navigable, and actionable.

Every recommendation should include:

- * Why it matters
- * What to do
- * When to do it
- * What evidence would show completion
- * What happens if it is not addressed

Build the system as a long-term physician-pathway tracker, not a frantic senior-year application generator.

The point is not merely to help a student get admitted.

The point is to help them become the kind of applicant whose record makes sense because the life behind it makes sense.

The strongest architectural choice here is the school-policy intelligence layer. A course may satisfy one school's requirement and fail another's because of online, AP, lab, community-college, residency, or international-credit rules. That needs to be database logic, not a generic checklist pretending the world is organized. AAMC itself directs applicants to school-specific policies, while AACOM notes broad coursework norms but tells applicants to verify individual college requirements.

For application tracking, build directly around the current application systems rather than trying to flatten them into one form. AMCAS has its own activity architecture, TMDSAS publishes an entry-year guide, AACOMAS has separate instructions and traffic rules, and OMSAS has province- and school-dependent rules.

Step 1:

USMLE Step 1 Master Blueprint, 2026

This is built around the official USMLE Step 1 content outline, not the AAMC outline. The AAMC framework is for the MCAT; Step 1 tests integrated basic science through clinical vignettes, mechanisms of disease, treatment effects, ethics, population health, and interpretation of data. Humanity, having invented them, naturally made them annoyingly different.

Step 1 is pass/fail, but the preparation still needs to be ruthless because the real outcome is not merely passing. It is building the base that determines how painful or efficient Step 2, clerkships, research literacy, and clinical reasoning become.

As of May 14, 2026, Step 1 is administered as fourteen 30-minute blocks in an eight-hour testing day, with up to 20 items per block and at least 55 minutes of break time.

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1. The Core Step 1 System

Your system should have five layers:

1. Learn the mechanism
2. Compress it into First Aid
3. Lock it with Anki
4. Stress-test it with questions
5. Repair weaknesses using an error system

Do not use every resource for every topic. That creates the illusion of productivity while your brain quietly turns into a browser with 47 tabs open.

Primary resource stack

Function	Primary Resource	Purpose
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Master index	First Aid 2026	Central map of all high-yield facts
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Pathology	Pathoma	Mechanism-first pathology
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Microbiology	Sketchy Micro	Organism recognition and recall
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Pharmacology	Sketchy Pharm	Drug class, MOA, toxicity
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Physiology/foundation	Boards & Beyond or Bootcamp	Mechanistic understanding
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Question bank UWorld Clinical integration and error discovery

Secondary question bank AMBOSS Targeted weak-area drilling

Spaced repetition AnKing Step Deck Long-term retention

Readiness testing NBME CBSSA Official-style performance checkpoints

Exam software USMLE Interactive Testing Experience Interface familiarization

First Aid should function as your master operating manual, not your only teacher. The 2026 edition is organized around core principles and organ systems, with high-yield facts, clinical images, mnemonics, and test-taking guidance.

Pathoma is best used for pathology mechanisms and pattern recognition. Its current package includes a 218-page text and more than 35 hours of lectures.

Sketchy is most valuable where pure memorization is unavoidable: microbiology and pharmacology. Sketchy Micro covers bacteria, viruses, fungi, and parasites through visual mnemonic scenes; Sketchy Pharm focuses on drug classes, mechanisms, indications, and adverse effects.

UWorld should be your main clinical reasoning engine. Its Step 1 QBank currently contains more than 3,600 questions and is continuously updated against the USMLE content outline.

The AnKing Step Deck is useful because it is tagged to major resources including First Aid, Boards & Beyond, UWorld, Pathoma, Sketchy, and more. It contains more than 30,000 cards, so you must use tags and selective unsuspension rather than attempting to brute-force the entire deck like a man trying to drink the ocean.

2. The Step 1 Competency Model

Every topic must be studied through the same clinical chain:

Normal physiology → disruption → pathogenesis → signs/symptoms → labs/imaging → complications → treatment mechanism → drug toxicity → differential diagnosis

For every disease, you should know:

1. What normally happens?
2. What changed?
3. Why did it change?
4. What does that produce clinically?
5. What laboratory pattern follows?
6. What disease looks similar?
7. What treatment corrects the mechanism?
8. What adverse effect can that treatment cause?
9. What is the classic vignette trap?

Example:

Primary hyperparathyroidism

* Normal: PTH raises serum calcium.

* Disruption: autonomous parathyroid hormone secretion.

* Mechanism: increased bone resorption, renal calcium reabsorption, phosphate wasting, increased calcitriol activation.

- * Findings: hypercalcemia, hypophosphatemia, elevated PTH, nephrolithiasis, bone pain.
- * Differential: familial hypocalciuric hypercalcemia has low urinary calcium.
- * Treatment logic: surgery when indicated; cinacalcet lowers PTH secretion through calcium-sensing receptor activation.
- * Common trap: confusing it with malignancy-mediated hypercalcemia, where PTH is suppressed.

That level of reasoning is what Step 1 actually rewards.

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3. The Non-Negotiable Topic Blueprint

The official USMLE outline organizes Step content by body system and process, including immune, blood, behavioral health, neuro, skin, musculoskeletal, cardiovascular, respiratory, gastrointestinal, renal, reproductive, endocrine, multisystem disease, biostatistics, public health, and social science.

A. Foundational Science Layer

This should be completed early and then revisited inside every organ system.

Biochemistry and molecular biology

Master:

- * DNA replication, repair, transcription, translation

- * Gene regulation and epigenetics
- * Mendelian inheritance
- * Pedigree analysis
- * Chromosomal disorders
- * Imprinting, anticipation, mosaicism
- * Hardy-Weinberg equilibrium
- * Oncogenes and tumor suppressor genes
- * Signal transduction pathways
- * Enzyme kinetics
- * Rate-limiting enzymes
- * Vitamin and cofactor deficiencies
- * Metabolic pathways:
 - * Glycolysis
 - * Gluconeogenesis
 - * TCA cycle
 - * Oxidative phosphorylation
 - * Pentose phosphate pathway
 - * Glycogen metabolism
 - * Fructose/galactose metabolism
 - * Fatty acid synthesis and oxidation
 - * Ketone metabolism
 - * Urea cycle
 - * Purine/pyrimidine synthesis

Resources:

- * Boards & Beyond or Bootcamp for first-pass mechanisms
- * First Aid biochemistry chapter for consolidation
- * AnKing tags for B&B/First Aid
- * UWorld biochemistry blocks
- * AMBOSS for difficult metabolism and genetics questions

High-yield traps:

- * Rate-limiting enzyme versus deficient enzyme
- * Autosomal dominant versus recessive inheritance
- * Mitochondrial inheritance
- * Vitamin deficiency versus enzyme deficiency
- * Glycogen storage diseases
- * Urea cycle versus organic acidemia
- * Lysosomal storage disorders
- * DNA repair disorder associations

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Immunology

Master:

- * Innate versus adaptive immunity
- * Antigen presentation
- * MHC I versus MHC II

- * T-cell activation
- * B-cell activation and class switching
- * Cytokine functions
- * Complement pathways
- * Hypersensitivity reactions I-IV
- * Immunodeficiency disorders
- * Autoimmune disease mechanisms
- * Transplant rejection
- * Graft-versus-host disease
- * Immunosuppressive drugs
- * Vaccines and immune response

Core disease set:

- * SCID
- * DiGeorge syndrome
- * Bruton's agammaglobulinemia
- * Common variable immunodeficiency
- * Hyper-IgM syndrome
- * Selective IgA deficiency
- * Wiskott-Aldrich syndrome
- * Chronic granulomatous disease
- * Leukocyte adhesion deficiency
- * Chediak-Higashi syndrome
- * Hereditary angioedema
- * Complement deficiencies

- * HIV/AIDS

The USMLE outline explicitly includes immunodeficiency, HIV/AIDS, hypersensitivity, transplantation, immune-drug adverse effects, and vaccine effects.

Resources:

- * B&B or Bootcamp Immunology
- * Pathoma Chapters 1-3
- * Sketchy Pharm immunosuppressants
- * AnKing immunology tags
- * UWorld immune blocks

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General pathology

This is one of the highest-return areas in Step 1.

Master Pathoma Chapters 1-3 until they are automatic:

- * Cell injury
- * Necrosis versus apoptosis
- * Reversible versus irreversible injury
- * Acute inflammation
- * Chronic inflammation

- * Granulomatous inflammation
- * Wound healing
- * Edema
- * Hyperemia and congestion
- * Thrombosis
- * Embolism
- * Infarction
- * Shock
- * Neoplasia
- * Tumor suppressors
- * Oncogenes
- * Paraneoplastic syndromes
- * Metastasis
- * Carcinogens
- * Tumor markers

Pathoma priority order:

1. Chapters 1-3
2. Hematopathology
3. Renal pathology
4. Endocrine pathology
5. Cardiac pathology
6. Respiratory pathology
7. GI pathology
8. Neuro pathology



Pharmacology principles

Master:

- * Receptors and signal transduction
- * Agonists, partial agonists, antagonists
- * Competitive versus noncompetitive inhibition
- * Potency versus efficacy
- * Therapeutic index
- * Zero-order versus first-order elimination
- * Clearance
- * Volume of distribution
- * Loading dose
- * Maintenance dose
- * Steady state
- * Cytochrome P450 inducers and inhibitors
- * Drug metabolism
- * Adverse drug reactions
- * Teratogenic medications

Resource sequence:

1. B&B or Bootcamp pharmacology foundation

2. Sketchy Pharm
3. First Aid drug tables
4. AnKing Sketchy Pharm cards
5. UWorld pharm questions

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4. Organ-System Blueprint

Cardiovascular

Physiology

- * Cardiac action potentials
- * Pressure-volume loops
- * Frank-Starling curves
- * Cardiac output
- * Venous return
- * Vascular resistance
- * Murmurs
- * Coronary circulation
- * Fetal circulation
- * Shock physiology

Pathology

- * Atherosclerosis
- * Hypertension
- * Ischemic heart disease
- * Myocardial infarction complications
- * Cardiomyopathies
- * Valvular disease
- * Endocarditis
- * Pericarditis
- * Aortic dissection
- * Aneurysms
- * Vasculitis
- * Congenital heart disease

Pharmacology

- * Antiarrhythmics
- * Antihypertensives
- * Antianginals
- * Heart failure drugs
- * Diuretics
- * Anticoagulants
- * Antiplatelet agents
- * Lipid-lowering drugs

Must recognize instantly

- * Murmur timing and radiation
- * Pressure-volume loop changes
- * MI wall complications by timeline
- * Aortic dissection associations
- * Heart failure drug mortality benefits
- * Antiarrhythmic class toxicities
- * Vasculitis vessel size associations

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Respiratory

Physiology

- * Lung volumes and capacities
- * Compliance
- * Surfactant
- * V/Q mismatch
- * Alveolar gas equation
- * Diffusion limitation
- * Oxygen dissociation curve
- * Acid-base compensation
- * Pulmonary function tests

Pathology

- * Asthma
- * COPD
- * Bronchiectasis
- * Interstitial lung disease
- * Pneumoconioses
- * ARDS
- * Pulmonary embolism
- * Pulmonary hypertension
- * Lung cancer
- * Pleural disease
- * Neonatal respiratory distress syndrome

Microbiology

- * Pneumonia organisms
- * Tuberculosis
- * Fungal pneumonias
- * Opportunistic infections
- * Viral respiratory infections

Pharmacology

- * Asthma drugs
- * COPD drugs
- * Tuberculosis drugs
- * Pulmonary hypertension medications

Must recognize instantly

- * Obstructive versus restrictive PFT patterns
- * A-a gradient logic
- * Hypoxemia mechanisms
- * Asthma/COPD medication hierarchy
- * TB drug toxicities
- * Small cell versus non-small cell carcinoma
- * Sarcoidosis versus TB versus fungal disease

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Renal

Physiology

- * Nephron transport
- * GFR regulation
- * Renin-angiotensin-aldosterone system
- * Sodium handling
- * Potassium handling
- * Acid-base physiology
- * Urine electrolytes
- * Concentration and dilution

Pathology

- * Nephritic syndromes
- * Nephrotic syndromes
- * Acute kidney injury
- * Chronic kidney disease
- * Tubulointerstitial disease
- * Renal tubular acidosis
- * Polycystic kidney disease
- * Renal cell carcinoma
- * Bladder cancer

Pharmacology

- * Diuretics
- * ACE inhibitors/ARBs
- * Potassium-altering drugs
- * Nephrotoxic drugs
- * Acid-base therapy

Must recognize instantly

- * Nephritic versus nephrotic patterns
- * RTA types I-IV
- * Fractional excretion of sodium
- * Prerenal versus intrinsic versus postrenal AKI

- * Diuretic site of action
- * Diuretic electrolyte effects
- * Acid-base disorders with compensation

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Gastrointestinal

Physiology

- * GI hormones
- * Gastric acid secretion
- * Pancreatic secretion
- * Bile physiology
- * Bilirubin metabolism
- * Nutrient absorption
- * Portal circulation

Pathology

- * Esophageal disorders
- * Peptic ulcer disease
- * Inflammatory bowel disease
- * Malabsorption
- * Celiac disease
- * Pancreatitis

- * Hepatitis
- * Cirrhosis
- * Portal hypertension
- * Gallbladder disease
- * Colorectal cancer
- * GI malignancy syndromes

Pharmacology

- * Acid suppression
- * Antiemetics
- * Laxatives
- * Inflammatory bowel disease medications
- * Hepatitis antivirals

Must recognize instantly

- * Crohn versus ulcerative colitis
- * Direct versus indirect bilirubin patterns
- * Viral hepatitis serologies
- * Pancreatitis complications
- * Cirrhosis complications
- * Gallstone location patterns
- * Colon cancer syndromes

Endocrine

Physiology

- * Hypothalamic-pituitary axes
- * Thyroid physiology
- * Adrenal steroid synthesis
- * Insulin and glucagon physiology
- * Calcium/PTH/vitamin D regulation
- * Reproductive hormone feedback loops

Pathology

- * Diabetes mellitus
- * DKA/HHS
- * Thyroid disease
- * Pituitary adenomas
- * Adrenal disorders
- * MEN syndromes
- * Calcium disorders
- * Osteoporosis
- * Metabolic syndrome

Pharmacology

- * Diabetes drugs
- * Thyroid drugs
- * Steroids
- * Osteoporosis drugs
- * Calcium/PTH agents

Must recognize instantly

- * Primary versus secondary endocrine dysfunction
- * DKA versus HHS
- * Graves versus Hashimoto thyroiditis
- * Cushing syndrome etiologies
- * Primary hyperaldosteronism versus secondary
- * MEN syndromes
- * Diabetes medication mechanisms and toxicities

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Reproductive, embryology, and development

The official outline now includes human development as its own major domain, including childhood, adolescence, adulthood, older adulthood, screening, nutrition, prevention, and psychosocial development.

Embryology

- * Germ layer derivatives

- * Neural crest derivatives
- * Pharyngeal arches
- * Congenital heart defects
- * Neural tube defects
- * Renal development
- * GI rotation
- * Reproductive development

Female reproductive

- * Menstrual cycle
- * Pregnancy physiology
- * Placental disorders
- * Ovarian tumors
- * Uterine pathology
- * Breast pathology
- * Cervical pathology
- * STIs

Male reproductive

- * Testicular tumors
- * Prostate pathology
- * Erectile dysfunction drugs
- * Infertility
- * Androgen disorders

Must recognize instantly

- * Pregnancy hormone changes
- * Placenta previa versus abruption
- * Ovarian tumor markers
- * Testicular tumor markers
- * Turner versus Klinefelter versus androgen insensitivity
- * Neural crest associations
- * Pharyngeal pouch/arch derivatives

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Neurology and special senses

Neuroanatomy

- * Spinal cord tracts
- * Brainstem lesions
- * Cranial nerves
- * Cerebral blood supply
- * Basal ganglia
- * Cerebellum
- * Visual pathway
- * Hearing pathway
- * Autonomic nervous system

Neuropathology

- * Stroke syndromes
- * Demyelinating disease
- * Neurodegenerative disease
- * Seizures
- * Brain tumors
- * CNS infections
- * Peripheral neuropathy
- * Neuromuscular junction disorders
- * Muscle disease

Pharmacology

- * Antiepileptics
- * Parkinson drugs
- * Alzheimer drugs
- * Antipsychotics
- * Antidepressants
- * Anxiolytics
- * Opioids
- * Local anesthetics
- * Autonomic drugs

Must recognize instantly

- * Stroke localization
- * Brainstem “crossed findings”
- * Spinal cord lesion patterns
- * Parkinson versus Huntington versus Wilson
- * Myasthenia gravis versus Lambert-Eaton
- * Seizure drug toxicities
- * Antipsychotic receptor effects
- * Visual field deficits

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Musculoskeletal, rheumatology, dermatology

Musculoskeletal

- * Bone remodeling
- * Bone tumors
- * Fracture healing
- * Muscular dystrophies
- * Myopathies
- * Neuromuscular junction disease

Rheumatology

- * Rheumatoid arthritis

- * SLE
- * Sjögren syndrome
- * Systemic sclerosis
- * Polymyositis/dermatomyositis
- * Vasculitides
- * Seronegative spondyloarthropathies
- * Crystal arthropathies

Dermatology

- * Primary lesions
- * Blistering disorders
- * Psoriasis
- * Acne
- * Skin cancers
- * Drug eruptions
- * Neurocutaneous syndromes

Must recognize instantly

- * Antibody and complement patterns
- * Vasculitis association map
- * Bullous disease level of skin separation
- * Bone tumor age/location patterns
- * Muscle disease enzyme patterns
- * Connective tissue disease antibody associations



Hematology and oncology

Hematology

- * Anemia workups
- * Hemolysis
- * RBC morphology
- * Coagulation cascade
- * Platelet disorders
- * Transfusion reactions

Oncology

- * Leukemias
- * Lymphomas
- * Myeloproliferative disorders
- * Plasma-cell disorders
- * Solid tumor genetics
- * Tumor suppressor genes
- * Oncogenes
- * Chemotherapy mechanisms

The official outline specifically includes leukemias, lymphomas, myeloma, anemia categories, thrombocytopenia, hypercoagulable states, coagulation disorders, transfusion reactions, and hematologic drug adverse effects.

Must recognize instantly

- * Microcytic anemia algorithm
- * Hemolytic anemia markers
- * Acute leukemia morphology
- * Lymphoma translocations
- * Coagulation study patterns
- * HIT versus TTP versus HUS versus DIC
- * Chemotherapy toxicities
- * Paraneoplastic syndromes

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5. Microbiology Blueprint

Bacteria

Know every organism through this template:

- * Gram-positive or gram-negative
- * Shape
- * Oxygen requirements
- * Virulence factor

- * Toxin
- * Transmission
- * Disease pattern
- * Diagnostic test
- * First-line treatment
- * Drug resistance
- * Classic complication

Mandatory groups:

- * Staphylococci
- * Streptococci
- * Enterococci
- * Bacillus
- * Clostridia
- * Corynebacterium
- * Listeria
- * Neisseria
- * Haemophilus
- * Bordetella
- * Legionella
- * Pseudomonas
- * Enteric gram-negatives
- * Vibrio
- * Campylobacter
- * Helicobacter pylori

- * Chlamydia
- * Rickettsia
- * Mycoplasma
- * Spirochetes
- * Mycobacteria

Viruses

Know:

- * Genome
- * Envelope status
- * Replication site
- * Transmission
- * Latency location
- * Cancer association
- * Congenital effects
- * Diagnostic test
- * Antiviral treatment

Fungi

Know:

- * Yeast versus mold
- * Dimorphic fungi

- * Geography
- * Immune status association
- * Morphology
- * Drug treatment

Parasites

Know:

- * Lifecycle
- * Geographic association
- * Vector
- * Organ involvement
- * Diagnostic morphology
- * Treatment

Sketchy Micro is a reasonable primary mnemonic system for this domain because it is designed specifically around organism characteristics and clinical associations.

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6. Behavioral Science, Ethics, Biostatistics, and Public Health

These are not “easy points.” They are where people lose points because they procrastinate them until dedicated, then become shocked that arithmetic still exists.

The official outline includes biostatistics, epidemiology/population health, medical literature interpretation, and social sciences.

Biostatistics

Master:

- * Sensitivity
- * Specificity
- * Positive predictive value
- * Negative predictive value
- * Prevalence
- * Relative risk
- * Odds ratio
- * Absolute risk reduction
- * Relative risk reduction
- * Number needed to treat
- * Number needed to harm
- * Type I and Type II error
- * Power
- * Confidence intervals
- * P values
- * Bias types
- * Study designs
- * Meta-analysis
- * Intention-to-treat analysis

- * Kaplan-Meier curves

Ethics and communication

Master:

- * Capacity versus competence

- * Informed consent

- * Surrogate decision-making

- * Confidentiality

- * Mandatory reporting

- * End-of-life care

- * Medical errors

- * Impaired colleagues

- * Professional boundaries

- * Decision-making in minors

- * Research ethics

- * Cultural humility

- * Health disparities

Nutrition

USMLE increased emphasis on nutrition science beginning in June 2026, including nutrition's role in health and chronic disease prevention.

Master:

- * Macronutrient metabolism
- * Enteral versus parenteral nutrition basics
- * Malnutrition
- * Refeeding syndrome
- * Vitamin and mineral deficiency
- * Obesity physiology
- * Diabetes nutrition
- * Cardiovascular nutrition
- * Pregnancy nutrition
- * Alcohol-related deficiency states
- * Bariatric surgery nutritional complications

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7. Resource-to-Topic Mapping

Topic Learn Memorize Apply

General pathology Pathoma AnKing UWorld

Immunology B&B/Bootcamp AnKing UWorld

Microbiology Sketchy Micro Sketchy-tagged AnKing UWorld

Pharmacology Sketchy Pharm Sketchy-tagged AnKing UWorld

Cardio B&B/Bootcamp + Pathoma First Aid + AnKing UWorld

Renal B&B/Bootcamp First Aid + AnKing UWorld

Endocrine B&B/Bootcamp + Pathoma First Aid + AnKing UWorld

Hematology Pathoma First Aid + AnKing UWorld

Neuro B&B/Bootcamp First Aid + AnKing UWorld

Biostatistics UWorld/AMBOSS review Personal formula cards Timed blocks

Ethics UWorld/AMBOSS Error cards Mixed blocks

Anatomy B&B/Bootcamp First Aid + AnKing UWorld

AMBOSS is best used as a targeted secondary question bank, especially for weak areas, difficult tables, ethics, and image-heavy questions. Its Step 1 platform includes more than 2,700 questions, a reference library, study plans, and Anki integration.

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8. The Daily Study Engine

During systems-based coursework

Daily minimum structure

1. Anki reviews

- * Complete all due reviews.
- * Do not let review debt snowball.
- * Review cards before adding more cards.

2. New learning

- * Watch the assigned B&B, Bootcamp, Pathoma, or Sketchy content.
- * Read the linked First Aid pages.
- * Unsuspend only the cards related to what you learned.

3. Questions

- * Early phase: 10-20 tutor-mode questions per topic.

- * Later phase: 20-40 timed questions per day.
- * Review every incorrect question and every lucky correct answer.

4. Error correction

- * Identify the failure type.
- * Make a concise Anki card only when the error represents a recurring gap.

Error categories

Error Type Meaning Fix

Knowledge gap You never learned it Targeted content review

Recall failure You learned it but forgot Anki card/tagged review

Mechanism gap You memorized facts without understanding Rebuild physiology

Reading error You missed a qualifier Slow down, identify task

Differential failure You knew diseases but not distinctions Comparison table

Test-taking failure You changed correct answer or overthought Review reasoning pattern

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9. The Question-Bank Protocol

Early phase

- * Use tutor mode.
- * Work system-specific.
- * Read explanations deeply.
- * Do not obsess over percentage correct.

- * Build reasoning.

Middle phase

- * Mix systems gradually.
- * Move into timed blocks.
- * Start identifying repeated weak domains.
- * Add targeted AMBOSS blocks.

Dedicated phase

- * Predominantly timed, mixed blocks.
- * Simulate Step 1 pace.
- * Review blocks the same day.
- * Keep Anki reviews alive.
- * Do not spend six hours making beautiful notes from one missed question. You are studying medicine, not producing illuminated manuscripts.

UWorld's Step 1 QBank supports custom tests, performance analytics, notebooks, flashcards, and a study planner.



10. The NBME Assessment System

NBME CBSSAs should serve as your main readiness checkpoints because they are designed for Step 1 readiness and provide estimated probability of passing, equated percent correct, content-area performance, explanations, and longitudinal trend data.

Suggested assessment sequence

Before dedicated

- * Take one baseline NBME.
- * Do not panic over the score.
- * Use it to identify your bottom three systems.

Mid-dedicated

- * Take an NBME every 1-2 weeks.
- * Repair weaknesses between exams.
- * Track recurring deficits.

Final 2-3 weeks

- * Take 2-3 additional NBMEs.
- * Complete the official Free 120.
- * Use UWorld self-assessments as additional data, not sole truth.

NBME's standard-paced CBSSA uses four sections of 50 questions with 75 minutes per section; it also has a self-paced option for detailed review.



11. The Dedicated Period Blueprint

Eight-week dedicated model

Weeks 1-2: Repair foundations

- * General pathology
- * Immunology
- * Biochemistry/genetics
- * Pharmacology principles
- * Biostatistics
- * Weakest organ system
- * 40-60 questions/day

Weeks 3-4: High-weight organ systems

- * Cardiology
- * Renal
- * Respiratory
- * GI
- * Endocrine
- * Hematology/oncology
- * 60-80 questions/day

Weeks 5-6: Consolidation and mixed blocks

- * Neurology
- * Reproductive
- * Musculoskeletal/rheumatology
- * Dermatology
- * Behavioral science
- * Microbiology/pharmacology review
- * 80 questions/day, mostly mixed

Weeks 7-8: Exam-mode preparation

- * Mixed timed blocks
- * NBME review
- * Free 120
- * Error log repair
- * High-yield review:
 - * Pathoma Chapters 1-3
 - * Sketchy weak organisms/drugs
 - * First Aid rapid pass
 - * Biostatistics formulas
 - * Ethics algorithms
 - * Neuroanatomy
 - * Renal physiology
 - * Endocrine axes



12. The “Definition of Done” for Every Topic

A topic is not complete because you watched a video.

It is complete only when you can:

- * Explain the normal physiology without notes.
- * Explain the pathologic disruption.
- * Differentiate it from its closest mimics.
- * Predict the laboratory findings.
- * Predict treatment mechanism.
- * Predict medication adverse effects.
- * Answer 70%+ of targeted question-bank questions.
- * Recall it in Anki several times over spaced intervals.
- * Recognize it in a mixed clinical vignette.

For a disease like nephrotic syndrome, “done” means you can distinguish:

- * Minimal change disease
- * FSGS
- * Membranous nephropathy
- * Membranoproliferative glomerulonephritis
- * Diabetic nephropathy
- * Amyloidosis

You should know the trigger, pathology, light microscopy, electron microscopy, immunofluorescence, laboratory profile, complications, and common associations.

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13. The First Aid Annotation Rule

Use First Aid as your final compressed knowledge base.

Annotate only:

- * UWorld explanations that reveal a recurrent gap
- * NBME concepts that appear repeatedly
- * Clarifying mechanism sentences
- * One-line distinctions between confusing diseases
- * Drug adverse effects you repeatedly miss
- * Your school-specific high-yield material if it is board relevant

Do not annotate every explanation. That turns First Aid into a landfill with tabs.

Your First Aid pages should become:

- * Clean enough to review rapidly
- * Dense enough to trigger recall
- * Personalized enough to represent your weak points



14. Anki Operating Rules

Use AnKing selectively

Prioritize cards tagged to:

- * First Aid
- * Pathoma
- * Sketchy Micro
- * Sketchy Pharm
- * Boards & Beyond or Bootcamp
- * UWorld missed-question tags
- * NBME weak systems

The AnKing Step Deck is maintained through AnkiHub and includes resource tags and UWorld question-ID linkage tools.

Rules

- * Never skip reviews for multiple days.
- * Unsuspend cards after learning content.
- * Do not add 800 new cards because motivation briefly possessed you at 1:00 AM.
- * Use filtered decks for weak systems.

- * Convert only repeated UWorld/NBME mistakes into custom cards.
- * Suspend cards that are low-yield, redundant, or no longer serving you.

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15. Exam-Day Preparation

The official USMLE materials include sample questions, an interactive testing experience, and a Prometric practice-session option. Use them. Exam software should not be a surprise on the day when your cortisol is already trying to escape through your ears.

Final-week priorities

- * No massive new resources.
- * No full reread of every textbook.
- * Review weak systems.
- * Do mixed timed blocks.
- * Practice break strategy.
- * Practice nutrition, hydration, caffeine timing, and bathroom timing.
- * Maintain sleep.
- * Review rapid-recognition material:
 - * Murmurs
 - * Acid-base
 - * Renal tubular acidosis
 - * Biostatistics
 - * Neuroanatomy
 - * Immunodeficiencies

- * Hematologic malignancies
- * Antibiotics
- * Autonomic drugs
- * Endocrine feedback loops
- * Congenital syndromes

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16. Your Minimal “Do Not Negotiate” Stack

For your actual Step 1 preparation, I would keep the active stack to this:

- * First Aid 2026
- * UWorld
- * AnKing
- * Pathoma
- * Sketchy Micro
- * Sketchy Pharm
- * Boards & Beyond or Bootcamp
- * NBME CBSSAs
- * AMBOSS only for targeted repair

That is enough. More resources do not create more mastery. They create more guilt, more tabs, and occasionally a beautiful color-coded spreadsheet that has never answered a question correctly.

The winning loop is:

Learn mechanism → confirm in First Aid → unlock Anki → do questions → classify error → repair → revisit under spacing → test under time.

That is the Step 1 blueprint.

MCAT:

The MCAT needs a different architecture than Step 1. It is not a miniature medical-school curriculum with a few passages stapled on afterward, because apparently standardized testing needed one more trick. It is an integration exam: AAMC content + experimental reasoning + data interpretation + timing + CARS judgment under pressure.

The AAMC's official outline should be the non-negotiable spine because it defines the tested foundational concepts, content categories, and scientific reasoning skills. Jack Westin's outline can serve as the navigable topic-and-practice layer mapped onto that spine. AAMC official materials should be reserved as the final authority for test logic and calibration.

Use the following as your master prompt for Claude, Codex, or another research-capable system.

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MCAT GOD BLUEPRINT: FULL-SCALE CONTENT, SKILL, RESOURCE, AND EXECUTION SYSTEM

ROLE

You are building the most comprehensive, technically rigorous, evidence-backed MCAT preparation blueprint possible for a student targeting elite performance.

This is not a generic MCAT study schedule.

This is a full operational blueprint that converts the entire MCAT into:

1. A master content map.
2. A scientific-reasoning map.
3. A CARS reasoning system.
4. A resource-routing system.
5. An Anki and spaced-repetition system.
6. A question-bank and passage-training system.
7. A full-length exam calibration system.
8. An error-analysis and weakness-repair system.
9. A timeline engine for multiple study durations.
10. A database-ready knowledge architecture that can later be integrated into an app such as Noctyrium.

The final output must be detailed enough that a serious student could execute it for 3, 4, 6, 9, or 12 months without needing to invent a study system halfway through preparation.

Do not produce vague motivational advice, generic “study 6 hours per day” statements, or lecture-style summaries.

Build a surgical training system.

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PRIMARY OBJECTIVE

Construct a full MCAT “God Blueprint” based on:

1. The current official AAMC MCAT Content Outline.
2. Current official AAMC section descriptions, skill distributions, and practice materials.
3. Jack Westin’s MCAT content outline and topic pages.
4. Khan Academy MCAT resources where appropriate.
5. High-quality commercial resources when useful, including:
 - * UWorld
 - * Kaplan
 - * Blueprint
 - * Altius
 - * Jack Westin
 - * Examrackers
 - * Princeton Review
 - * Anki / AnKing-related MCAT decks
6. Official AAMC practice materials as the final calibration standard.
7. Evidence-based learning science:
 - * active recall
 - * spaced repetition
 - * interleaving

- * retrieval practice
- * deliberate practice
- * error categorization
- * cumulative review
- * metacognitive calibration

The AAMC framework must always override third-party resource organization.

Third-party resources are tools. They are not the curriculum.

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SOURCE HIERARCHY

Use this hierarchy when building recommendations:

Tier 1: Official AAMC Authority

Use official AAMC sources for:

- * Current test structure.
- * Section timing.
- * Number of questions.
- * Tested foundational concepts.
- * Content categories.
- * Scientific Inquiry and Reasoning Skills.

- * CARS skill categories.
- * Official practice materials.
- * Official full-length exams.
- * Official Question Packs.
- * Official Section Banks.
- * Official CARS Diagnostic Tool.
- * Official Independent Question Bank.
- * Official free questions.
- * Current AAMC practice-exam inventory.
- * Current official changes to MCAT preparation products.

The AAMC explicitly organizes science testing around foundational concepts, content categories, and four scientific inquiry/reasoning skills: knowledge of scientific concepts, scientific reasoning/problem-solving, research design/execution, and data-based/statistical reasoning.

The blueprint must treat these skills as separate trainable competencies, not vague “critical thinking.”

Tier 2: Jack Westin Content Navigation Layer

Use Jack Westin’s MCAT outline as:

- * A searchable topic navigator.
- * A detailed outline expansion tool.
- * A bridge from AAMC category to subtopic.
- * A source of topic-level practice and explanations.

* A CARS practice source during early and intermediate preparation.

Jack Westin should not replace official AAMC materials for final reasoning calibration.

Tier 3: Learning and Remediation Resources

Use Khan Academy, textbooks, videos, and concise review books for remediation when a student has a true content deficit.

Assign each resource a specific role.

Example:

Resource Type	Correct Use
Khan Academy	Foundational explanation, weak-topic repair, Psych/Soc clarification
Kaplan books	Structured broad content review
UWorld	Passage reasoning, data interpretation, difficult applied science
Jack Westin Daily CARS	content outline navigation, early passage exposure
AAMC materials	Final logic calibration and official-style mastery
Anki	Long-term retention, formulas, definitions, mechanisms, missed-question repair
Full-length exams	Endurance, timing, score calibration, systems stress testing

Do not recommend a resource merely because it is popular.

State what exact problem it solves.

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REQUIRED OUTPUT FORMAT

Build the final blueprint in the following order.

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PART I: MCAT EXAM ARCHITECTURE

Provide a precise overview of the current MCAT structure.

For each section include:

- * Official section name.
- * Number of questions.
- * Number of passages.
- * Number of discrete questions.
- * Timing.
- * Break structure.
- * Content composition.
- * Foundational concepts tested.
- * Scientific reasoning skills tested.
- * Typical cognitive demands.
- * Most common traps.
- * What distinguishes a 500-level performance from a 510-level performance.

- * What distinguishes a 510-level performance from a 520+ performance.
- * What cannot be fixed through content review alone.

Include:

1. Chemical and Physical Foundations of Biological Systems.
2. Critical Analysis and Reasoning Skills.
3. Biological and Biochemical Foundations of Living Systems.
4. Psychological, Social, and Biological Foundations of Behavior.

CARS must be handled separately because it tests passage-based comprehension, reasoning within the text, and reasoning beyond the text rather than outside factual knowledge. AAMC describes CARS passages as roughly 500–600 words and states that required information is contained within the passage and questions.

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PART II: MASTER AAMC CONTENT MAP

Construct a full content map organized first by AAMC foundational concept, then by content category, then by subtopic.

Do not organize the blueprint only by college course name.

Use this hierarchy:

MCAT Section

Foundational Concept

AAMC Content Category

Topic

Subtopic

Required knowledge

Required equations

Required mechanisms

Required diagrams

Required experiments

Common passage contexts

High-frequency traps

Required practice type

Best resources

Anki requirement



PART III: FOUNDATIONAL CONCEPT BLUEPRINT

Build a separate deep-dive section for every AAMC foundational concept.

Foundational Concept 1

Biomolecules have unique properties that determine how they contribute to the structure and function of cells and how they participate in the processes necessary to maintain life.

Include:

- * Amino acids.
- * Protein structure.
- * Enzymes.
- * Enzyme kinetics.
- * Inhibition.
- * Carbohydrates.
- * Lipids.
- * Nucleotides.
- * DNA/RNA structure.
- * DNA replication.
- * Transcription.
- * Translation.
- * Gene regulation.
- * Recombinant DNA techniques.
- * Lab methods.
- * Molecular biology experiments.
- * Gel electrophoresis.
- * PCR.
- * Western blot.
- * Southern blot.
- * Northern blot.
- * CRISPR basics if currently relevant to AAMC-level conceptual testing.
- * Operons.
- * Epigenetics.

- * Viruses.
- * Cellular energetics.
- * Glycolysis.
- * Gluconeogenesis.
- * Glycogen metabolism.
- * Pentose phosphate pathway.
- * Citric acid cycle.
- * Electron transport chain.
- * Oxidative phosphorylation.
- * Fatty-acid metabolism.
- * Ketone bodies.
- * Amino-acid metabolism.
- * Hormonal metabolic regulation.

For every biochemical pathway include:

- * Location.
- * Purpose.
- * Inputs.
- * Outputs.
- * Rate-limiting enzyme.
- * Regulation.
- * Fed-state behavior.
- * Fasting-state behavior.
- * Insulin effects.
- * Glucagon effects.

- * Epinephrine effects.
- * Common disease or experimental contexts.
- * High-yield graph patterns.
- * How the pathway appears in MCAT passages.

Do not turn this into a medical-school biochemistry lecture.

Focus on what the MCAT can reasonably test through passages, figures, graphs, experiments, and discrete questions.

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Foundational Concept 2

Highly organized assemblies of molecules, cells, and organs interact to carry out the functions of living organisms.

Include:

- * Cell biology.
- * Organelles.
- * Membrane transport.
- * Signal transduction.
- * Cell cycle.
- * Mitosis.
- * Meiosis.
- * Apoptosis.

- * Stem cells.
- * Cellular specialization.
- * Tissue types.
- * Nervous system.
- * Endocrine system.
- * Cardiovascular system.
- * Respiratory system.
- * Renal system.
- * Digestive system.
- * Immune system.
- * Musculoskeletal system.
- * Reproductive system.
- * Development.
- * Homeostasis.
- * Feedback loops.
- * Thermoregulation.
- * Fluid balance.
- * Acid-base balance.
- * Hormonal physiology.

For each organ system include:

- * Core anatomy.
- * Major physiologic purpose.
- * Inputs and outputs.
- * Major hormones and signaling molecules.

- * Feedback systems.
- * High-yield equations.
- * Common pathology-adjacent contexts.
- * Experimental manipulations.
- * Graph interpretation patterns.
- * Common MCAT distractors.

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Foundational Concept 3

Complex systems of tissues and organs sense the internal and external environments and respond to them.

Include:

- * Sensory systems.
- * Motor control.
- * Nervous system integration.
- * Endocrine signaling.
- * Immune response.
- * Reproduction.
- * Development.
- * Evolution.
- * Ecology.
- * Population genetics.

- * Natural selection.
- * Hardy-Weinberg equilibrium.
- * Adaptation.
- * Species interactions.
- * Experimental evolution concepts.

For population genetics include:

- * Hardy-Weinberg assumptions.
- * Allele frequency.
- * Genotype frequency.
- * Founder effect.
- * Bottleneck effect.
- * Genetic drift.
- * Gene flow.
- * Sexual selection.
- * Kin selection.
- * Inclusive fitness.
- * Evolutionary tradeoffs.

Include equations, assumptions, and question patterns.

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Foundational Concept 4

Chemical and physical principles underlie the mechanisms operating in the human body.

Include:

- * Atomic structure.
- * Periodic trends.
- * Bonding.
- * Intermolecular forces.
- * Stoichiometry.
- * Gases.
- * Solutions.
- * Acids and bases.
- * Buffers.
- * Equilibrium.
- * Thermodynamics.
- * Kinetics.
- * Electrochemistry.
- * Redox.
- * Fluids.
- * Hydrostatics.
- * Hydrodynamics.
- * Circuits.
- * Electrostatics.
- * Magnetism.
- * Waves.
- * Sound.

- * Optics.
- * Light.
- * Nuclear chemistry.
- * Radioactivity.
- * Organic chemistry.
- * Functional groups.
- * Stereochemistry.
- * Isomers.
- * Nucleophiles.
- * Electrophiles.
- * SN1.
- * SN2.
- * E1.
- * E2.
- * Carbonyl chemistry.
- * Carboxylic acids and derivatives.
- * Alcohols.
- * Aldehydes.
- * Ketones.
- * Amines.
- * Spectroscopy.
- * Laboratory separation methods.

For every physics topic include:

- * Core conceptual model.

- * Primary equation.
- * Derived equations.
- * Units.
- * Dimensional analysis shortcuts.
- * Estimation methods.
- * Common graph forms.
- * Biological application.
- * Experimental setup examples.
- * Common traps.

For every chemistry topic include:

- * Conceptual model.
- * Governing equations.
- * Reaction patterns.
- * Laboratory contexts.
- * Biological relevance.
- * Common distractors.
- * Required mental math.

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Foundational Concept 5

The principles that govern chemical interactions and transformations form the basis for a broader understanding of the molecular dynamics of living systems.

Include:

- * Thermodynamics.
- * Kinetics.
- * Equilibrium.
- * Acid-base chemistry.
- * Buffers.
- * Enzymes.
- * Bioenergetics.
- * Oxidation-reduction.
- * Electrochemistry.
- * Chemical separation.
- * Spectroscopy.
- * Analytical chemistry.
- * Laboratory techniques.
- * Protein purification.
- * Chromatography.
- * Centrifugation.
- * Dialysis.
- * Electrophoresis.
- * Immunoassays.
- * Spectrophotometry.

For laboratory methods, create a table:

Method Principle Separates/Measures Output Interpretation Common Passage Use
Common Trap

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Foundational Concepts 6–10: Psychology, Sociology, Behavior, and Society

Build an extremely detailed Psych/Soc system.

Organize it into:

- * Sensation and perception.

- * Consciousness.

- * Learning.

- * Memory.

- * Cognition.

- * Language.

- * Motivation.

- * Emotion.

- * Personality.

- * Psychological disorders.

- * Stress.

- * Social processes.

- * Social cognition.

- * Social behavior.

- * Attitudes.

- * Prejudice.
- * Discrimination.
- * Group behavior.
- * Culture.
- * Socialization.
- * Demographics.
- * Social structure.
- * Stratification.
- * Poverty.
- * Health disparities.
- * Healthcare systems.
- * Social determinants of health.
- * Population health.
- * Epidemiology.
- * Research methods.
- * Ethics.
- * Statistics.

For every Psych/Soc term include:

- * Exact definition.
- * Closely confused terms.
- * Opposite or competing concepts.
- * Classic experimental example.
- * MCAT-style vignette.
- * Common wording traps.

- * Associated Anki card type.
- * Whether the concept is best learned through:
 - * memorization,
 - * comparison table,
 - * scenario practice,
 - * flashcard,
 - * passage application.

Create “confusion clusters” such as:

Proactive interference vs retroactive interference

Negative reinforcement vs punishment

Fundamental attribution error vs actor-observer bias

Social facilitation vs social loafing

Assimilation vs accommodation

Prevalence vs incidence

Reliability vs validity

Sensitivity vs specificity

Relative risk vs odds ratio

Confounding vs effect modification

For each cluster, build a distinction table.

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PART IV: SCIENTIFIC INQUIRY AND REASONING SKILLS SYSTEM

Build a dedicated training framework for the four AAMC scientific reasoning skills:

1. Knowledge of Scientific Concepts and Principles.
2. Scientific Reasoning and Problem-Solving.
3. Reasoning About the Design and Execution of Research.
4. Data-Based and Statistical Reasoning.

For each skill provide:

- * Definition.
- * What it looks like in a passage.
- * Typical question stems.
- * How students fail.
- * How high scorers think.
- * Drills.
- * Timed drills.
- * Untimed drills.
- * Error categories.
- * Review workflow.
- * Progress metrics.

Create a “Passage Decoding Protocol.”

Example format:

Step 1: Identify the research question.

Step 2: Identify the hypothesis.

Step 3: Mark independent variables.

Step 4: Mark dependent variables.

Step 5: Identify controls.

Step 6: Identify confounders.

Step 7: Translate each figure.

Step 8: Predict what the experiment should show.

Step 9: Answer only from evidence.

Step 10: Eliminate answers that exceed the data.

Include separate decoding protocols for:

- * Biochemistry passages.

- * Biology passages.

- * Physiology passages.

- * Chemistry passages.

- * Physics passages.

- * Psychology experiments.

- * Sociology studies.

- * Epidemiology studies.

- * Genetics experiments.

- * Molecular-biology experiments.

PART V: DATA INTERPRETATION AND RESEARCH METHODS BOOTCAMP

Build a full MCAT data-literacy curriculum.

Include:

- * Independent variable.
- * Dependent variable.
- * Confounder.
- * Mediator.
- * Moderator.
- * Control group.
- * Positive control.
- * Negative control.
- * Blinding.
- * Double blinding.
- * Randomization.
- * Selection bias.
- * Recall bias.
- * Observer bias.
- * Attrition bias.
- * Publication bias.
- * Sampling bias.
- * Cross-sectional study.
- * Case-control study.
- * Cohort study.

- * Randomized controlled trial.
- * Meta-analysis.
- * Systematic review.
- * Correlation.
- * Causation.
- * Internal validity.
- * External validity.
- * Construct validity.
- * Reliability.
- * Statistical significance.
- * Clinical significance.
- * p-values.
- * Confidence intervals.
- * Type I error.
- * Type II error.
- * Power.
- * Sensitivity.
- * Specificity.
- * Positive predictive value.
- * Negative predictive value.
- * Prevalence.
- * Incidence.
- * Relative risk.
- * Odds ratio.
- * Absolute risk reduction.
- * Number needed to treat.

Include:

1. Definitions.
2. Equations.
3. Figure interpretation.
4. Common MCAT question formats.
5. Mini-vignettes.
6. Error patterns.
7. Flashcard templates.
8. Passage drills.

The AAMC specifically expects test takers to interpret patterns in tables, figures, and graphs and reason from data to conclusions.



PART VI: MATH, PHYSICS, AND QUANTITATIVE REASONING SYSTEM

Create an MCAT-specific quantitative toolkit.

Include:

- * Scientific notation.
- * Exponent rules.
- * Logarithms.

- * Half-life.
- * Ratios.
- * Proportions.
- * Percent change.
- * Unit conversion.
- * Dimensional analysis.
- * Estimation.
- * Mental arithmetic.
- * Significant figures.
- * Graph slopes.
- * Linear graphs.
- * Semilog graphs.
- * Log-log graphs.
- * Area under curves.
- * Probability.
- * Basic statistics.
- * Mean.
- * Median.
- * Mode.
- * Standard deviation.
- * Normal distributions.
- * Z-scores.
- * Correlation coefficient.
- * Scatterplots.
- * Interpreting error bars.

For every equation include:

Equation Variables Units Conceptual Meaning Common Rearrangements Biological
Application Trap

Build a “must memorize” formula list and a “derive when needed” formula list.

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PART VII: CARS GOD SYSTEM

Build a complete CARS training system.

Use AAMC’s three CARS skill domains:

1. Foundations of Comprehension.
2. Reasoning Within the Text.
3. Reasoning Beyond the Text.

AAMC describes the CARS section as 53 passage-based questions in 90 minutes, with approximately half humanities and half social-science passages.

For CARS, include:

Passage Method

Create a repeatable method for:

- * Identifying thesis.
- * Identifying author attitude.
- * Identifying paragraph function.
- * Identifying competing viewpoints.
- * Identifying concessions.
- * Identifying examples versus claims.
- * Identifying scope.
- * Identifying assumptions.
- * Predicting the author's next move.
- * Mapping passage structure without over-highlighting.

Question-Type System

Create decision trees for:

- * Main idea.
- * Author's tone.
- * Function questions.
- * Inference.
- * Strengthen.
- * Weaken.
- * New information.
- * Application.
- * Analogy.
- * "According to the passage."

- * “The author would most likely agree.”
- * “Which statement best supports.”
- * “Which claim undermines.”

Wrong-Answer Taxonomy

Classify CARS wrong answers into:

- * Too extreme.
- * True but irrelevant.
- * Opposite of author’s view.
- * Outside the passage.
- * Half-right.
- * Wrong scope.
- * Wrong relationship.
- * Unsupported inference.
- * Distorted wording.
- * Trap answer using familiar language.

CARS Review Protocol

For every missed question require:

1. What was the author’s thesis?
2. What was the relevant paragraph’s purpose?
3. What exact wording in the passage supported the correct answer?

4. Why did my selected answer feel attractive?
5. What trap category did I fall for?
6. What rule will prevent this mistake next time?

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PART VIII: RESOURCE ROUTING SYSTEM

Create a resource decision matrix.

For every content category, specify:

- * Primary content source.
- * Secondary explanation source.
- * Primary practice source.
- * Advanced practice source.
- * Official AAMC reinforcement source.
- * Anki source.
- * Flashcard style.
- * When to stop using the resource.

Example format:

Topic	Learn	Reinforce	Practice	Official Calibration	Anki	Stop Rule
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Include resources such as:

- * AAMC Content Outline Course.
- * AAMC Full-Length Exams.
- * AAMC Section Banks.
- * AAMC Question Packs.
- * AAMC CARS Diagnostic Tool.
- * Jack Westin Outline.
- * Jack Westin CARS.
- * Khan Academy.
- * UWorld.
- * Kaplan.
- * Blueprint.
- * Altius.
- * Examkrackers.
- * Anki.
- * Self-made flashcards.

The current AAMC practice ecosystem includes Question Packs, Section Banks, a CARS Diagnostic Tool, an Independent Question Bank, and official full-length practice exams. The official online bundle currently contains 2,710 unique questions, and the AAMC currently lists seven official practice exams, including two free exams.

Do not tell the student to buy everything.

Build tiers:

Minimal Budget Plan

Moderate Budget Plan

High-Resource Plan

Maximum-Score Plan

For each tier state:

- * What to buy.

- * What to avoid.

- * What can be replaced with free resources.

- * What cannot be replaced.

- * What should be saved for the final phase.

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PART IX: ANKI AND MEMORY ARCHITECTURE

Build a complete MCAT Anki strategy.

Include:

Core Principles

- * Anki is for durable recall.

- * Passages are for application.

- * Full-length exams are for integration and calibration.

- * Do not confuse card volume with mastery.

* Do not create cards for every sentence in a textbook.

* Build cards from weak points, confusion clusters, equations, mechanisms, and missed questions.

Deck Structure

MCAT

Biology

Biochemistry

General Chemistry

Organic Chemistry

Physics

Psychology

Sociology

CARS Vocabulary and Reasoning

Equations

Amino Acids

Metabolism

Lab Methods

Statistics

Error Log Cards

Full-Length Misses

Card Types

Include templates for:

- * Basic fact recall.
- * Cloze deletion.
- * Compare-and-contrast.
- * Diagram labeling.
- * Mechanism sequencing.
- * Equation manipulation.
- * “Why is this answer wrong?” cards.
- * Passage-derived cards.
- * Common-trap cards.
- * Psych/Soc scenario cards.
- * Lab-method cards.
- * Amino-acid cards.
- * Metabolic pathway cards.

Daily Anki Rules

Create separate rules for:

- * Content-review phase.
- * Practice-heavy phase.
- * Full-length phase.
- * Final two weeks.

Include:

- * New-card caps.
- * Review caps.
- * Missed-card triage.
- * Leech handling.
- * When to suspend cards.
- * When to convert a card into a passage drill instead.

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PART X: QUESTION-BANK SYSTEM

Build a question-bank progression model.

Divide preparation into:

1. Untimed learning questions.
2. Semi-timed topical passage sets.
3. Timed mixed blocks.
4. Section-level timed blocks.
5. Full-length simulation.
6. Official AAMC calibration.

For every question set, require an error log with:

Question ID	Section	Topic	Skill	Error Type	Why Wrong	Correct Logic	Fix Anki Card?	Re-test Date
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Use the following error categories:

Content gap

Equation gap

Definition gap

Passage misread

Figure misread

Graph misread

Experimental design error

Statistical reasoning error

Timing error

Careless arithmetic

CARS scope error

CARS inference error

CARS tone error

Changed answer incorrectly

Panic / stamina failure

Create a corrective action for every error type.

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PART XI: FULL-LENGTH EXAM SYSTEM

Build a full-length exam protocol.

For every full length include:

Before the Exam

- * Sleep standard.
- * Wake-up timing.
- * Food plan.
- * Caffeine plan.
- * Break plan.
- * Workspace requirements.
- * Phone restrictions.
- * Scratch-paper system.
- * Timing checkpoints.

During the Exam

- * Section pacing.
- * Passage triage.
- * Flagging rules.
- * Guessing rules.
- * When not to reread.
- * When to move on.
- * How to recover after a difficult passage.
- * How to prevent one bad question from infecting the next six, because the exam is already unpleasant enough without emotional contagion.

After the Exam

Require three review passes:

Pass 1: Score-Level Review

- * Section score.
- * Timing.
- * Accuracy.
- * Confidence.
- * Fatigue.
- * Emotional state.

Pass 2: Question-Level Review

- * Review every missed question.
- * Review every guessed question.
- * Review every correct question answered with low confidence.
- * Review every question that took too long.

Pass 3: Systems Review

- * Identify recurring weakness clusters.
- * Identify whether the issue is content, reasoning, timing, stamina, or confidence.
- * Build a repair plan before the next full length.

Create a full-length dashboard:

Exam	Date	C/P CARS	B/B P/S Total	Timing Issue	Main Weakness	Repair Plan
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PART XII: STUDY PHASES

Build a complete phase system.

Phase 0: Diagnostic and Baseline

Include:

- * Diagnostic exam.
- * Course-prerequisite audit.
- * Content-gap audit.
- * CARS baseline.
- * Math baseline.
- * Reading-speed baseline.
- * Schedule audit.
- * Sleep audit.
- * Resource inventory.
- * Target score analysis.

Output:

Strengths

Weaknesses

Unknowns

High-risk sections

High-yield repair areas

Timeline recommendation

Resource recommendation

Phase 1: Foundation Build

Goals:

- * Cover all AAMC foundational concepts.
- * Establish Anki.
- * Begin daily CARS.
- * Begin untimed passage practice.
- * Build equation and Psych/Soc fluency.

Phase 2: Applied Content and Passage Reasoning

Goals:

- * Shift from reading to questions.
- * Begin mixed practice.

- * Develop experimental-reasoning skill.
- * Train graph interpretation.
- * Repair weak systems.

Phase 3: Timed Integration

Goals:

- * Timed passage blocks.
- * Mixed-topic blocks.
- * Weekly or biweekly full lengths.
- * Detailed error-log use.
- * Weakness-cluster repair.

Phase 4: AAMC Calibration

Goals:

- * Prioritize official AAMC materials.
- * Learn official logic.
- * Use official CARS.
- * Use Section Banks and Question Packs strategically.
- * Take official full lengths under real conditions.

Phase 5: Final Consolidation

Goals:

- * No broad new content.
- * Targeted Anki.
- * Formula refresh.
- * Error-log review.
- * CARS maintenance.
- * Sleep stabilization.
- * Exam-day rehearsal.

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PART XIII: TIMELINE ENGINE

Create separate schedules for:

1. Twelve-month plan.
2. Nine-month plan.
3. Six-month plan.
4. Four-month plan.
5. Three-month intensive plan.
6. Eight-week emergency plan.

For every timeline provide:

- * Weekly goals.

- * Section priorities.
- * Number of Anki reviews.
- * Number of new cards.
- * CARS frequency.
- * Questions per day.
- * Passage blocks per week.
- * Full-length frequency.
- * Official AAMC resource timing.
- * Catch-up protocol.
- * Burnout prevention rules.

Do not simply scale hours.

Different timelines require different resource compression strategies.

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PART XIV: HIGH-YIELD SYSTEMS AND CONFUSION CLUSTERS

Build special “must-master” modules for:

Biology/Biochemistry

- * Amino acids.
- * Enzymes.
- * Metabolism.

- * Genetics.
- * Molecular biology.
- * Cell signaling.
- * Immunology.
- * Organ systems.
- * Experimental techniques.

Chemistry/Physics

- * Acids and bases.
- * Buffers.
- * Electrochemistry.
- * Thermodynamics.
- * Fluids.
- * Circuits.
- * Optics.
- * Gases.
- * Equilibrium.
- * Organic mechanisms.
- * Spectroscopy.

Psychology/Sociology

- * Learning.
- * Memory.
- * Social theories.

- * Development.
- * Personality.
- * Disorders.
- * Identity.
- * Culture.
- * Demographics.
- * Epidemiology.
- * Research design.

CARS

- * Thesis identification.
- * Tone.
- * Scope.
- * Inference.
- * Function.
- * Author intention.
- * Wrong-answer elimination.

For each module include:

- * Why it is high yield.
- * What must be memorized.
- * What must be understood conceptually.
- * What must be applied through passages.
- * Most common traps.

- * Recommended drills.
- * Minimum mastery threshold.

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PART XV: PERFORMANCE METRICS

Build a measurable performance dashboard.

Track:

- * Daily Anki completion.
- * CARS accuracy.
- * CARS timing.
- * Science passage accuracy.
- * Discrete-question accuracy.
- * Data interpretation accuracy.
- * Experimental design accuracy.
- * Graph interpretation accuracy.
- * Math error rate.
- * Content-category accuracy.
- * Full-length section scores.
- * Confidence calibration.
- * Timing failures.
- * Sleep quality.
- * Burnout risk.

Include “red flag” thresholds.

Example:

If CARS accuracy remains below target for 10 consecutive passage sets:

Stop adding volume.

Diagnose question-type pattern.

Return to passage mapping and review.

If physics accuracy is low but equation recall is high:

Diagnose passage translation and unit reasoning.

If Psych/Soc accuracy is low despite large Anki volume:

Diagnose scenario application and confusing-term clusters.

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PART XVI: OUTPUT TABLES REQUIRED

Produce the following final tables.

Table A: Master Content Blueprint

Section	Foundational Concept	Content Category	Topic	Subtopic	Priority	Resource
Practice Type	Anki Type					

Table B: Equation Database

Topic	Equation	Variables	Units	Biological Application	Trap
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Table C: Lab Methods Database

Method	Principle	Output	Interpretation	Common MCAT Use	Trap
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Table D: Psych/Soc Definition Database

Term	Definition	Confused With	Vignette	Flashcard Type
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Table E: CARS Question System

Question Type	What It Tests	Method	Wrong Answer Traps	Review Questions
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Table F: Resource Map

Resource	Best For	Not Best For	When to Use	When to Stop
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Table G: Error Log Taxonomy

Error Type	Diagnostic Clue	Corrective Action	Re-test Method
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Table H: Full-Length Dashboard

Exam	C/P CARS	B/B P/S Total	Main Error Cluster	Repair Plan
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PART XVII: QUALITY CONTROL REQUIREMENTS

Before finalizing the blueprint:

1. Verify all current MCAT facts through official AAMC sources.
2. Verify all current AAMC product names and practice-exam inventory.
3. Separate official facts from third-party recommendations.
4. Do not claim a topic is “high yield” without explaining why.
5. Do not recommend more resources than a student can realistically use.
6. Do not present third-party heuristics as official AAMC rules.
7. Do not use outdated MCAT information.
8. Do not include USMLE-level detail that exceeds MCAT relevance.
9. Do not turn the MCAT into a memorization contest.
10. Do not underweight CARS.
11. Do not treat full lengths as mere score checks.
12. Do not recommend using all AAMC materials too early.
13. Cite every external source used.
14. Include direct source links in a final reference appendix.
15. Flag any claim that is based on expert consensus rather than official AAMC guidance.



FINAL DELIVERABLE

The finished product must read like a hybrid of:

- * an AAMC-aligned curriculum map,
- * a technical operations manual,
- * an Anki architecture,
- * a question-bank system,
- * a diagnostic engine,
- * a full-length review protocol,
- * and a high-performance training plan.

It should be possible to turn the final blueprint into:

1. A Notion workspace.
2. A spreadsheet database.
3. A Noctyrium course category.
4. A daily study dashboard.
5. An Anki tagging system.
6. A personalized MCAT scheduler.
7. A weak-area recommendation engine.

Do not produce a short answer.

Do not omit foundational concepts.

Do not merely list resources.

Build the actual machine.

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A useful strategic correction: do not let the system treat “content review” as the main event. The MCAT rewards knowledge, but the score rises when you can translate unfamiliar passages, analyze experiments, read graphs, reject seductive distractors, and keep functioning after two hours of CARS humanities prose written by someone who clearly resented punctuation.

The official AAMC material should dominate the final phase because its full-lengths replicate test-day functionality and use scaled scores; its official prep library also includes targeted sets for science and CARS.